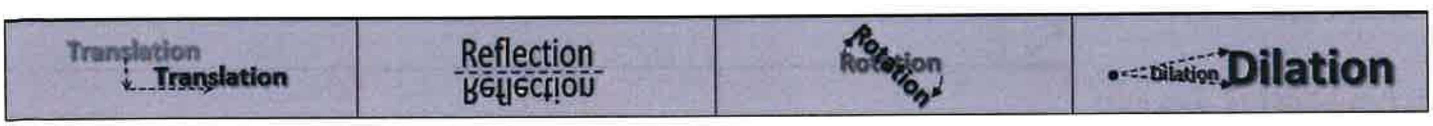
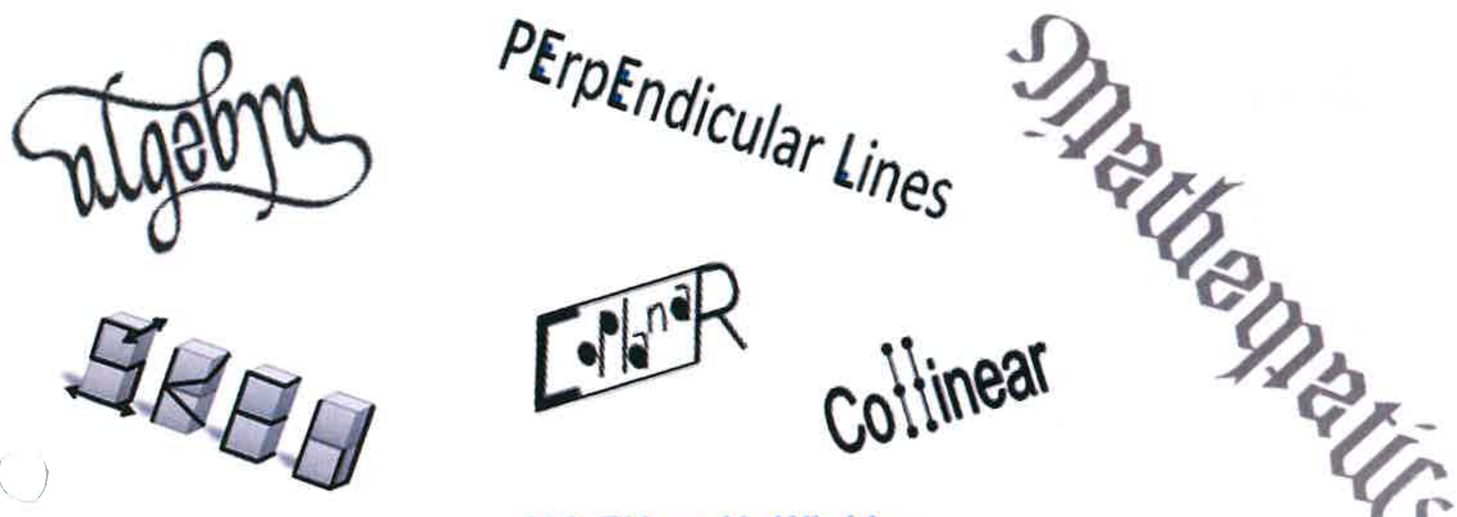
Geometry

Unit 1: Congruence, Angles, and Proofs (Part 1)





MAtTHew M. Winking

Sec 1.1 –Transformation in the Coordinate Plane

Geometric Definitions

Name:

Define the following Geometric terms:

point:

LINE:

SEGMENT:

RAY:

Plane:

Collinear:

CoplanaR:

ANGLE:

Parallel Lines:

PErpEndicular Lines:

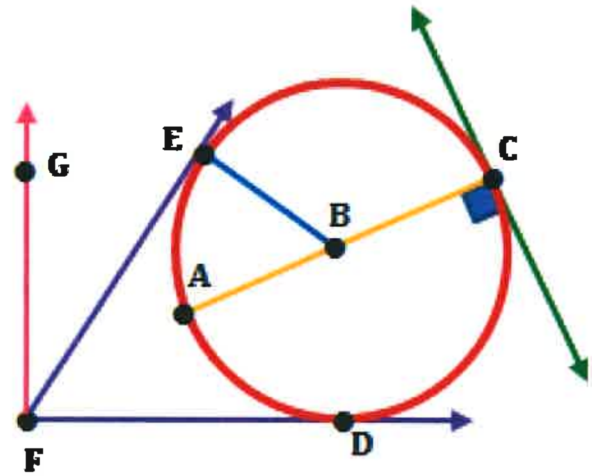
SKEW:

Circle:

Use the word bank below to describe each object or set of objects.

Line	Point	Perpendicular	Segment		Plane
	Parallel	Diameter	Ray	Radius	
Circle		Angle	Skew	Coplanar	Collinear

1. How would you best describe the **purple geometric shape** shown in the diagram at the right?



2. How would you best describe the **red geometric shape** shown in the diagram at the right?

3. How would you best describe the **center of the circle** shown in the diagram at the right?

4. How would you best describe the pink geometric shape shown in the diagram above?

5. How would you best describe the **blue geometric shape** shown in the diagram above?

6. How would you best describe the **green geometric shape** shown in the diagram above?

7. How would you best describe the orange geometric shape shown in the diagram above?

8. How would you best describe the relation between the set of geometric shapes **Point A, Point B, and Point C**?

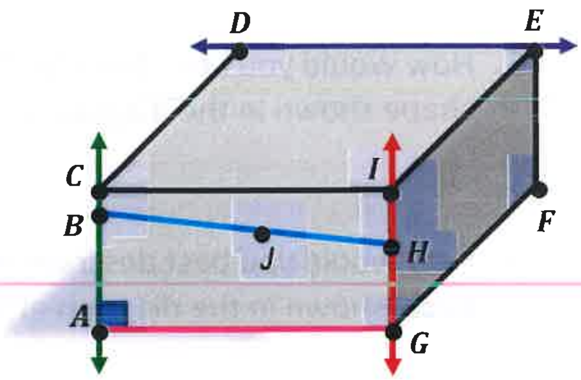
9. How would you best describe the relation between the set of **green** and orange geometric shapes?

Use the word bank below to describe each object or set of objects.

Line Point Perpendicular Segment Plane
Parallel Diameter Ray Radius
Circle Angle Skew Coplanar Collinear

The geometric shape shown in the diagram below is a 3-dimensional rectangular prism.

10. How would you best describe the relationship between the line $\overleftrightarrow{AC}$ and the line $\overleftrightarrow{GI}$?



11. How would you best describe the relationship between the line $\overleftrightarrow{AC}$ and the segment $\overline{AG}$?

12. How would you best describe the relationship between the line $\overleftrightarrow{AC}$ and the line $\overleftrightarrow{DE}$?

13. How would you best describe the set of **Point A, Point B, Point J and Point H**?

14. How would you best describe the set of **Point G, Point H, and Point I**?

15. (True or False) Any 3 distinct points are always coplanar.

16. (True or False) Any 2 distinct circles are always coplanar.

17. (True or False) If 2 lines intersect once then the lines are coplanar.

18. (True or False) Two lines that are skew can sometimes intersect.

19. (True or False) An angle and a circle can have more than 3 intersections.

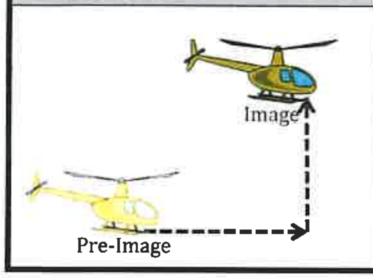
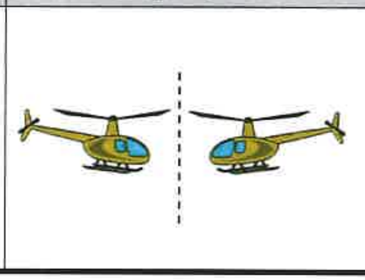
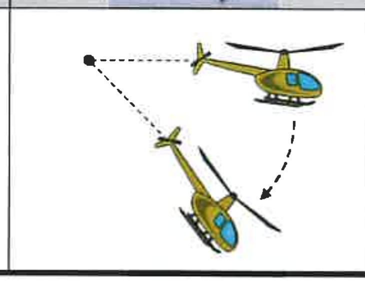
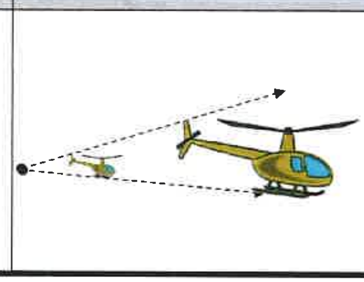
20. (True or False) Two distinct circles can have more than 2 intersections.

21. (True or False) Any given line and point are always coplanar.

Sec 1.2 Transformations in the Coordinate Plane

Identifying Types of Transformations Name:

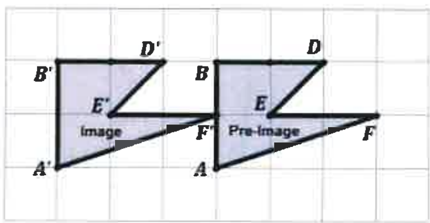
Transformation Types:

Translation	Reflection	Rotation	Dilation
			

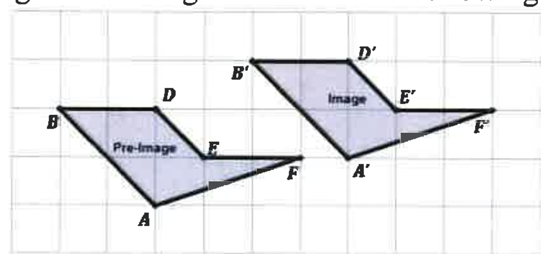
Translations

1. Describe the transformation in rectangular units from the Pre-Image to the Image for each of the following:

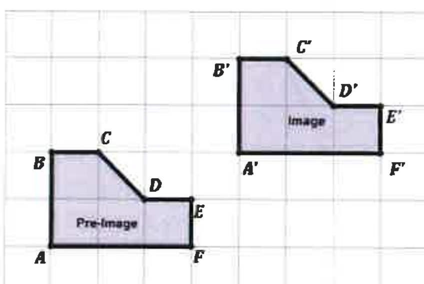
A.



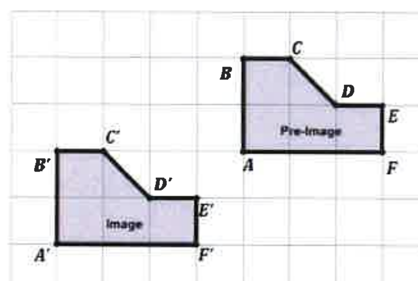
B.



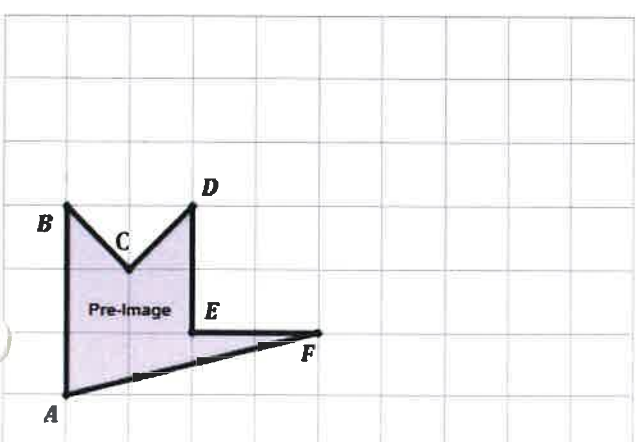
C.



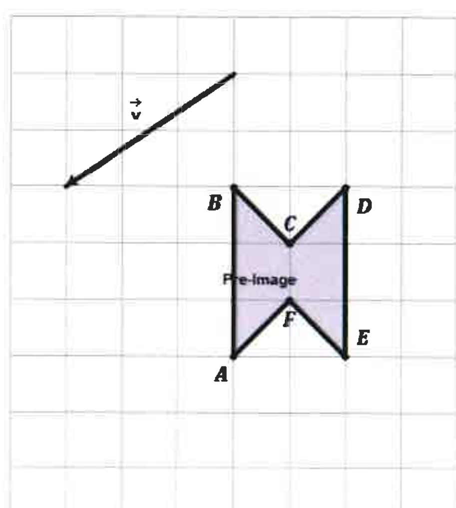
D.



2. Translate the following object right 4 units and up 1 unit. Label each vertex appropriately.



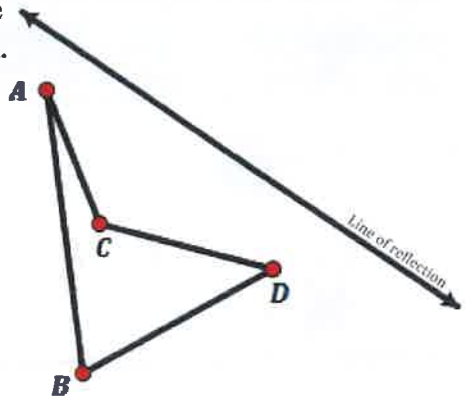
3. Translate the following object by the vector $\vec{v}$.



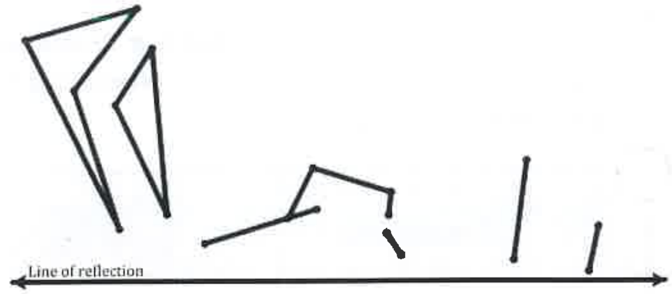
Reflections

4. Create a reflection of the Pre-image over the line of reflection.

a.

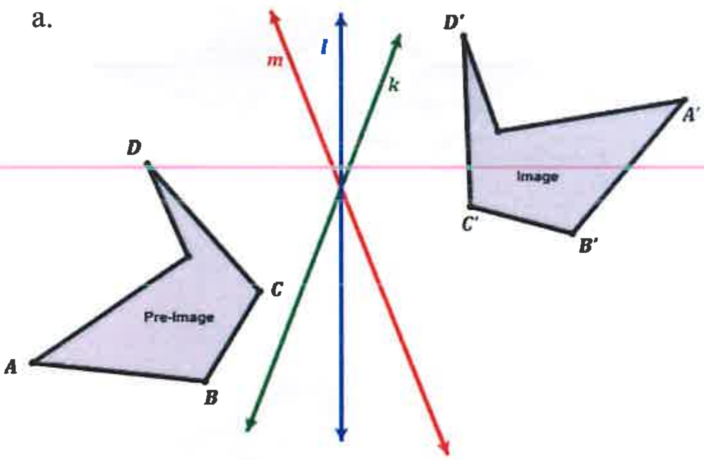


b.

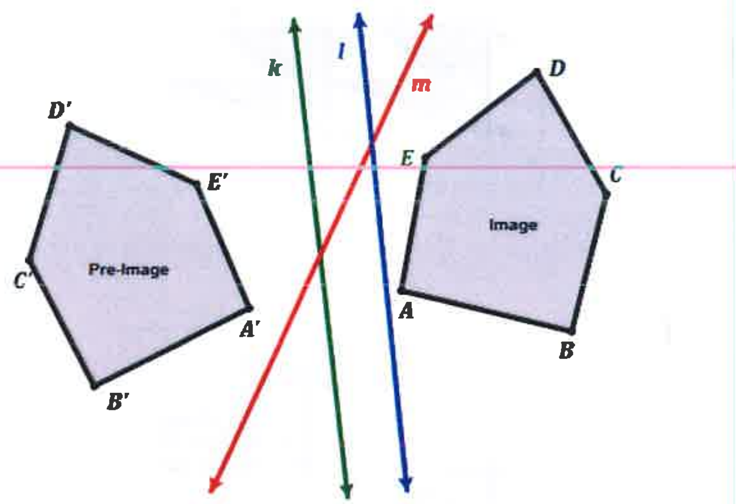


5. Determine which is the correct line of reflection in each diagram between the Image and Pre-Image?

a.

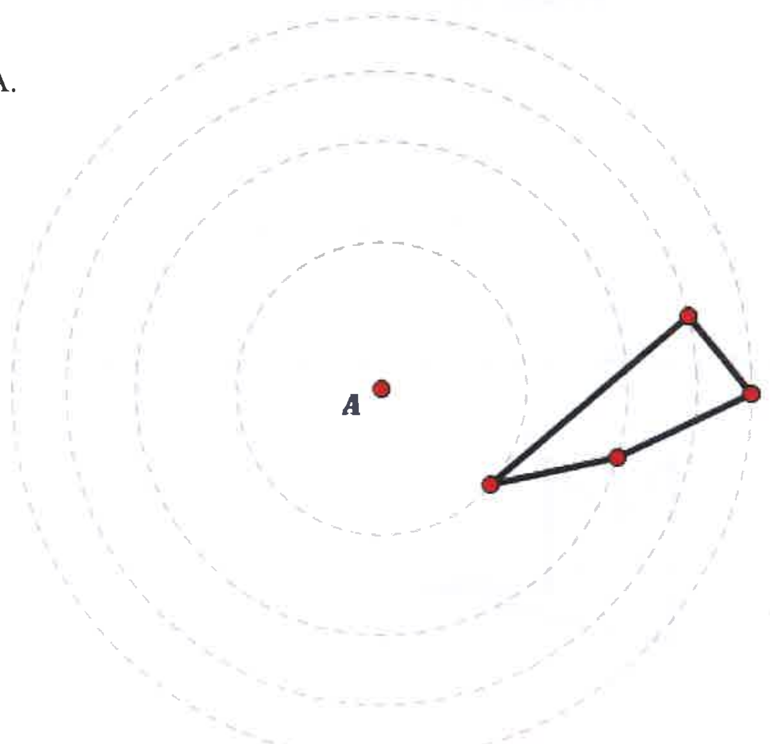


b.



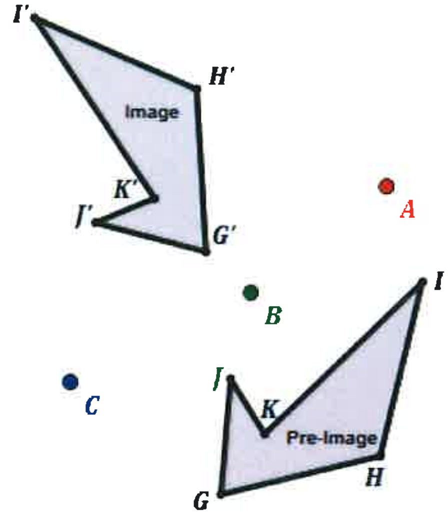
Rotations

6. Rotate the following polygon 110° about the point A.

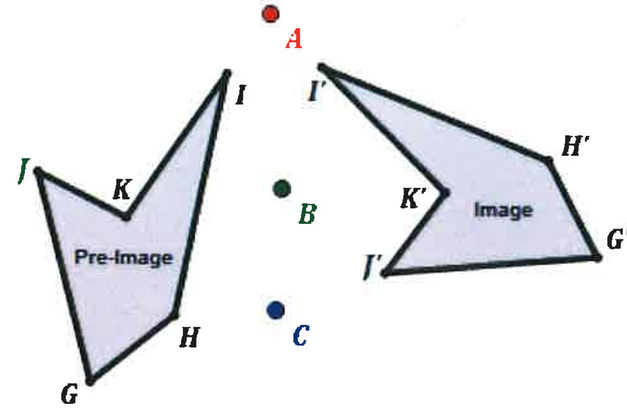


7. Determine which is the correct center if an 80° rotation was used?

a.

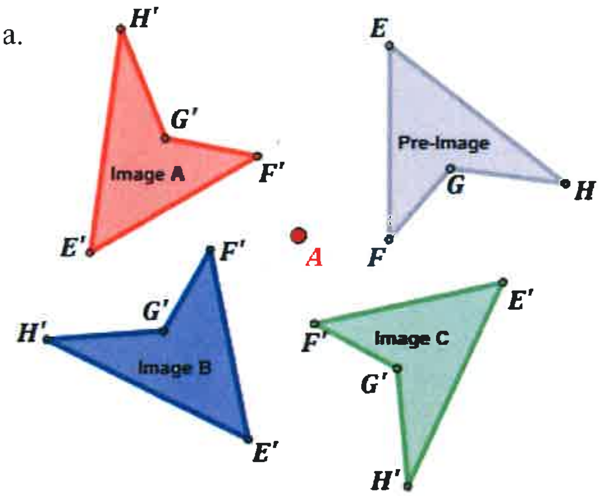


b.

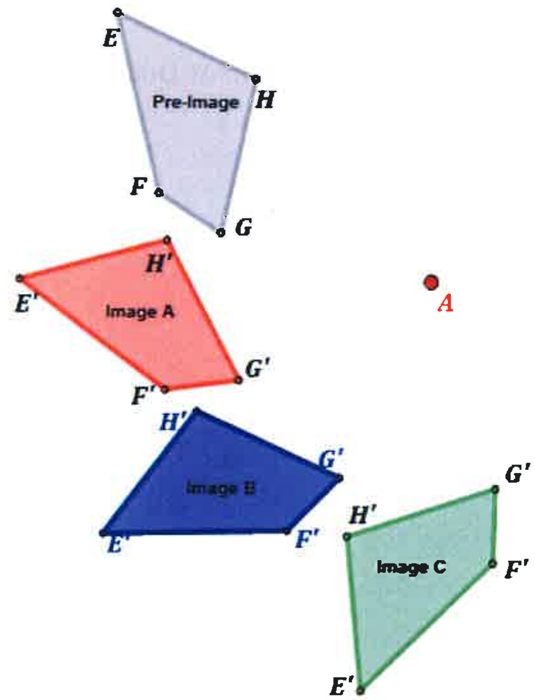


8. Which Image is a rotation of 120° about the point A?

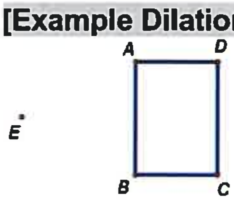
a.



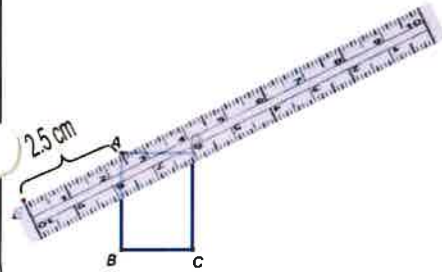
b



[Example Dilation]: Dilate the □ABCD by a factor of 2.0 from point E.



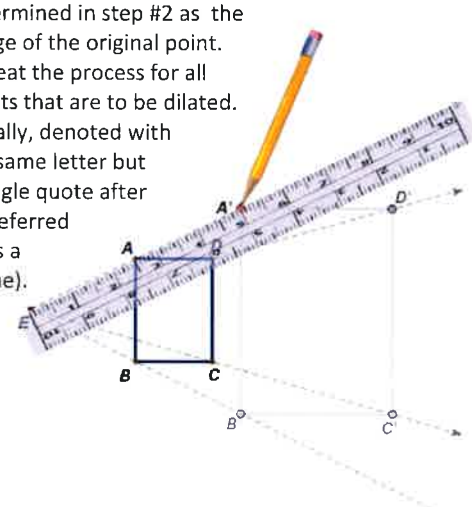
<u>Step I</u>: Measure the distance from the point of dilation to a point to be dilated (preferably using centimeters).



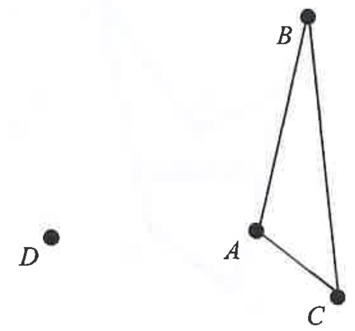
<u>Step 2</u>: Multiply the measured distance by the scale factor.

$2.5cm \times 2.0 = 5\ cm$

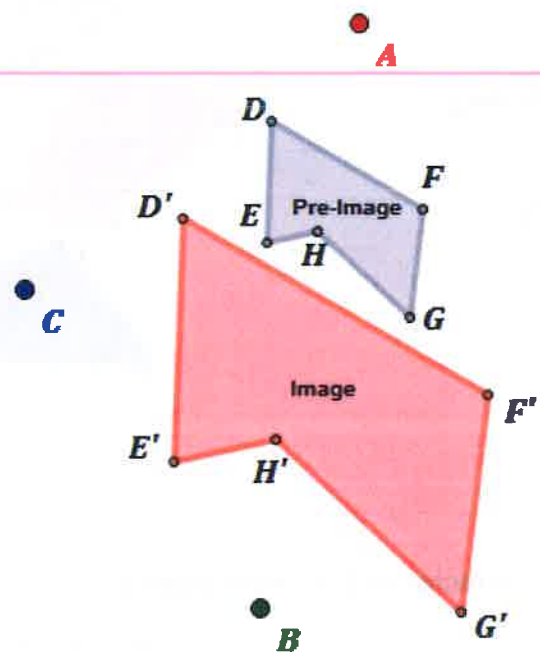
<u>Step I</u>: With the ruler in the same place as it was in step #1, mark a point at the measured distance determined in step #2 as the image of the original point. Repeat the process for all points that are to be dilated. Usually, denoted with the same letter but a single quote after it (referred to as a prime).



9. Dilate the Δ ABC by a factor of $\frac{3}{2}$ from point D.



10. Which is the correct point of Dilation if the pre-image was dilated by a factor of 2?

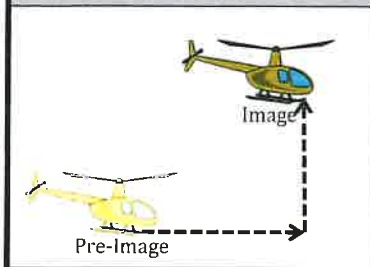
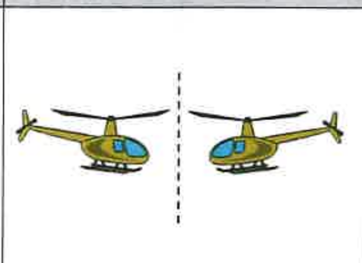
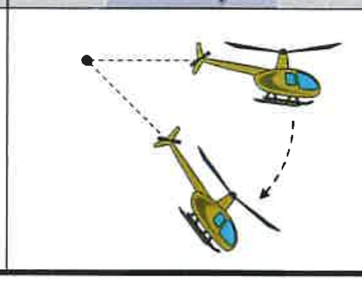
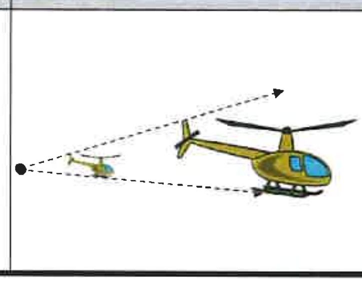


11. Which of the four transformations are **isometries**?

Sec 1.3 Transformations in the Coordinate Plane

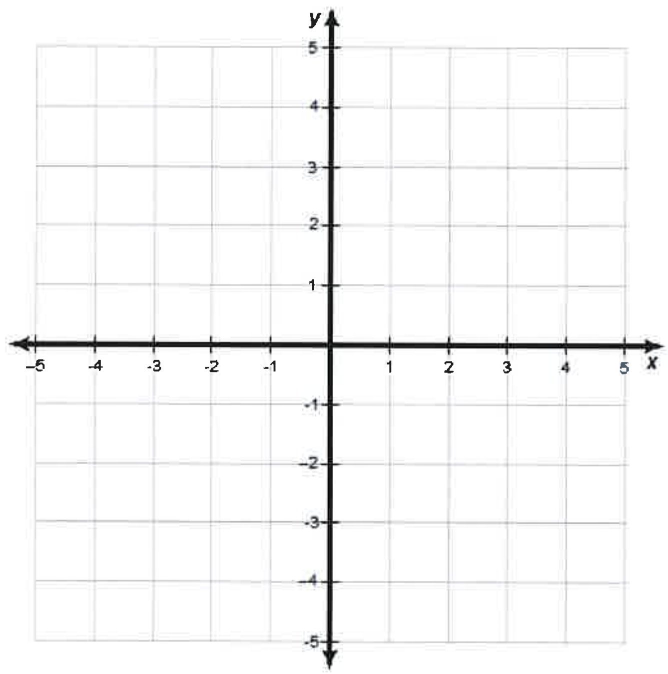
Coordinate Transformations Name:

Transformation Types:

Translation	Reflection	Rotation	Dilation
			

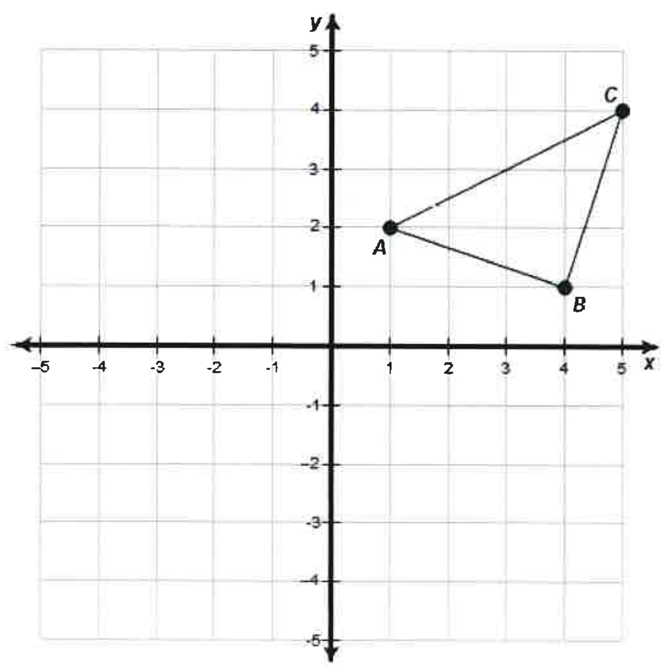
Translations

1. Plot the points and create the pre-image triangle with vertices M(– 2, 3), N(–3, 5), and P (–5,2). Then, determine the coordinates M', N', and P' of the image triangle that has been translated right 6 and down 3. Explain what you did to each of the coordinates.



Reflections

2. List the coordinates of the triangle ABC

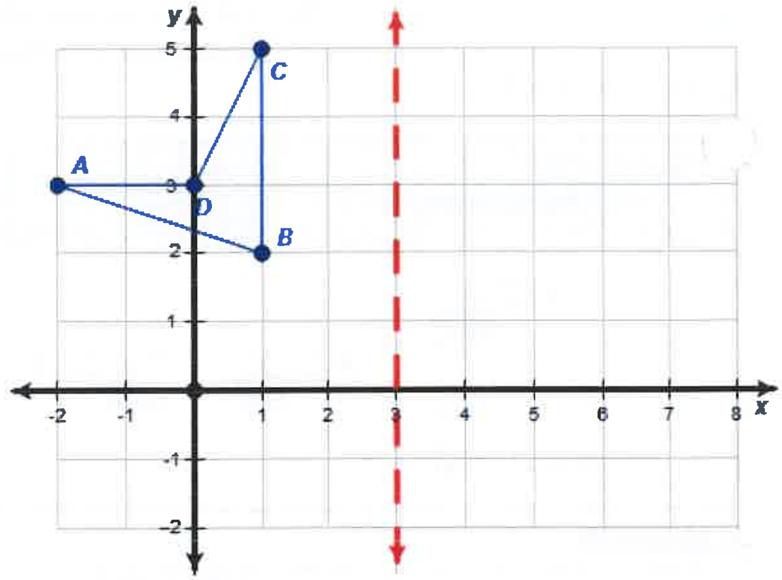


3. Reflect the triangle ABC over the **y-axis** and list the coordinates of the vertex A', B', and C'. Describe what happened to the coordinates from the pre-image to the image.

4. Reflect the triangle ABC over the **x-axis** and list the coordinates of the vertex A', B', and C'. Describe what happened to the coordinates from the pre-image to the image.

Reflections

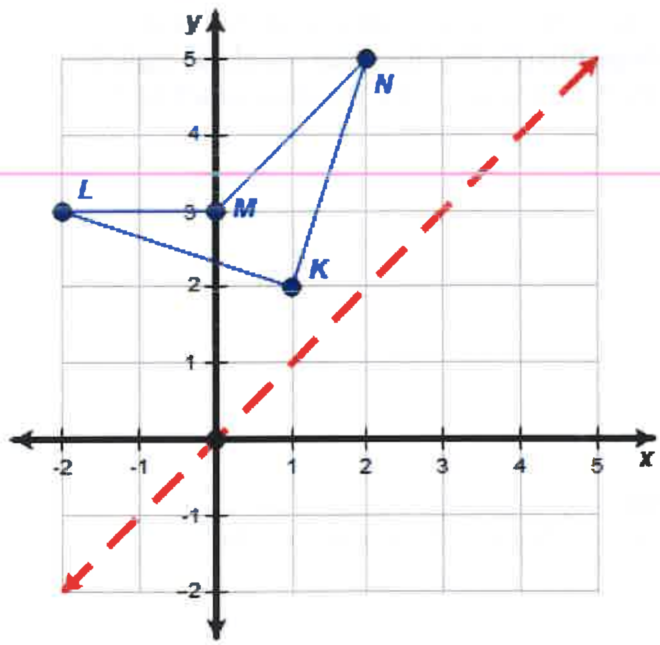
5. List the coordinates of the quadrilateral ABCD



6. Reflect the quadrilateral ABCD over the line **x = 3** and list the coordinates of the vertex A', B', C', and D'.

Reflections

7. List the coordinates of the quadrilateral KLMN



8. Reflect the quadrilateral KLMN over the line **y = x** and list the coordinates of the vertex K', L', M', and N'. Describe what happened to the coordinates from the pre-image to the image.

Reflections

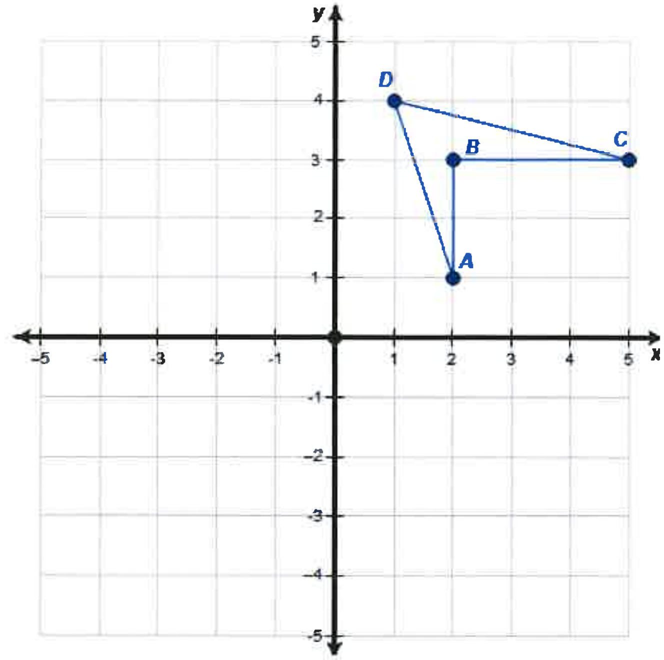
9. If the point A is located at (– 3, 2) and A' is the image of A after being reflected over the **x-axis**, what are the coordinates of A'?

10. If the point B is located at (– 4, – 1) and B' is the image of B after being reflected over the **y-axis**, what are the coordinates of B'?

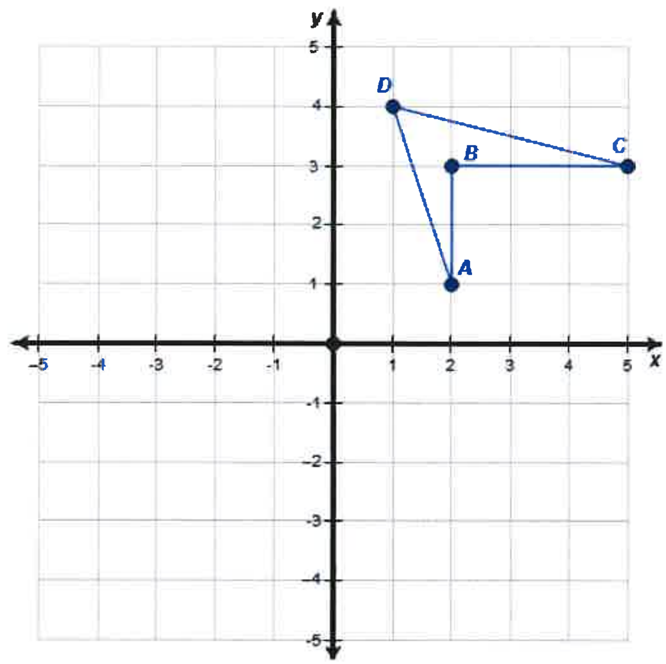
11. If the point C is located at (2, – 3) and C' is the image of C after being reflected over the line **y = x**, what are the coordinates of C'?

Rotations

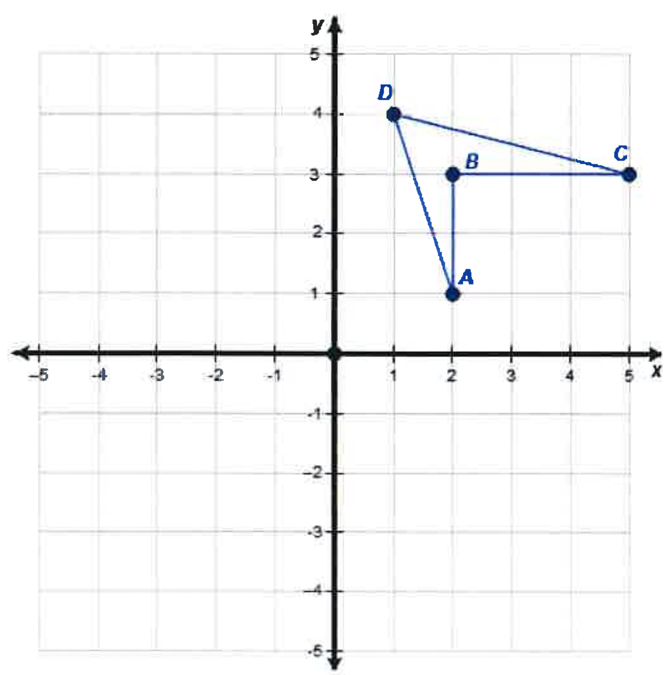
12. List the coordinates of the quadrilateral ABCD.



13. Rotate the quadrilateral **ABCD about the origin by 90°** and list the coordinates of the vertex A', B', C', and D'. Describe what happened to the coordinates from the pre-image to the image.



14. Rotate the quadrilateral **ABCD about the origin by 180°** and list the coordinates of the vertex A', B', C', and D'. Describe what happened to the coordinates from the pre-image to the image.

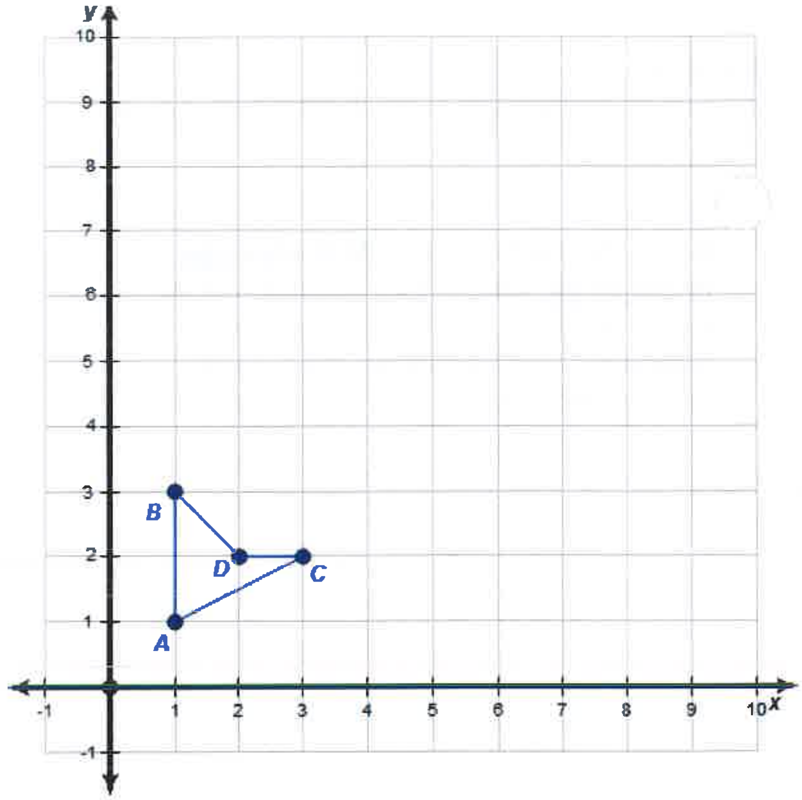


15. Rotate the quadrilateral **ABCD about the origin by 270°** and list the coordinates of the vertex A', B', C', and D'. Describe what happened to the coordinates from the pre-image to the image.

16. If the point A is located at (– 3, 2) and A' is the image of A after being rotated about the origin by 270. What are the coordinates of A'?

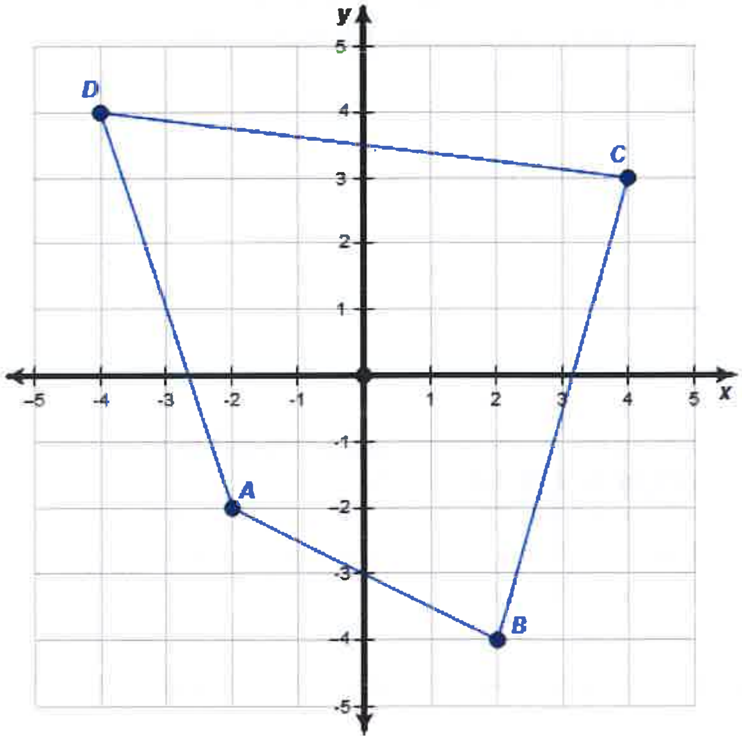
Dilations

17. List the coordinates of the quadrilateral ABCD.



18. Dilate the quadrilateral ABCD **by a scale factor of 3 from the origin** and list the coordinates of the vertex A', B', C', and D'. Describe what happened to the coordinates from the pre-image to the image.

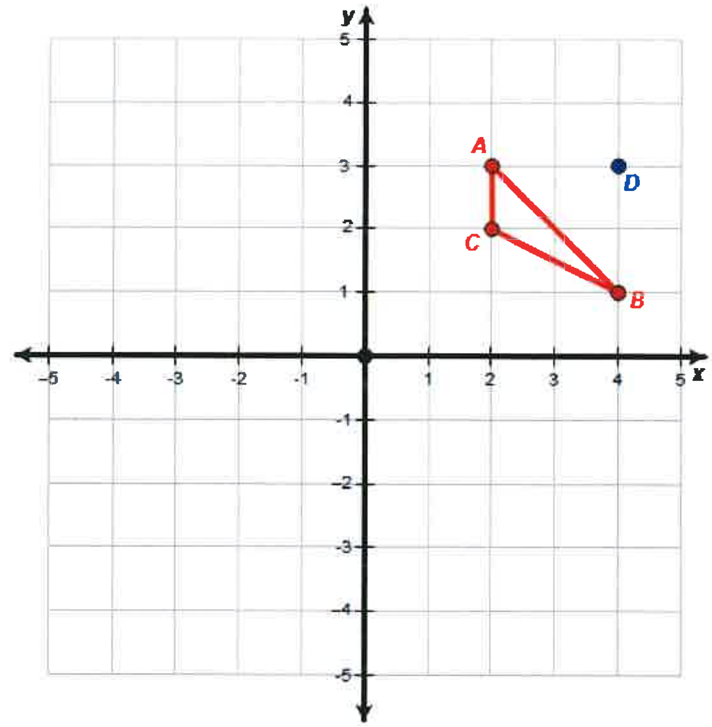
19. List the coordinates of the quadrilateral ABCD.



20. Dilate the quadrilateral ABCD **by a scale factor of ½ from the origin** and list the coordinates of the vertex A', B', C', and D'. Describe what happened to the coordinates from the pre-image to the image.

21. If the point A is located at (3, – 2) and A' is the image of A after being dilated by a scale factor of 5 from the origin. What are the coordinates of A'?

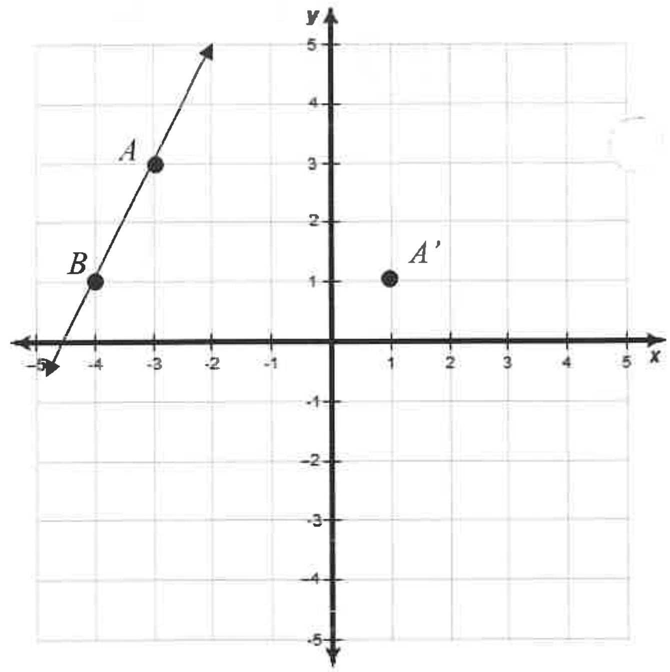
22. Dilate the triangle ABC **by a scale factor of 4 from the point D** and list the coordinates of the vertex A', B', and C'.



Create a list of all of the basic coordinate transformation rules:

Translations

23. Point A (-3, 3) is on $\overleftrightarrow{AB}$. A translation moves the point A to its image A'(1,1).

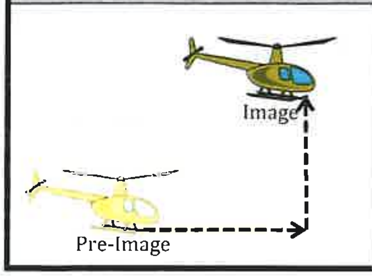
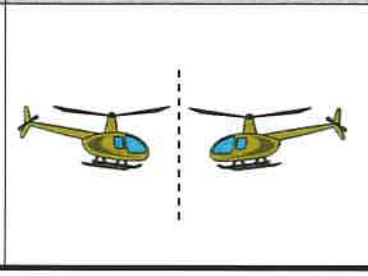
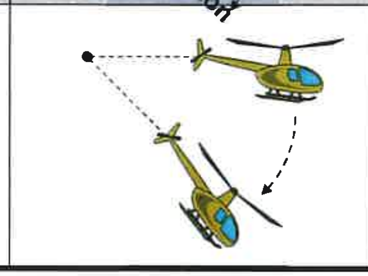
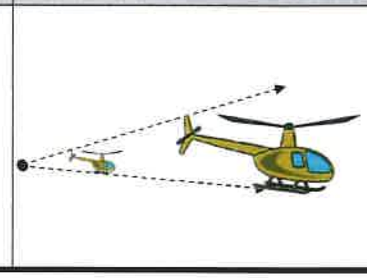


What is the distance, in units, between any point on $\overleftrightarrow{AB}$ and its image?

Sec 1.4 Transformations in the Coordinate Plane

Compound Transformations Name:

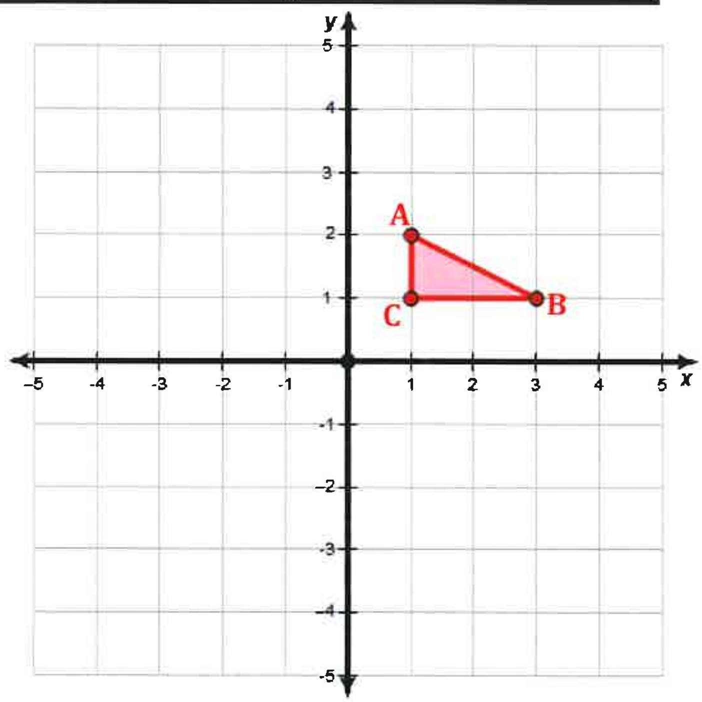
Transformation Types:

Translation	Reflection	Rotation	Dilation
			

1. Consider the pre-image triangle with vertices A(1,2), B(3,1), and C(1,1).

   a. Rotate the pre-image triangle ABC 90° about the origin and label this triangle A'B'C'

   b. Reflect the triangle A'B'C' over the x-axis and label this triangle A"B"C".

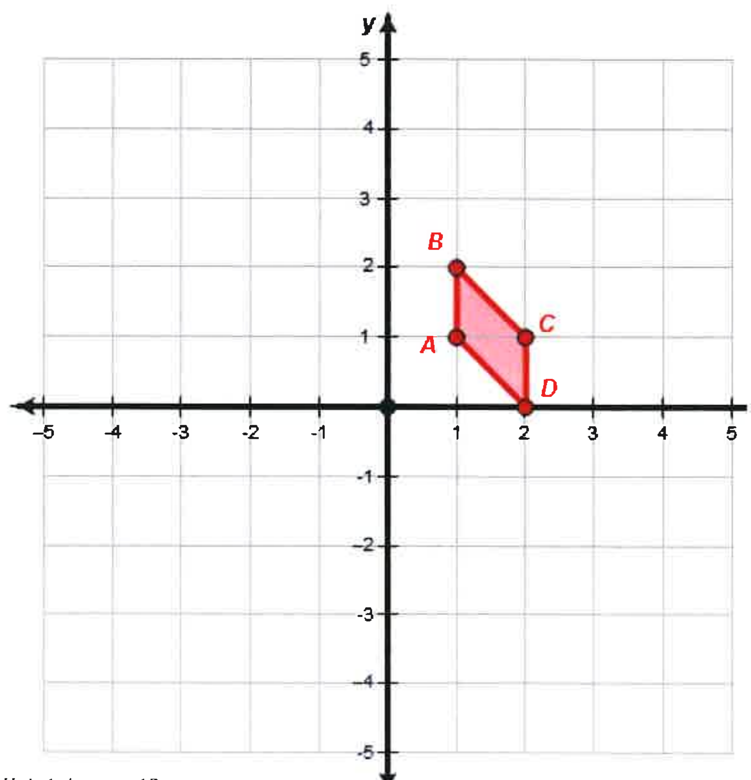


2. Consider the pre-image quadrilateral with vertices A(1,1), B(1,2), C(2,1), and D(2,0)

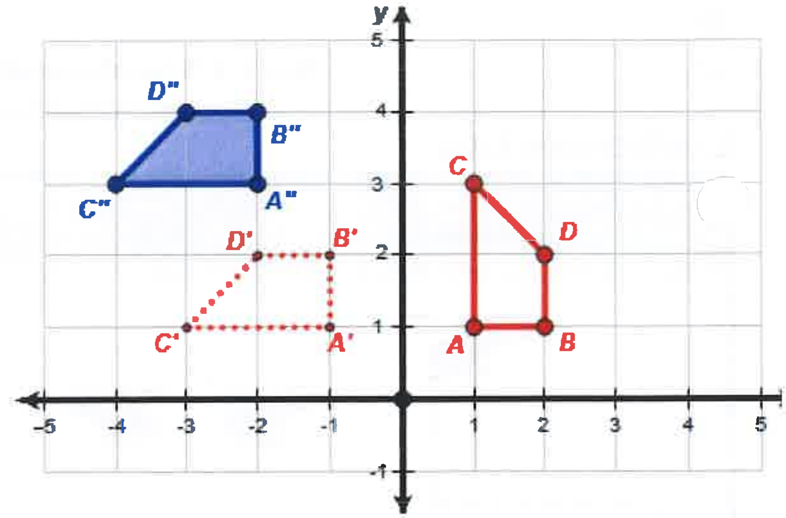
   a. Dilate the quadrilateral by a factor of 2 from the origin and label the image quadrilateral A'B'C'D'.

   b. Translate the quadrilateral image A'B'C'D' down 5 units and left 1 unit. Label this new image A"B"C"D".

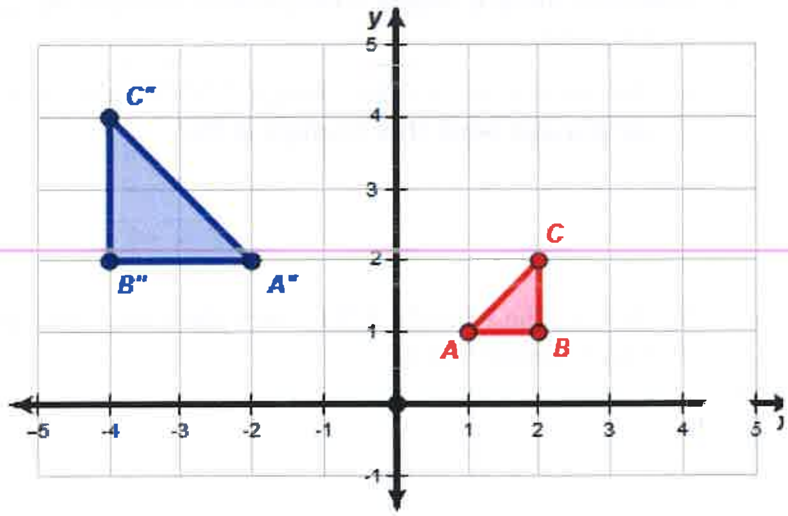
   c. Reflect the quadrilateral image A"B"C"D" over the y-axis and label this image A"'B"'C"'D"'



3. Describe two transformation that would map quadrilateral ABCD onto quadrilateral A"B"C"D"

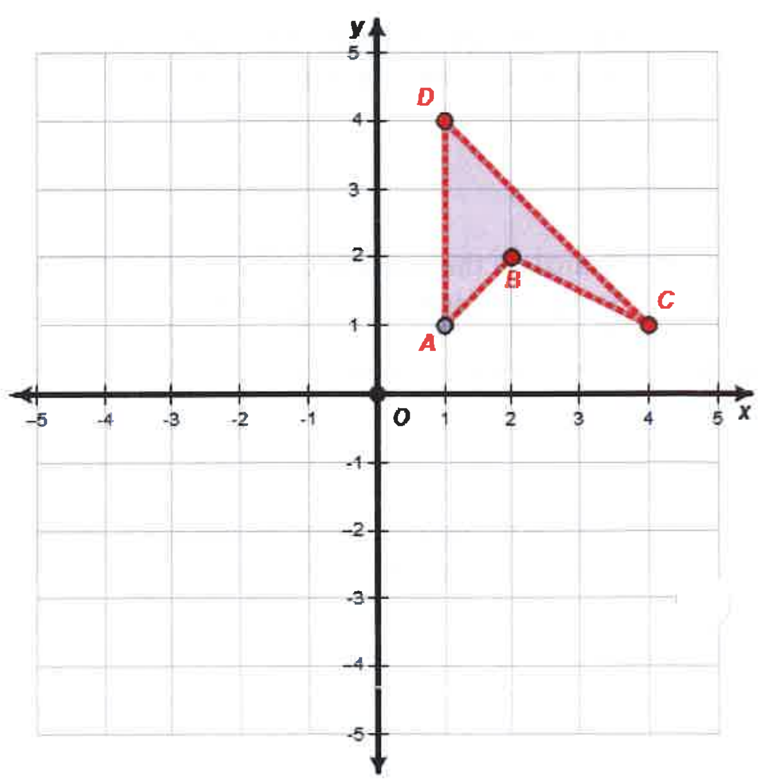


4. Describe two transformation that would map triangle ABC onto triangle A"B"C"

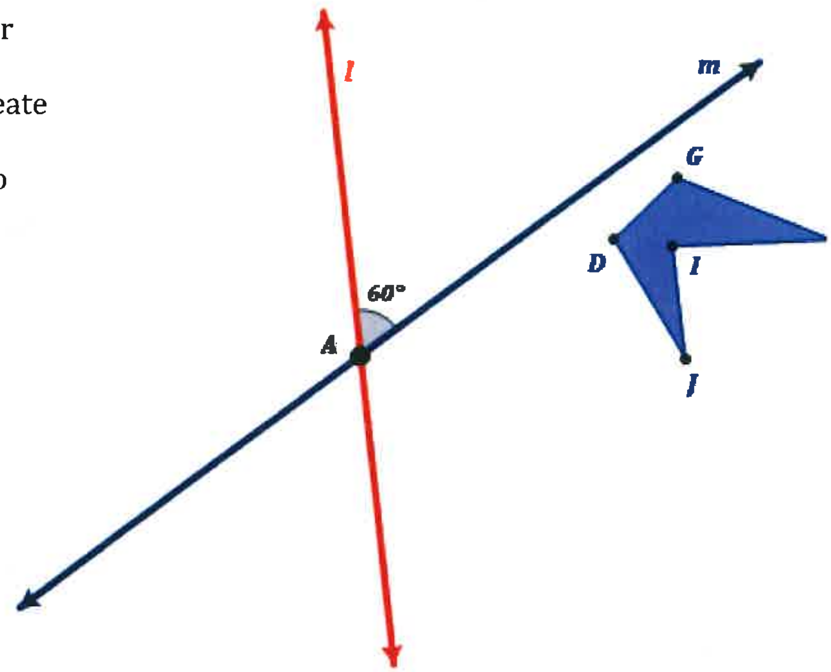


5. First reflect quadrilateral ABCD over the y-axis and label the triangle A'B'C'D'. Then, reflect quadrilateral A'B'C'D' over the x-axis and label this quadrilateral A"B"C"D".

6. Anytime you use a double reflection, there should be a rotation about the intersection of the reflection lines that maps the pre-image onto the final image. In this example what is the amount of the rotation?

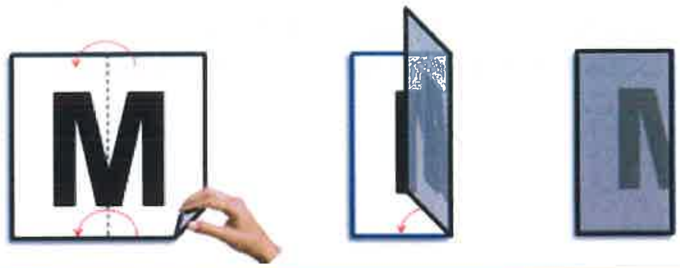


7. Given that pentagon DGHIJ is first reflected over ***line m*** to create the image D'G'H'I'J'. Then, the image D'G'H'I'J' is reflected over the ***line l*** to create the image D"G"H"I"J". What is a different transformation that would also map DGHIJ onto D"G"H"I"J"?

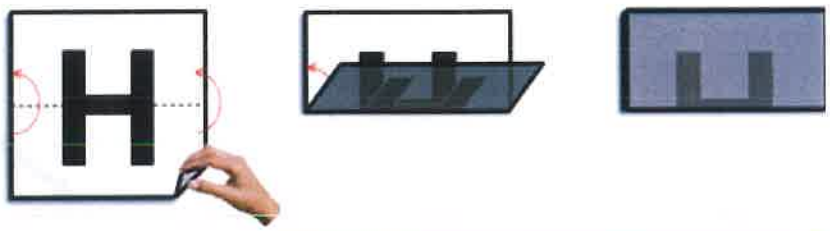


Symmetries

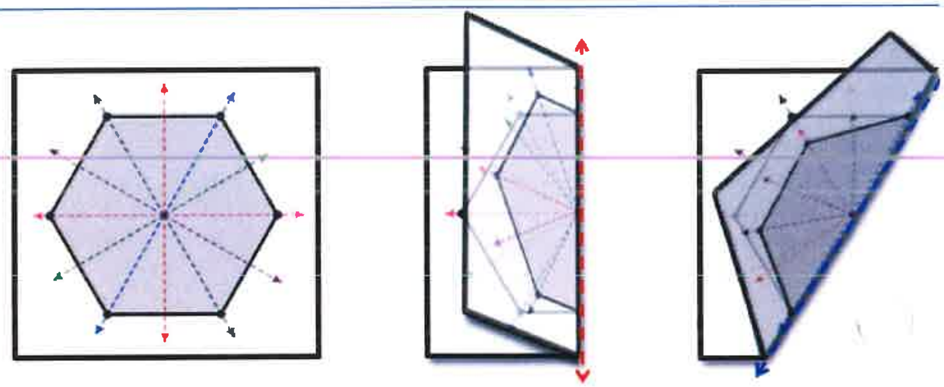
A shape has a **vertical line of symmetry** if you can fold it in half vertically and have the halves match up. On a scrap sheet of paper try folding the letter "M" in half.



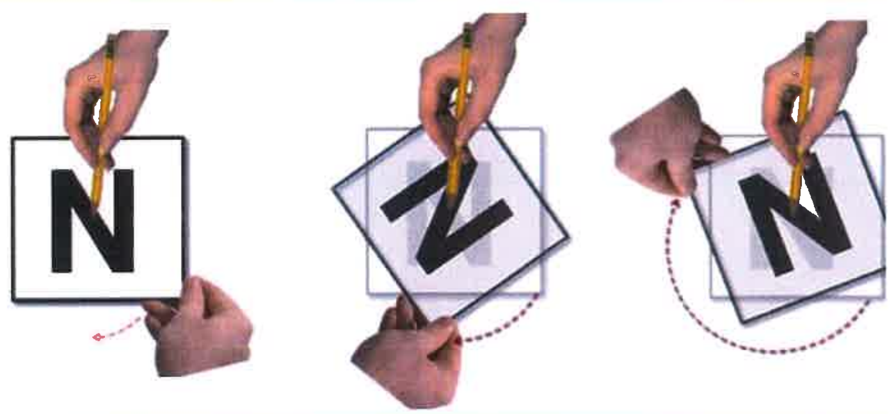
A shape has a **horizontal line of symmetry** if you can fold it in half horizontally and have the halves match up. On a scrap sheet of paper try folding the letter "H" in half.



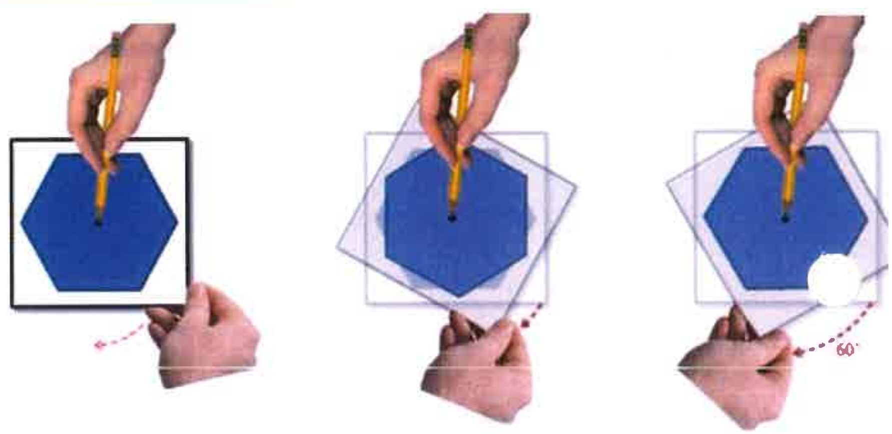
A general **line of symmetry** is any line in which you can fold the shape in half and it maps onto itself. Consider the hexagon at the right has 6 lines of symmetry.



A shape has **point symmetry** if you can rotate the shape 180˚ from a center point and have it look the same as it did before you rotated it. On a scrap sheet of paper try rotating a letter "N" as shown a the right.



A shape has **rotational symmetry** if you can rotate the shape by a set degree from a center point and have it look the same as it did before you rotated it. On a scrap sheet of paper try rotating a hexagon (which has 60˚ rotational symmetry.)



1. Provide 2 Letters of the Alphabet that have a **vertical line of symmetry**.

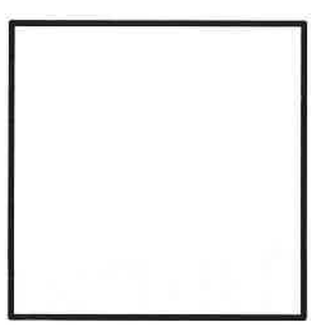
2. Provide 2 Letters of the Alphabet that have a **horizontal line of symmetry**.

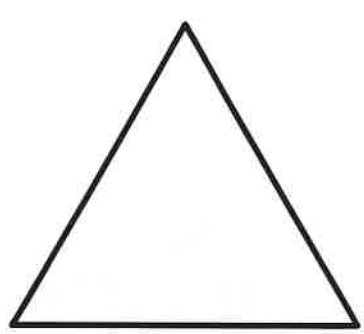
3. Provide 2 Letters of the Alphabet that have both a **horizontal line of symmetry** and **vertical line of symmetry**.

4. Provide a letter of the Alphabet that has **point symmetry** but **NOT** a **vertical line of symmetry.**

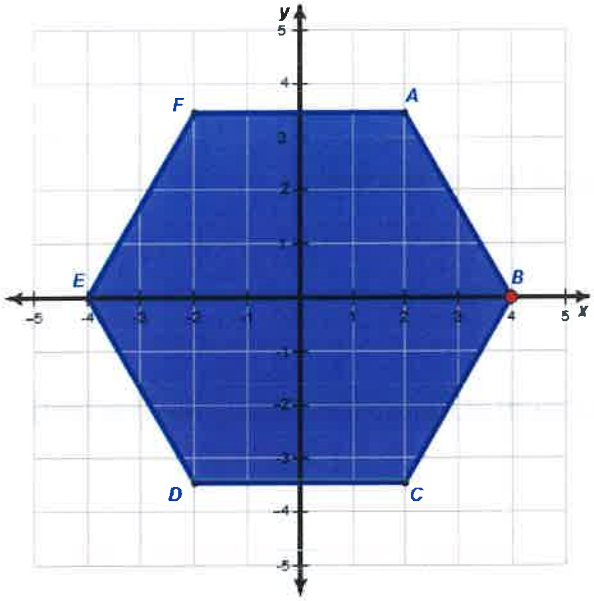
5. Which letter depending on how it's written could have **infinite lines of symmetry**?

6. Draw all the lines of symmetry for the following regular shapes.

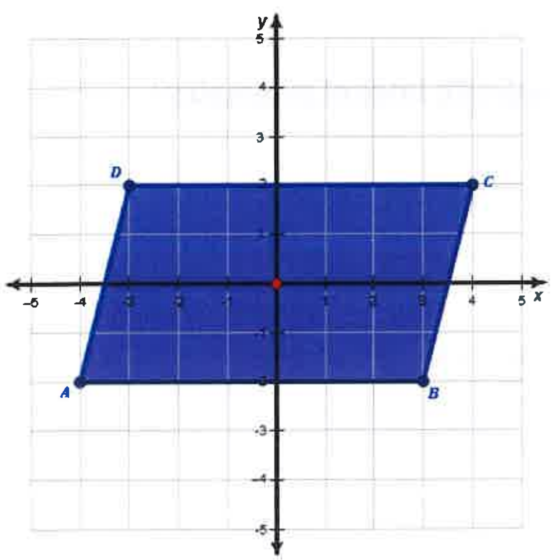




7. Describe in detail at least three transformations that would map hexagon ABCDEF onto itself.



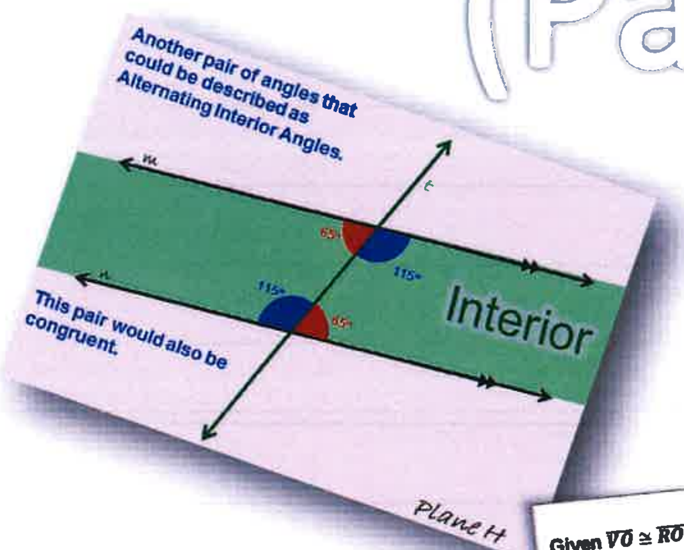
8. Describe any symmetries that parallelogram ABCD might have.

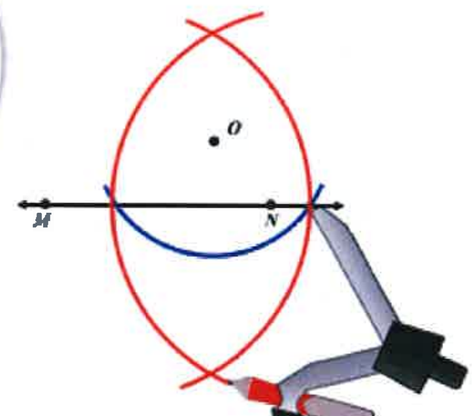


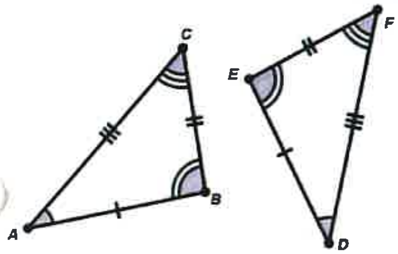
9. Ambigrams are usually words created with point symmetry.

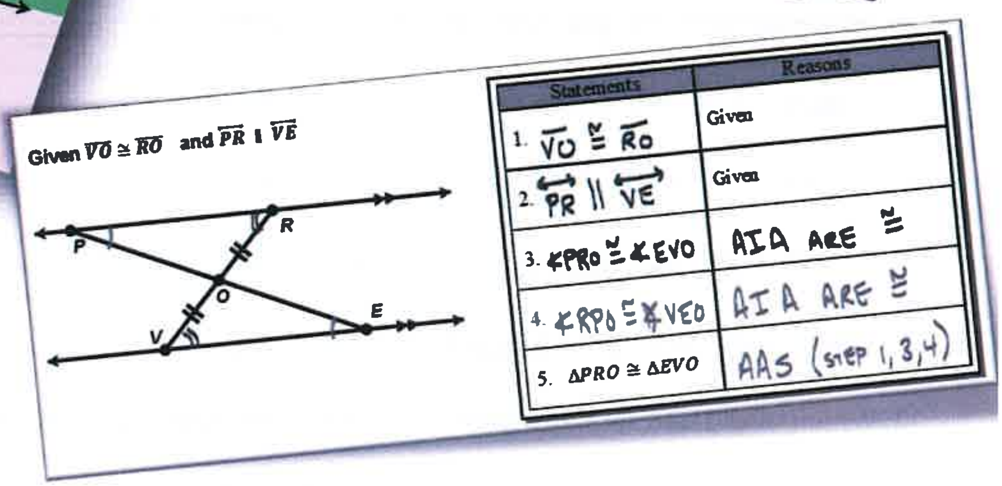
Geometry

Unit 1: Congruence, Angles, and Proofs (Part 2)








Given $\overline{VO} \cong \overline{RO}$ and $\overline{PR} \parallel \overline{VE}$

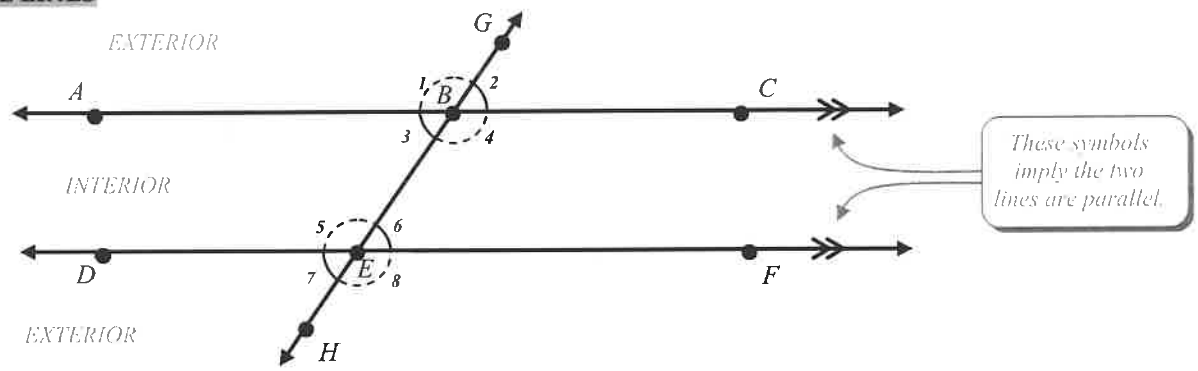
Statements	Reasons
1. $\overline{VO} \cong \overline{RO}$	Given
2. $\overleftrightarrow{PR} \parallel \overleftrightarrow{VE}$	Given
3. $\angle PRO \cong \angle EVO$	AIA ARE $\cong$
4. $\angle RPO \cong \angle VEO$	AIA ARE $\cong$
5. $\Delta PRO \cong \Delta EVO$	AAS (STEP 1, 3, 4)

MAtTHew M. Winking

Sec 1.6 Geometry – Parallel Lines and Angles

Name:

PARALLEL LINES

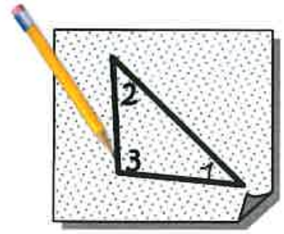


WORD BANK:

$\measuredangle GBC$ | Consecutive Interior Angles | Alternate Interior Angles | $\measuredangle 5$ | $\measuredangle DEH$ | Transversal | Congruent | $\measuredangle 4$ | $\measuredangle HEF$ | Vertical Angles | Alternate Exterior Angles | Corresponding Angles | Consecutive Exterior Angles | $\measuredangle ABG$ | Supplementary

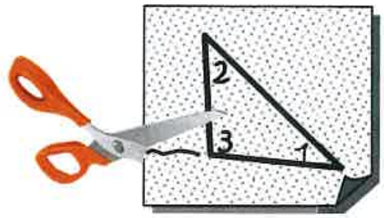
1. Give an alternate name for angle $\measuredangle 2$ using 3 points: ________________
2. Angles $\measuredangle ABE$ and $\measuredangle CBG$ can best be described as: ________________
3. Angles $\measuredangle 6$ and $\measuredangle 3$ can best be described as: ________________
4. The line $\overleftrightarrow{GH}$ can best be described as a: ________________
5. Which angle corresponds to $\measuredangle DEB$: ________________
6. Angles $\measuredangle FEB$ and $\measuredangle CBE$ can best be described as: ________________
7. Angles $\measuredangle 1$ and $\measuredangle 8$ can best be described as: ________________
8. Which angle is an alternate interior angle with $\measuredangle CBE$: ________________
9. Angles $\measuredangle GBC$ and $\measuredangle BEF$ can best be described as: ________________
10. Angles $\measuredangle 2$ and $\measuredangle 8$ can best be described as: ________________
11. Which angle is an alternate exterior angle with $\measuredangle ABG$: ________________
12. Which angle is a vertical angle to $\measuredangle ABG$: ________________
13. Which angle can be described as consecutive exterior angle with $\measuredangle 1$: ________________
14. Any two angles that sum to 180˚ can be described as ________________ angles.

TRIANGLE's INTERIOR ANGLE SUM

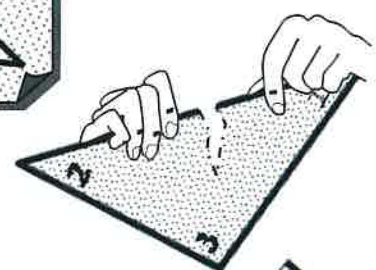


1. a. **First, Create a random triangle on a piece of patty papers.**

   b. **Using your pencil, write a number inside each interior angle a label.**



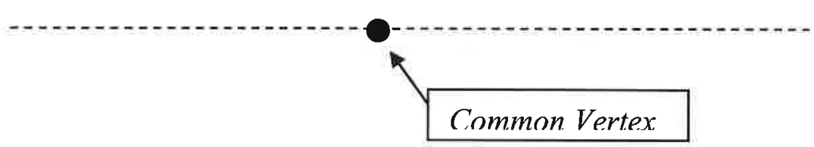
   c. **Next, cut out the triangle.**



   d. **Finally, tear off or cut each of the angles from the triangle**

   e. **Using tape, carefully put all 3 angles next to one another so that they all have the same vertex and the edges are touching but they aren't overlapping**

Paste or Tape your 3 vertices here:

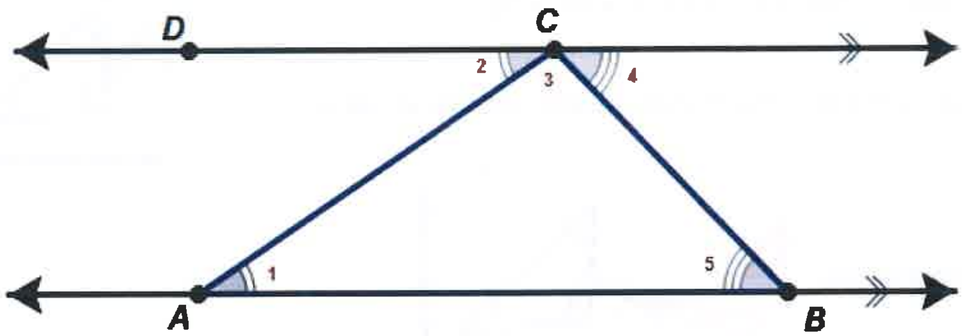


2. **What is the measure of a straight angle or the angle that creates a line by using two opposite rays from a common vertex?**

3. **Collectively does the sum of your 3 interior angles of a triangle form a straight angle? What about others in your class?**

4. **Make a conjecture about the sum of the interior angles of a triangle. Do you think your conjecture will always be true? (please explain using complete sentences)**

5. More formally, why do the 3 interior angles of any triangle sum to 180˚?



Consider $\triangle ABC$. The segment $\overline{AB}$ is extended into a line and a parallel line is constructed through the opposite vertex. So, $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$.

a. Why is $\measuredangle 1 \cong \measuredangle 2$?____

b. Why is $\measuredangle 5 \cong \measuredangle 4$?____

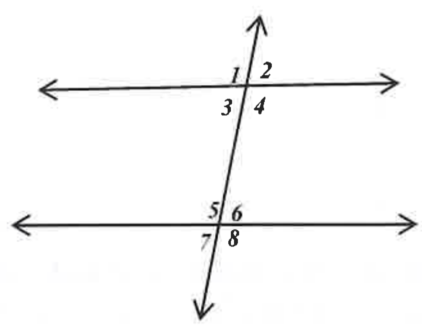
c. Why is $m\measuredangle 2 + m\measuredangle 3 + m\measuredangle 4 = 180°$?______________________________

d. Using substitution we can replace $m\measuredangle 2$ with $m\measuredangle 1$ and $m\measuredangle 4$ with $m\measuredangle 5$ to show that the interior angles of a triangle must always sum to 180°.

$$(\qquad) + m\measuredangle 3 + (\qquad) = 180°$$

Write the angle number in the ____ and then write the letter that corresponds with the number based on the code at the bottom in the box.

7. Angle 2 and Angle ___ ☐ are alternate exterior angles.
8. Angle 7 and Angle ___ ☐ are alternate exterior angles.
9. Angle 4 and Angle ___ ☐ are corresponding angles.
10. Angle 5 and Angle___ ☐ are consecutive interior angles.
11. Angle 3 and Angle___ ☐ are alternate interior angles.
12. Angle 7 and Angle___ ☐ are consecutive exterior angles.
13. Angle 6 and Angle___ ☐ are vertical angles.
14. Angle 2 and Angle___ ☐ are a linear pair and on the same side of the transversal.
15. Angle 1 and Angle___ ☐ are corresponding angles.

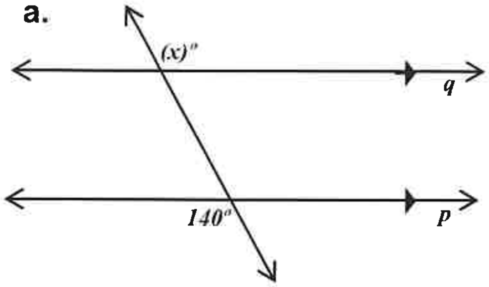


1=D	2=U	3= L	4 = A	5 = N	6 = I	7 = E	8 = C

What type of Geometry is this?____________________________

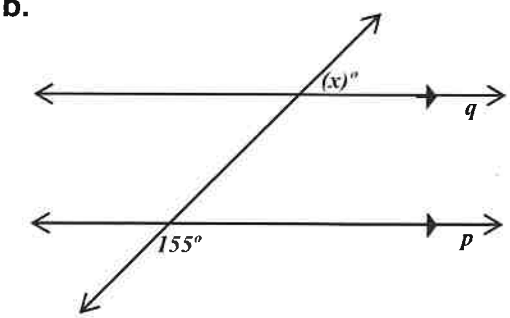
16. Given lines p and q are parallel, find the value of *x* that makes each diagram true.

a.



$x =$

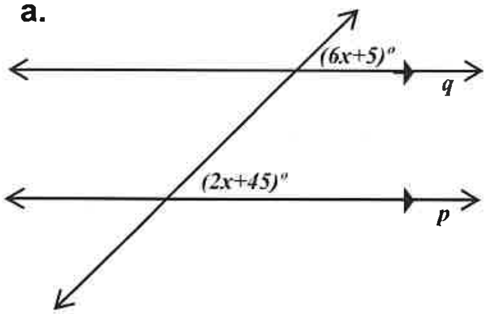
b.



$x =$

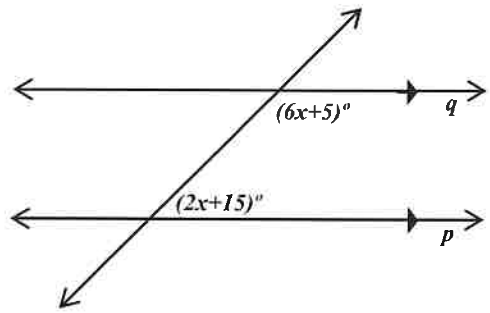
17. Given lines p and q are parallel, find the value of *x* that makes each diagram true.

a.



$x =$

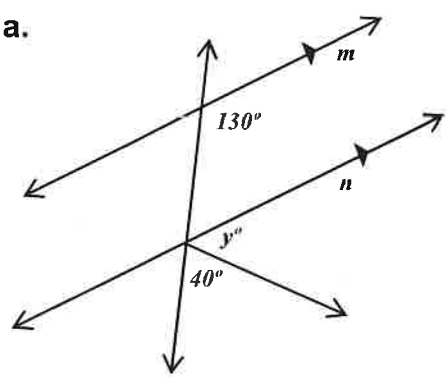
b.



$x =$

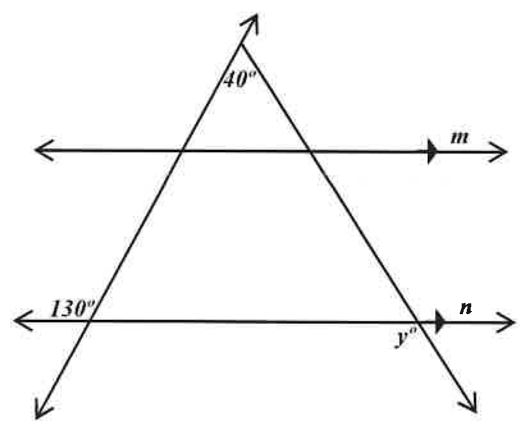
18. Given lines m and n are parallel, find the value *y* of that makes each diagram true.

a.



$y =$

b.



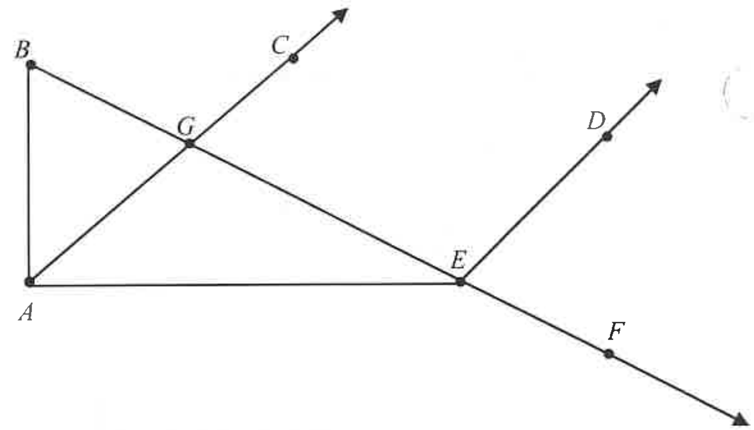
$y =$

19. ANGLE PUZZLE. Find $m\measuredangle AEF$

Given:

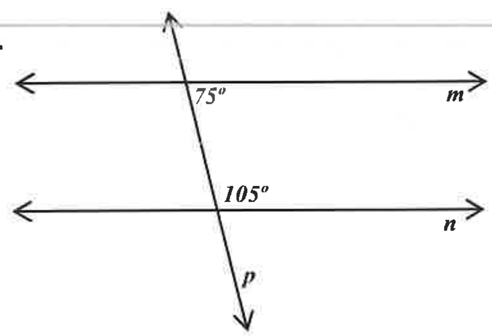
- $m\measuredangle DEF = 85°$
- $m\measuredangle ABG = 50°$
- $\measuredangle BAE$ is a right angle
- $\measuredangle CGE$ and $\measuredangle DEG$ are supplementary

$m\measuredangle AEF =$

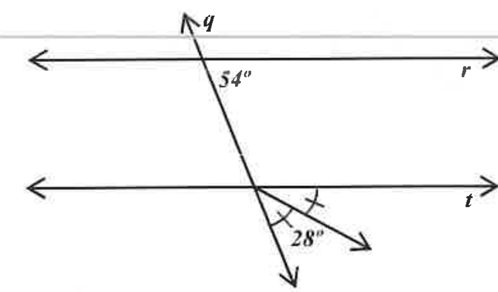


20. Converse of AIA, AEA, CIA, CEA. Which sets of lines are parallel and explain why?

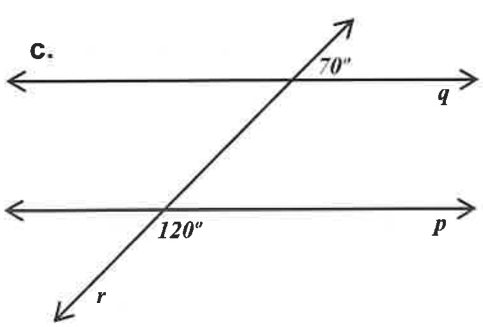
a.



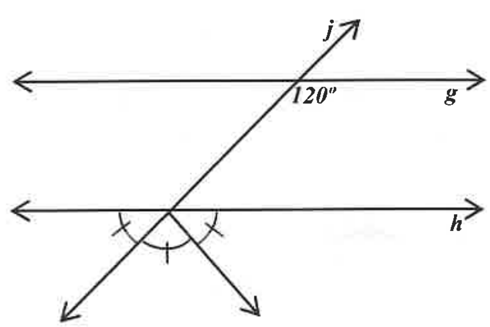
b.



c.



d.



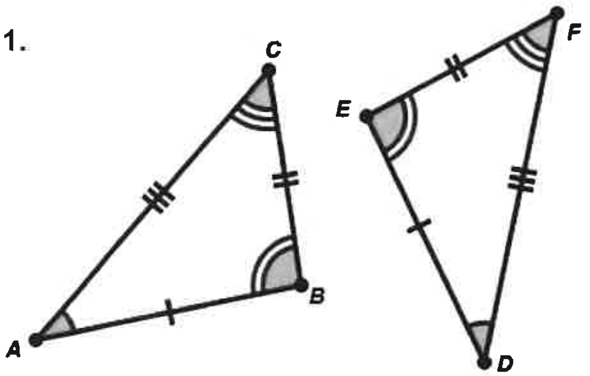
Sec 1.7 Geometry – Congruence

Name:

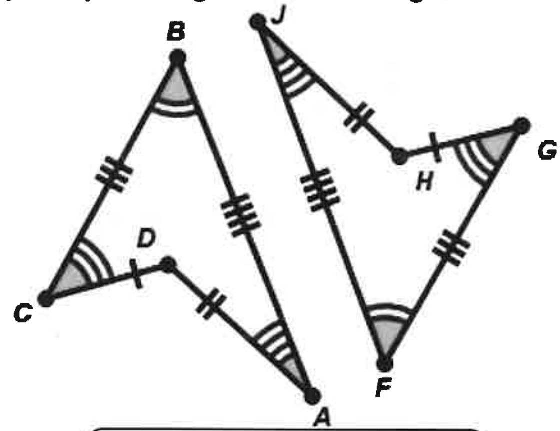
Any two congruent figures can be mapped onto one another using a series of rigid or isometric transformation (reflections, rotations, and translations). ***– See GSP Lab (Transformations) –***

Each of the following pairs of figures shown below are congruent. Write a congruence statement for each and tell whether or not a reflection would be needed to map the pre-image onto the image.

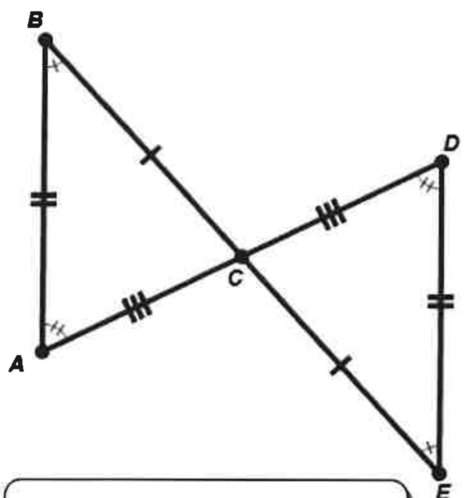
1.



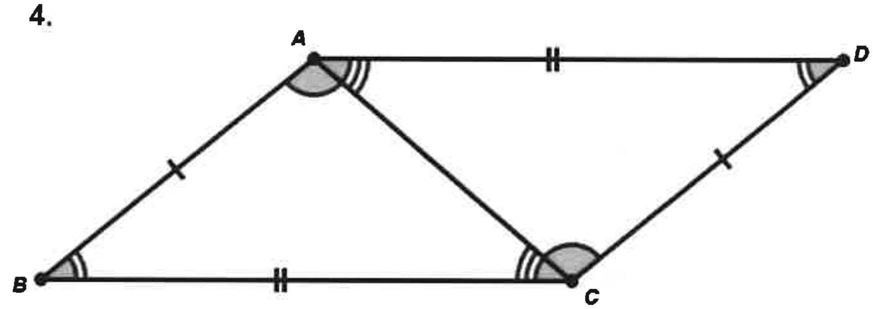
2.



3.

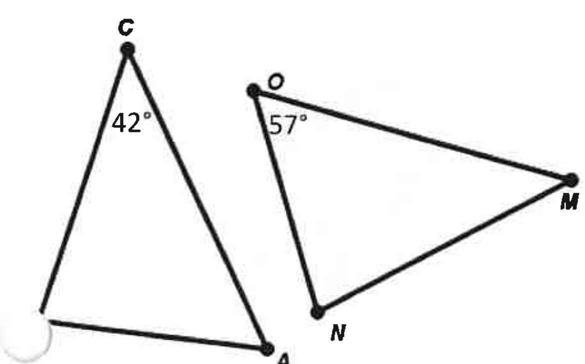


4.



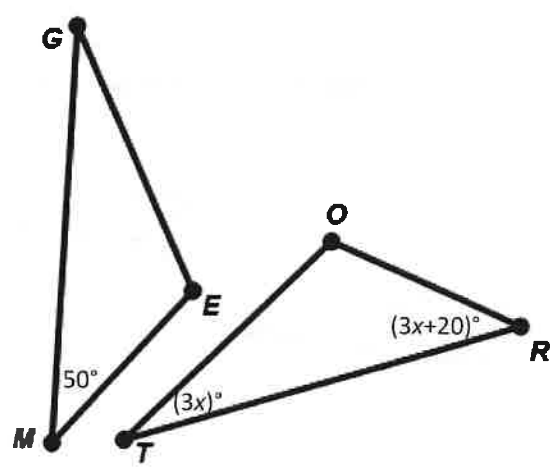
Given the following congruencies find the requested unknown angle.

5. $\Delta ABC \cong \Delta ONM$



$m\angle MNO =$

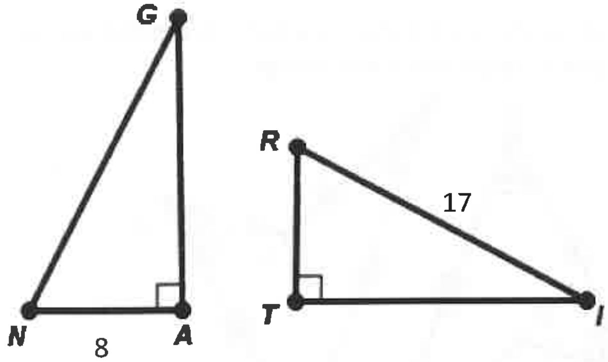
6. $\Delta GEM \cong \Delta TOR$



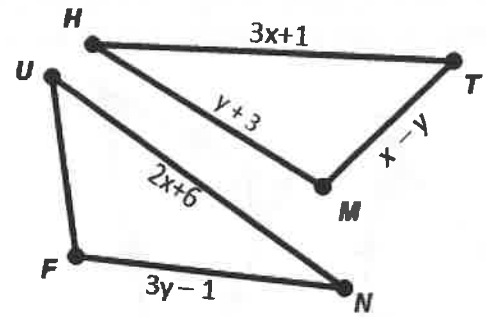
$m\angle GEM =$

Given the following congruencies find the requested unknown side.

7. $\triangle TRI \cong \triangle ANG$



8. $\Delta MTH \cong \Delta FUN$

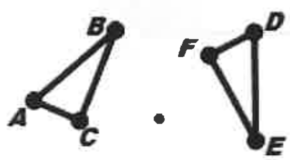


$GN =$

$UF =$

The following pairs of triangles are congruent. Provide a suggested transformation or series of transformations that can map one triangle onto the other congruent triangle. ***(In each diagram*** $\Delta ABC \cong \Delta DEF$ ***)***

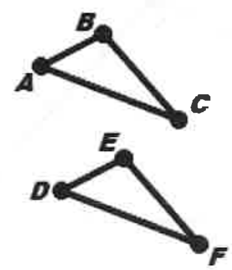
9.



Circle which transformation(s) could be used to map ΔABC onto ΔDEF.

Translation	**Reflection**
Rotation	**Dilation**

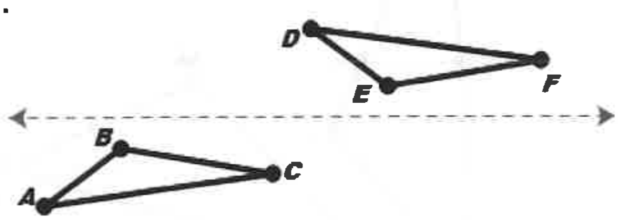
10.



Circle which transformation(s) could be used to map ΔABC onto ΔDEF.

Translation	**Reflection**
Rotation	**Dilation**

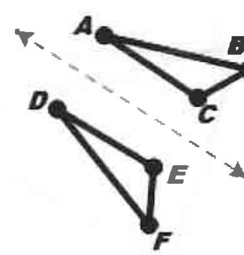
11.



Circle which transformation(s) could be used to map ΔABC onto ΔDEF.

Translation	**Reflection**
Rotation	**Dilation**

12.



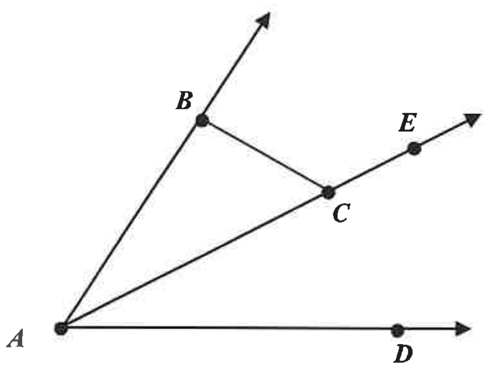
Circle which transformation(s) could be used to map ΔABC onto ΔDEF.

Translation	**Reflection**
Rotation	**Dilation**

Angle Puzzles. (angles are not drawn to scale)

9. Find $m\measuredangle BCE$

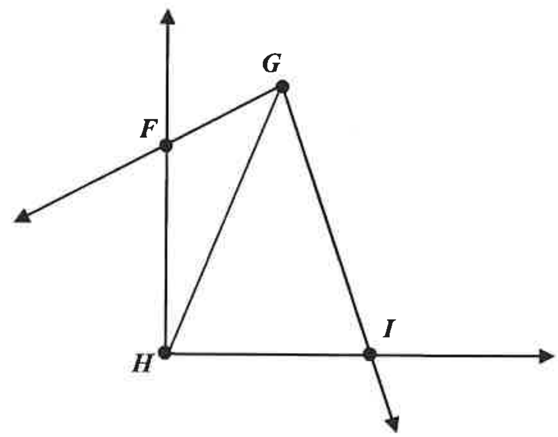
Given:
- $\overrightarrow{AC}$ bisects $\measuredangle DAB$
- $m\measuredangle DAB = 50°$
- $\measuredangle ABC$ is a right angle



$m\measuredangle BCE =$

10. Find $m\measuredangle GFH$

Given:
- $\overline{GH}$ bisects $\measuredangle FGI$
- $m\measuredangle FGI = 60°$
- $m\measuredangle GIH = 75°$
- $\measuredangle FHI$ is a right angle



$m\measuredangle GFH =$

Sec 1.8 Geometry – Triangle Proofs

Name:

COMMON POTENTIAL REASONS FOR PROOFS

Definition of Congruence: Having the exact same size and shape and there by having the exact same measures.
Definition of Midpoint: The point that divides a segment into two congruent segments.
Definition of Angle Bisector: The ray that divides an angle into two congruent angles.
Definition of Perpendicular Lines: Lines that intersect to form right angles or 90°
Definition of Supplementary Angles: Any two angles that have a sum of 180°
Definition of a Straight Line: An undefined term in geometry, a line is a straight path that has no thickness and extends forever. It also forms a straight angle which measures 180°

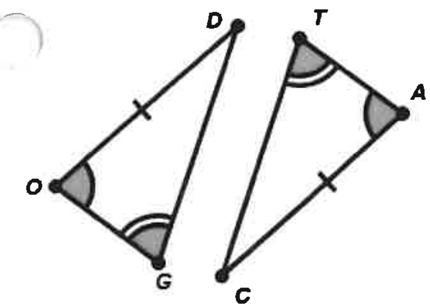
Reflexive Property of Equality: any measure is equal to itself (a = a)
Reflexive Property of Congruence: any figure is congruent to itself ($Figure\ A \cong Figure\ A$)
Addition Property of Equality: if a = b, then a + c = b + c
Subtraction Property of Equality: if a = b, then a – c = b – c
Multiplication Property of Equality: if a = b, then ac = bc
Division Property of Equality: if a = b, then $\frac{a}{c} = \frac{b}{c}$
Transitive Property: if a = b & b =c then a = c OR if a ≅ b & b ≅ c then a ≅ c.

Segment Addition Postulate: If point B is between Point A and C then AB + BC = AC
Angle Addition Postulate: If point S is in the interior of ∠PQR, then $m\angle PQS + m\angle SQR = m\angle PQR$
Side – Side – Side Postulate (SSS) : If three sides of one triangle are congruent to three sides of another triangle, then the triangles are congruent.
Side – Angle – Side Postulate (SAS): If two sides and the included angle of one triangle are congruent to two sides and the included angle of another triangle, then the triangles are congruent.
Angle – Side – Angle Postulate (ASA): If two angles and the included side of one triangle are congruent to two angles and the included side of another triangle, then the triangles are congruent.
Angle – Angle – Side Postulate (AAS) : If two angles and the non-included side of one triangle are congruent to two angles and the non-included side of another triangle, then the triangles are congruent
Hypotenuse – Leg Postulate (HL): If a hypotenuse and a leg of one right triangle are congruent to a hypotenuse and a leg of another right triangle, then the triangles are congruent

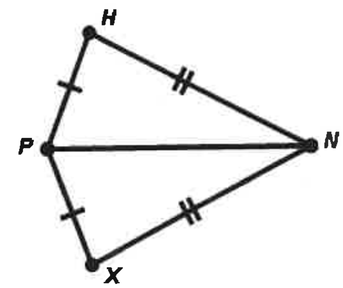
Right Angle Theorem (R.A.T.): All right angles are congruent.
Vertical Angle Theorem (V.A.T.): Vertical angles are congruent.
Triangle Sum Theorem: The three angles of a triangle sum to 180°
Linear Pair Theorem: If two angles form a linear pair then they are adjacent and are supplementary.
Third Angle Theorem: If two angles of one triangle are congruent to two angles of another triangle, then the third pair of angles are congruent.
Alternate Interior Angle Theorem (and converse): Alternate interior angles are congruent if and only if the transversal that passes through two lines that are parallel.
Alternate Exterior Angle Theorem (and converse): Alternate exterior angles are congruent if and only if the transversal that passes through two lines that are parallel.
Corresponding Angle Theorem (and converse) : Corresponding angles are congruent if and only if the transversal that passes through two lines that are parallel.
Same-Side Interior Angles Theorem (and converse) : Same Side Interior Angles are supplementary if and only if the transversal that passes through two lines that are parallel.
Pythagorean Theorem (and converse): A triangle is right triangle if and only if the given the length of the legs a and b and hypotenuse c have the relationship $a^2+b^2 = c^2$
Isosceles Triangle Theorem (and converse): A triangle is isosceles if and only if its base angles are congruent.
Triangle Mid-segment Theorem: A mid-segment of a triangle is parallel to a side of the triangle, and its length is half the length of that side.

CPCTC: Corresponding Parts of Congruent Triangles are Congruent by definition of congruence.

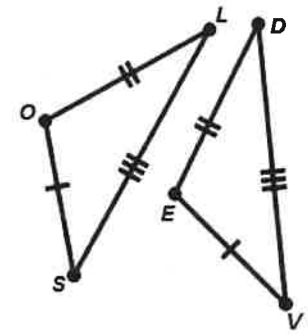
1. **Tell which of the following triangle provide enough information to show that they must be congruent. If they are congruent, state which theorem suggests they are congruent (SAS, ASA, SSS, AAS, HL) and write a congruence statement.**



Circle one of the following:
SSS SAS ASA AAS HL Not Enough Information
Congruence Statement if necessary:



Circle one of the following:
SSS SAS ASA AAS HL Not Enough Information
Congruence Statement if necessary:



Circle one of the following:
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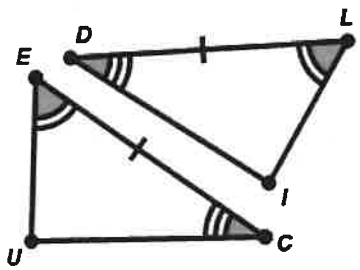
Congruence Statement if necessary:



Circle one of the following:
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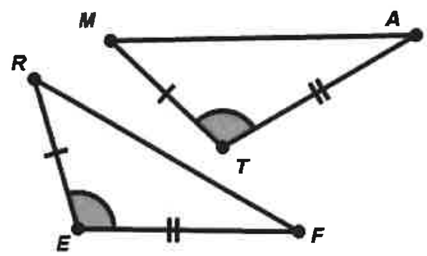
Congruence Statement if necessary:



Circle one of the following:
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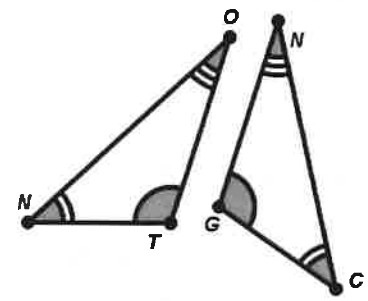
Congruence Statement if necessary:



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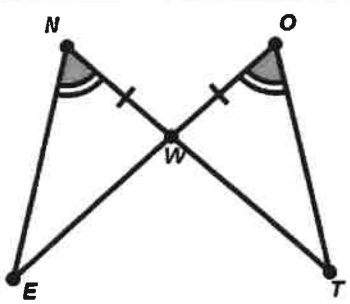
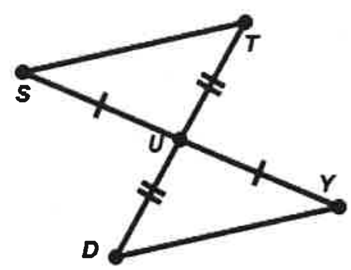
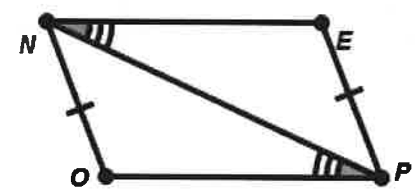
Congruence Statement if necessary:



Circle one of the following:
SSS SAS ASA AAS HL Not Enough Information
Congruence Statement if necessary:



Circle one of the following:
SSS SAS ASA AAS HL Not Enough Information
Congruence Statement if necessary:



Circle one of the following:
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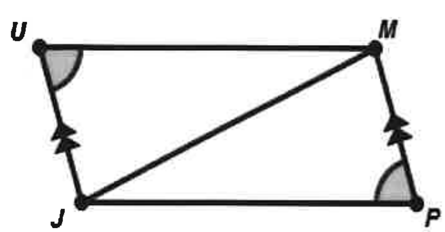
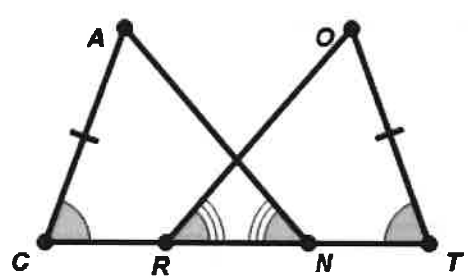
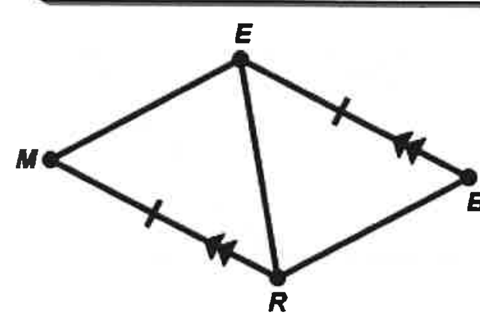
Congruence Statement if necessary:



Circle one of the following:
SSS SAS ASA AAS HL Not Enough Information
Congruence Statement if necessary:



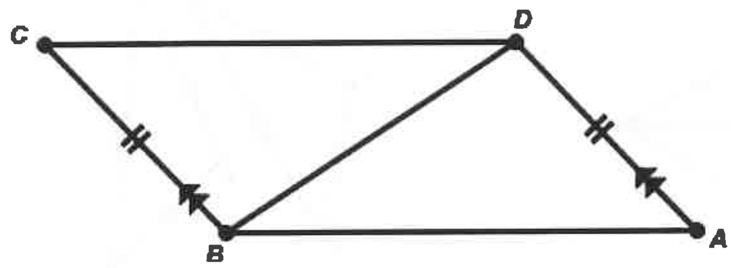
Circle one of the following:
SSS SAS ASA AAS HL Not Enough Information
Congruence Statement if necessary:



Circle one of the following:
SSS SAS ASA AAS HL Not Enough Information
Congruence Statement if necessary:

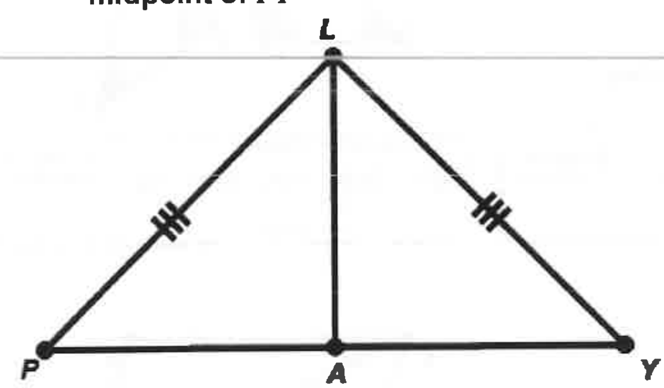
2. Prove which of the following triangles congruent if possible by filling in the missing blanks:

a. **Given $\overline{CB} \cong \overline{AD}$ and $\overleftrightarrow{CB} \parallel \overleftrightarrow{AD}$**



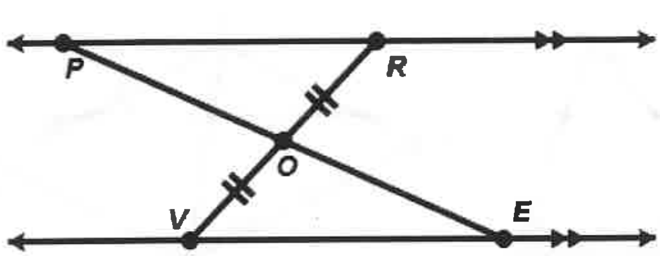
Statements	Reasons
1. $\overline{CB} \cong \overline{AD}$	
2. $\overleftrightarrow{CB} \parallel \overleftrightarrow{AD}$	
3. $\measuredangle CBD \cong \measuredangle ADB$	
4. $\overline{BD} \cong \overline{BD}$	
5. $\Delta BCD \cong \Delta DAB$	

b. **Given $\overline{PL} \cong \overline{YL}$ and Point A is the midpoint of $\overline{PY}$**



Statements	Reasons
1.	Given
2.	Given
3.	Definition of Midpoint
4.	Reflexive property of congruence
5.	By steps 1,3,4 and SSS

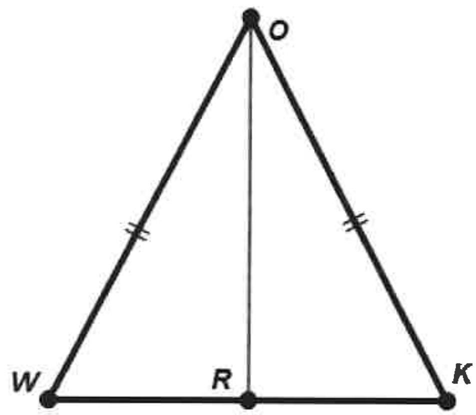
c. **Given $\overline{VO} \cong \overline{RO}$ and $\overleftrightarrow{PR} \parallel \overleftrightarrow{VE}$**



Statements	Reasons
1.	Given
2.	Given
3.	
4.	
5. $\Delta PRO \cong \Delta EVO$	

Prove the Isosceles Triangle Theorem and the rest of the suggested proofs.

d. Given ΔWOK is isosceles and point R is the midpoint of $\overline{WK}$



Statements	Reasons
1. ΔWOK is isosceles	
2. $\overline{WO} \cong \overline{KO}$	
3. R is the midpoint of $\overline{WK}$	
4. $\overline{WR} \cong \overline{KR}$	
5. $\overline{OR} \cong \overline{OR}$	
6. $\Delta WRO \cong \Delta KRO$	
7. $\measuredangle OWR \cong \measuredangle OKR$	

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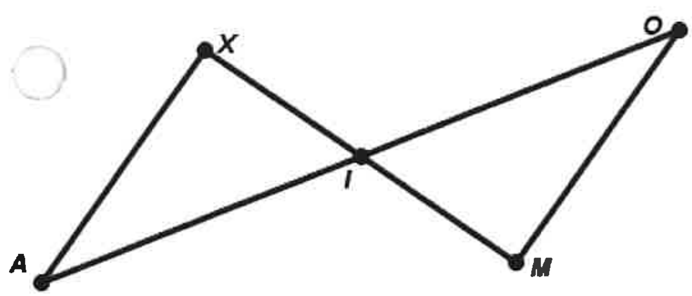
e. Given point I is the midpoint of $\overline{XM}$ and point I is the midpoint of $\overline{AO}$



Statements	Reasons
1. I is the midpoint of $\overline{XM}$	
2.	Definition of Midpoint
3. I is the midpoint of $\overline{AO}$	
4.	
5. $\measuredangle AIX \cong \measuredangle OIM$	
6. $\Delta AXI \cong \Delta OMI$	

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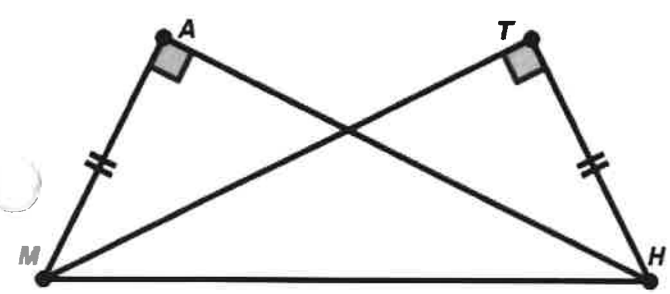
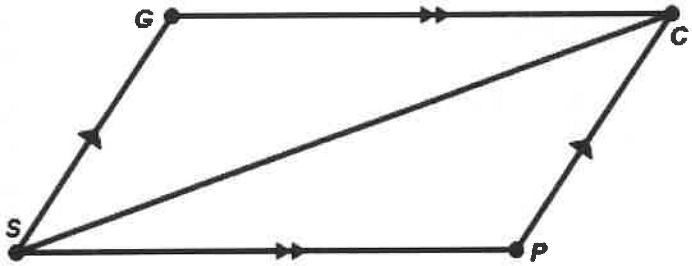
f. Given $\measuredangle MAH$ and $\measuredangle HTM$ are right angles and $\overline{MA} \cong \overline{TH}$



Statements	Reasons
1. $\measuredangle MAH$ & $\measuredangle HTM$ are right angles	
2. $\overline{MA} \cong \overline{TH}$	
3. $\overline{MH} \cong \overline{MH}$	
4. $\Delta MAH \cong \Delta HTM$	

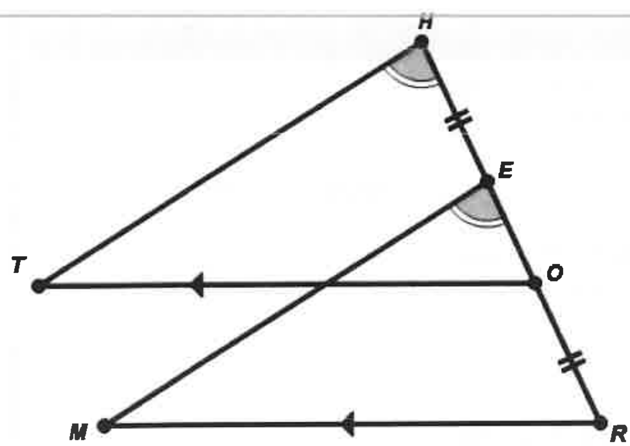
Prove the suggested proofs by filling in the missing blanks.

g. Given $\overleftrightarrow{GC} \parallel \overleftrightarrow{PS}$ and $\overleftrightarrow{GS} \parallel \overleftrightarrow{CP}$



Statements	Reasons
1. $\overleftrightarrow{GC} \parallel \overleftrightarrow{PS}$	
2.	
3. $\overleftrightarrow{GS} \parallel \overleftrightarrow{CP}$	
4.	
5.	
6. $\Delta GCS \cong \Delta PSC$	

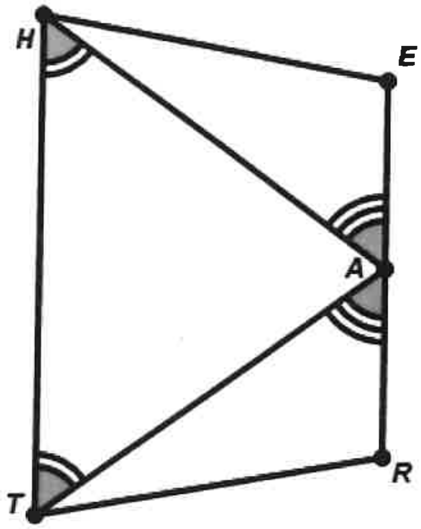
h. Given $\overline{RO} \cong \overline{HE}$, $\measuredangle THO \cong \measuredangle MER$ and $\overleftrightarrow{TO} \parallel \overleftrightarrow{MR}$



Statements	Reasons
1. $\overline{HE} \cong \overline{OR}$	
2. $RO = HE$	
3. $RO + OE = HE + EO$	
4. $RO + OE = RE$ and $HE + EO = HO$	
5. $RE = HO$	Substitution Property
6. $\overline{RE} \cong \overline{OR}$	
7. $\overleftrightarrow{TO} \parallel \overleftrightarrow{MR}$	
8. $\measuredangle MRE \cong \measuredangle TOH$	
9. $\measuredangle THO \cong \measuredangle MER$	
10. $\Delta THO \cong \Delta MER$	

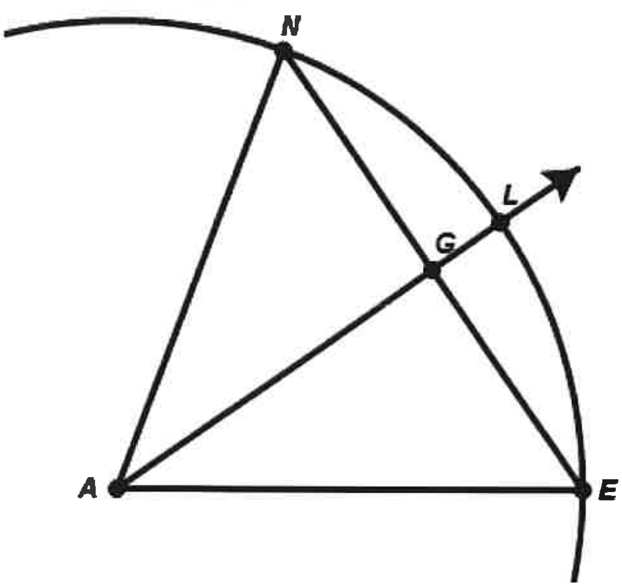
Prove the suggested proofs by filling in the missing blanks.

i. **Given $\measuredangle HTA \cong \measuredangle THA$, $\measuredangle TAR \cong \measuredangle HAE$, and Point A is the midpoint of $\overline{ER}$**



Statements	Reasons
1. $\measuredangle THA \cong \measuredangle HTA$	
2. ΔHAT is an isosceles triangle	
3. $\overline{HA} \cong \overline{TA}$	
4. A is the midpoint of $\overline{ER}$	
5. $\overline{EA} \cong \overline{AR}$	
6. $\measuredangle TAR \cong \measuredangle HAE$	
7. $\Delta TAR \cong \Delta HAE$	

j. **Given that $\overrightarrow{AG}$ bisects $\measuredangle NAE$. Also, $\overline{AN}$ and $\overline{AE}$ are radii of the same circle with center A.**



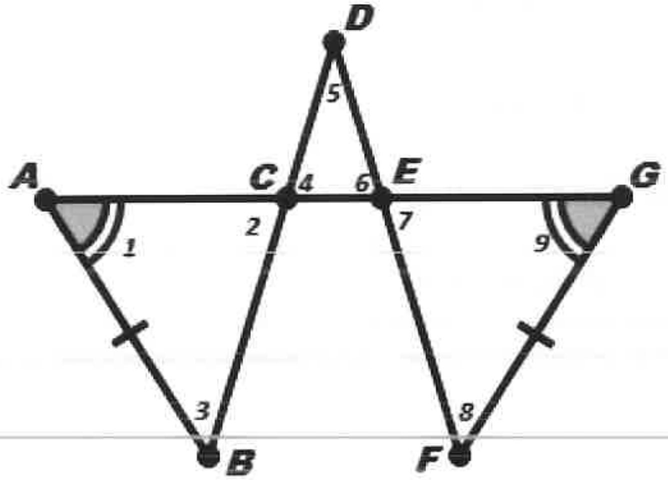
Statements	Reasons
1. $\overrightarrow{AG}$ bisects $\measuredangle NAE$	
2.	Definition of Angle Bisector
3.	Radii of the same circle are congruent.
4.	Reflexive property of congruence
5. $\Delta ANG \cong \Delta AEG$	

Prove the suggested proofs by filling in the missing blanks.

i. **Given:**

- $\measuredangle 1 \cong \measuredangle 9$
- $\overline{AB} \cong \overline{GF}$
- **ΔCDE forms an isosceles triangle with base $\overline{CE}$**

Prove: $\Delta ABC \cong \Delta GFE$



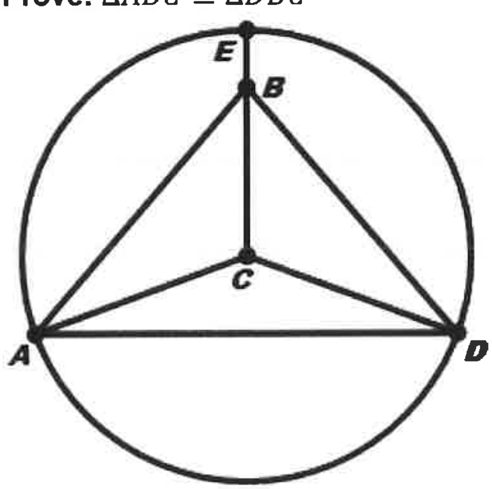
Statements	Reasons
1. $\measuredangle 1 \cong \measuredangle 9$	
2. $\overline{AB} \cong \overline{GF}$	
3. $\measuredangle 2 \cong \measuredangle 4$	
4. ΔCDE is an isosceles triangle	
5.	Isosceles Triangle Theorem
6. $\measuredangle 6 \cong \measuredangle 7$	
7. $\measuredangle 2 \cong \measuredangle 7$	
8.	

Prove the suggested proofs by filling in the missing blanks.

k. **Given:**

- **The circle has a center at point C**
- **ΔABE forms an isosceles triangle with base $\overline{AD}$**

Prove: $\Delta ABC \cong \Delta DBC$



Statements	Reasons
1. ΔABD is an isosceles triangle w/ base $\overline{AD}$	
2. $\overline{AE} \cong \overline{DB}$	
3. The circle is centered at point C	
4. $\overline{AC} \cong \overline{DC}$	
5.	Reflexive Property of Congruence
6. $\Delta ABC \cong \Delta DBC$	

Name: ________________________ Class: ___________________ Date: __________ ID: A

Section-01-01-GeometricDefinitions

Multiple Choice
Identify the choice that best completes the statement or answers the question.

____ 1. In Euclidean geometric terms, which of the below would be the best description of a **LINE**?

a. It is a straight object that begins at a point and extends forever in one direction.

b. It is a straight object that extends infinitely in opposite directions without any width or thickness.

c. It is a straight object that consists of all of the points that are equal distance from a common point.

d. It is a straight object that consists of all of the points that exist between two different points.

____ 2. In Euclidean geometric terms, which of the below would be the best description of a **PLANE**?

a. It could be described as a flat object with infinite length, infinite width, and infinite thickness

b. It could be described as a flat object with infinite length and infinite width but no thickness.

c. It could be described as a flat object with no width, no length, and no thickness.

d. It could be described as an object with propellers.

____ 3. Which word best describes a set of 3 or more different points that are all on the same line.

a. Accurate

b. Line-up

c. Collinear

d. Sightline

Name: ______________________ ID: A

____ 4. Which word best describes two lines that are not in the same plane?

a. Long Distance

b. Parallel

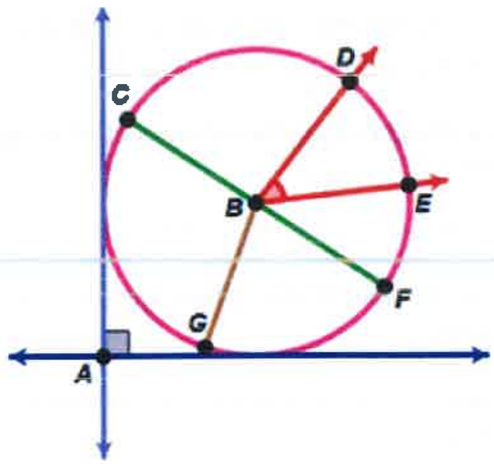
c. Skew

d. Uncommon

e. Perpendicular

____ 5.

Consider the diagram at the right. Which of the words below would geometrically best describe the relationship between **the two blue** objects that pass through point A?

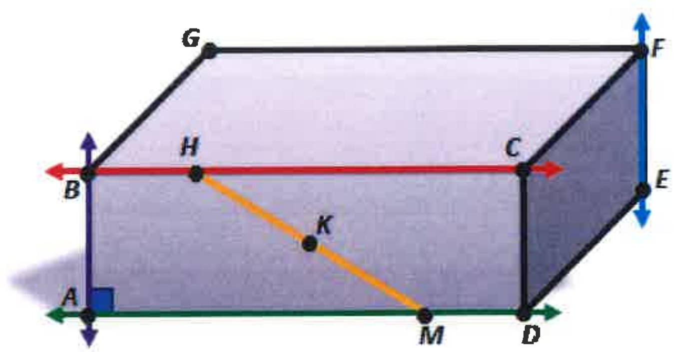


a. Upright
b. Perpendicular
c. Parallel
d. Skew

____ 6.

Assume in the figure shown at the right that the solid is a rectangular prism.

How would you **best** describe the relationship between the **green** ($\overleftrightarrow{AD}$) and **red** ($\overleftrightarrow{BC}$) lines?

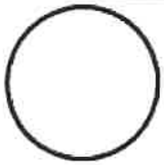


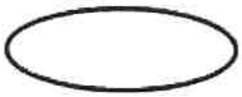
a. Aligned
b. Parallel
c. Perpendicular
d. Skew

____ 7. Which of the below figures most likely represents a set of points equidistant from a point in a plane?

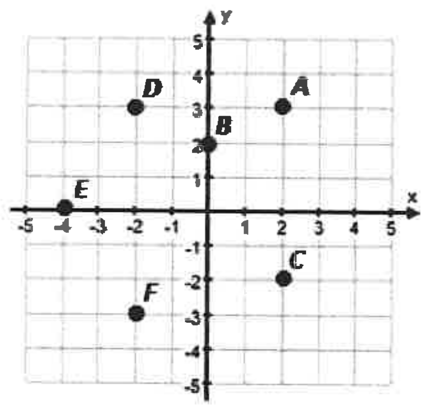
a. 

b. 

c. 

d. 

____ 8. Which group or set of points of the below choices are **collinear**?



a. Point A, Point B, Point C

b. Point C, Point E, Point F

c. Point B, Point C, Point D

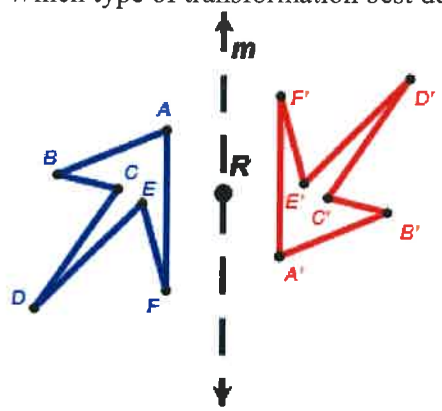
d. Point A, Point B, Point E

Name: ________________________ Class: ___________________ Date: __________ ID: A

Section-01-02-Transformation Types

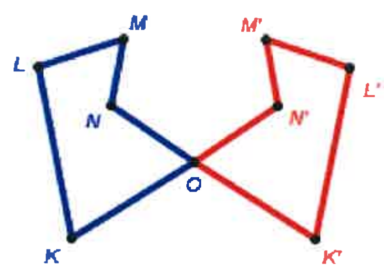
Multiple Choice
Identify the choice that best completes the statement or answers the question.

____ 1. Which type of transformation best describes the **pre-image** to the **image** shown below?



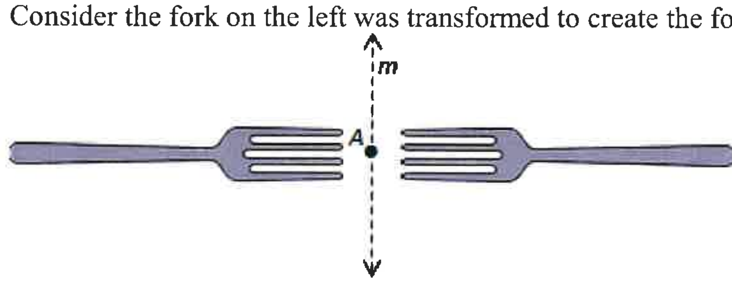
a. Translation by vector m.
b. Reflection over line m.
c. Rotation about point R of 90°.
d. Rotation about point R of 180°.

____ 2. Tell whether the transformation appears to be a **translation**. Explain.



a. Yes; all of the points have moved the same distance in the same direction.
b. No; not all of the points have moved the same distance.

____ 3. Consider the fork on the left was transformed to create the fork on the right.

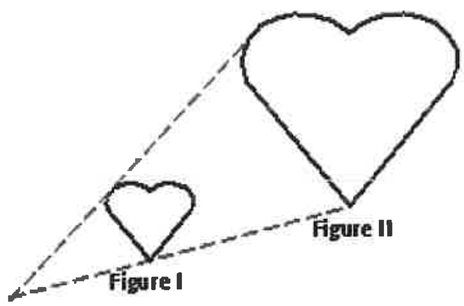


Which transformation that could **NOT** transform the picture of the fork on the left to the one on the right?
a. Rotation about point A by 180°.
b. Using a Translation.
c. Reflection over line M.
d. Dilation through point A

Name: ______________________ ID: A

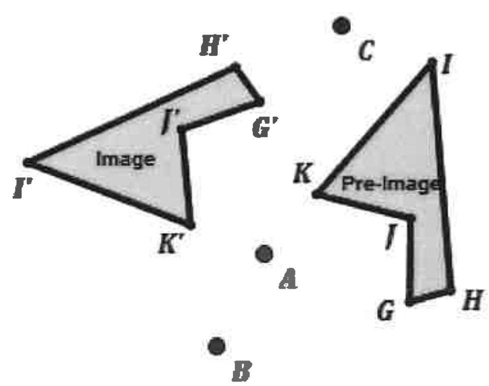
____ 4. Which statement correctly describes one of the **first** steps in the process of drawing a reflection?
a. Through each vertex draw a line perpendicular to the line of reflection.
b. Measure the distance from each vertex to the line of reflection. Locate the image of each vertex on the same side of the line of reflection and the same distance from it.
c. Connect the pre-images of the vertices.
d. Measure the distance from each vertex to the line of reflection. Locate the pre-image of each vertex on the opposite side of the line of reflection and the same distance from it.

____ 5. Name the transformation. (The pre-image is Figure I and the image is Figure II)



a. Translation
b. Dilation
c. Rotation
d. Reflection

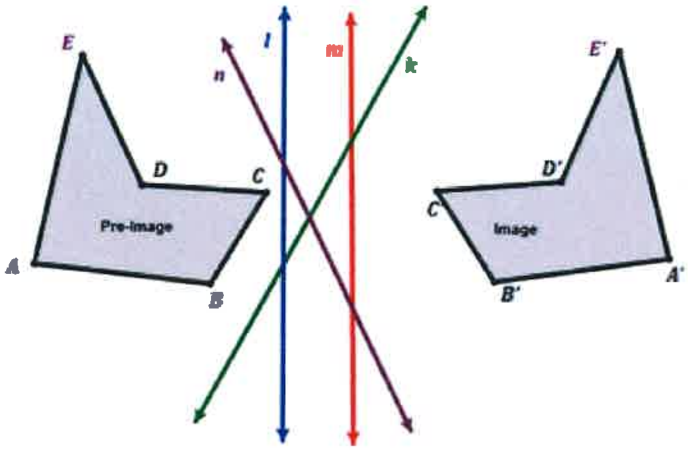
____ 6. Determine which is the correct center if a 100° rotation was used?



a. Point A
b. Point B
c. Point C

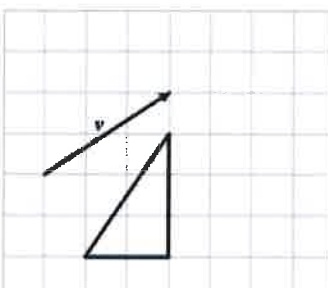
Name: ________________________ ID: A

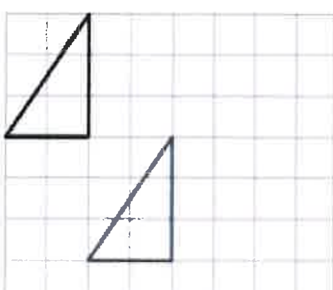
7. Which is the correct line of reflection for the Image and Pre-Image?

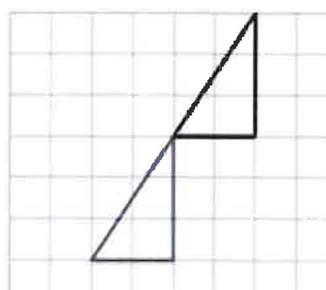


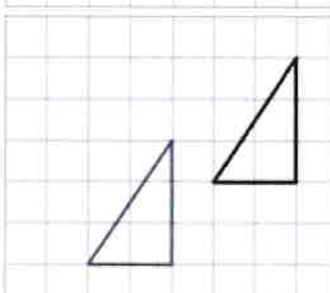
a. Line k
b. Line l
c. Line m
d. Line n

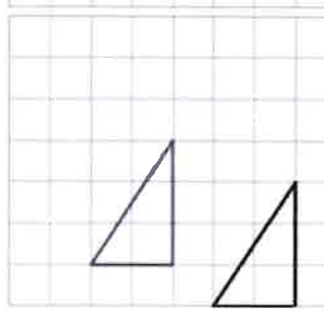
8. Find the translation of the triangle along $\vec{v}$.



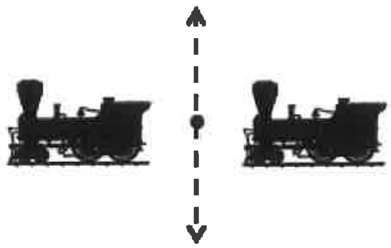
a. 

c. 

b. 

d. 

____ 9. What transformation is being demonstrated below?



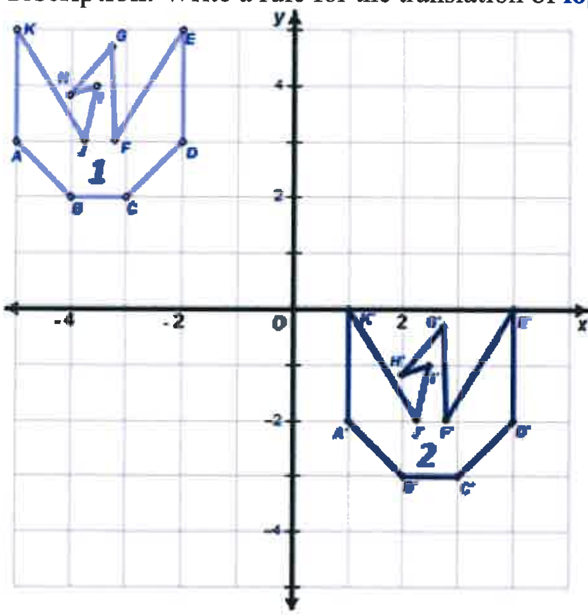
a. Translation
b. Rotation
c. Reflection
d. Dilation
e. Glide

Name: ______________________ Class: __________________ Date: __________ ID: A

Section-01-03-Coordinate Transforms

Multiple Choice
Identify the choice that best completes the statement or answers the question.

____ 1. A video visual effects animator creates an animation by translating a logo using a coordinate description. Write a rule for the translation of **logo 1** to **logo 2**.

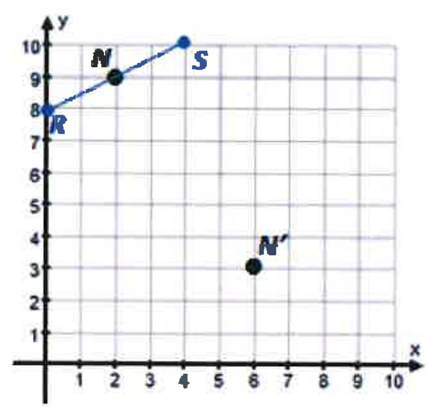


a. $(x, y) \rightarrow (x + 6, y - 5)$
b. $(x, y) \rightarrow (x + 5, y - 6)$
c. $(x, y) \rightarrow (x - 6, y + 5)$
d. $(x, y) \rightarrow (x - 5, y + 6)$

____ 2.

Point N (2,9) is on $\overline{RS}$. A translation moves point N to its image N'(6, 3).

What is the distance, in units, between any point on $\overline{RS}$ and its image?



a. $\sqrt{10}$
b. $\sqrt{40}$
c. $\sqrt{52}$
d. $\sqrt{130}$

____ 3. Consider Quadrilateral ABCD defined by the coordinates:

A: (2, 3)
B: (3, 1)
C: (- 4, 2)
D: (2, - 3)

If Quadrilateral ABCD were reflected over the x-axis, what would the coordinates of the newly created Quadrilateral Image A'B'C'D'?

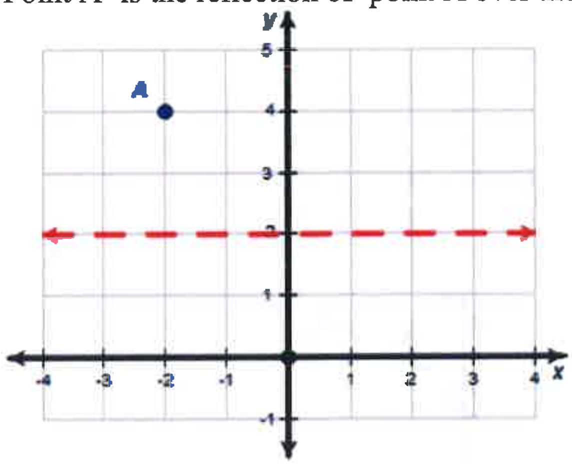
a. A: (- 2, 3)
B: (- 3, 1)
C: (4, 2)
D: (- 2, - 3)

b. A: (- 2, - 3)
B: (- 3, - 1)
C: (4, - 2)
D: (- 2, 3)

c. A: (2, - 3)
B: (3, - 1)
C: (- 4, - 2)
D: (2, 3)

d. A: (3, 2)
B: (1, 3)
C: (2, - 4)
D: (- 3, 2)

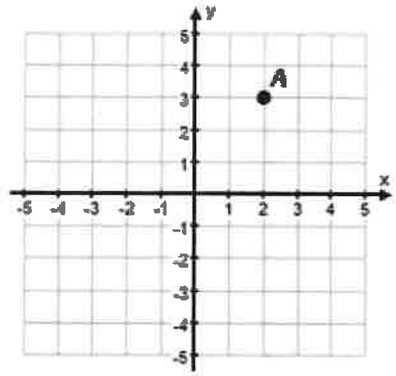
____ 4. Point A' is the reflection of point A over the line y =2. What are the coordinates of A'?



a. A'(2,4)
b. A'(- 2, 0)
c. A'(0, - 2)
d. A'(4, 2)

Name: ________________________ **ID: A**

____ 5. If the point A is located at (2, 3) and A' is the image of A after being **rotated about the origin by 90° (counter clockwise)**. What are the coordinates of A'?



a. A'(3, – 2)
b. A'(– 3, 2)
c. A'(– 3, – 2)
d. A'(2, – 3)

____ 6. Which process stated below would result in a dilation of coordinate points to the origin by a scale factor of 0.5?

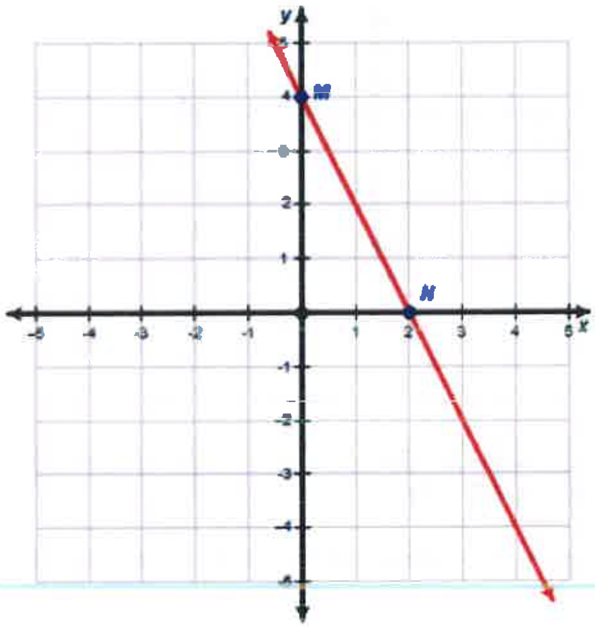
a. Multiply only the x-coordinate by 0.5 of each point to be dilated.
b. Multiply only the y-coordinate by 2 of each point to be dilated.
c. Multiply both the x-coordinate and the y-coordinate by 0.5 of each point to be dilated.
d. Multiply both the x-coordinate and the y-coordinate by 2 of each point to be dilated.

____ 7. Which process stated below would result in a translation to the up 3 of coordinate points?

a. Add 3 to just the x-coordinate of each point to be translated.
b. Subtract 3 from just the x-coordinate of each point to be translated.
c. Add 3 to just the y-coordinate of each point to be translated.
d. Subtract 3 from just the y-coordinate of each point to be translated.

Name: ____________________ **ID: A**

____ 8. In the graph below $\overleftrightarrow{MN}$ below is represented by the equation $y = -2x + 4$. If $\overleftrightarrow{MN}$ is reflected in the **y-axis** what would be the new value of x when y = 0.



a. - 4
b. - 2
c. 4
d. 2

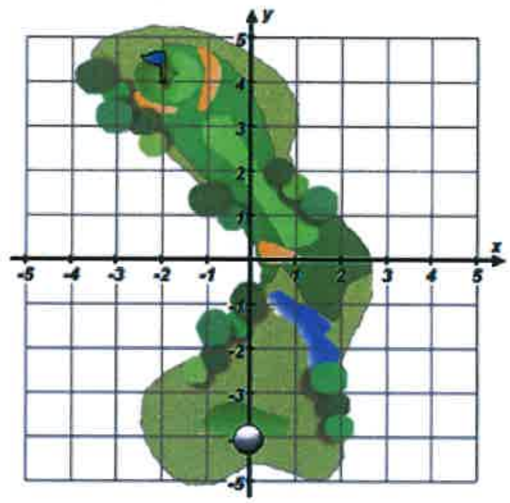
Name: ________________________ Class: ___________________ Date: __________ ID: A

Section-01-04-Compound Transforms

Multiple Choice

Identify the choice that best completes the statement or answers the question.

____ 1. A coordinate grid system is overlayed on a hole at a golf course. Jim starts with the golf ball at the coordinates (0, - 4).



Which sequence of transformations would sink the ball in the hole located at the point (-2, 4)?

a. First, **rotate** the golf ball 90° counter clockwise about the origin.
Then, **translate** the golf ball along the vector $\langle 6,4 \rangle$.

b. First, **reflect** the golf ball over the line $y=-1$.
Then, **translate** the golf ball along the vector $\langle -2,2 \rangle$.

c. First, **translate** the golf ball along the vector $\langle 1,5 \rangle$.
First, **reflect** the golf ball over the line y-axis.

d. First, **reflect** the golf ball over the line x-axis.
Then, **translate** the golf ball along the vector $\langle 2,-2 \rangle$.

____ 2. Consider the point C located at the coordinates (x, y). The point C' coordinates are located at $(x+1,\ y-3)$. Finally, the point C" coordinates are located at $(2\cdot(x+1),\ 2\cdot(y-3))$.

Describe the transformations that map point C onto the point C".

a. First, translate the point C to the right 1 and down 3. Then, dilate that point by a scale factor of $\frac{1}{2}$ to the origin.

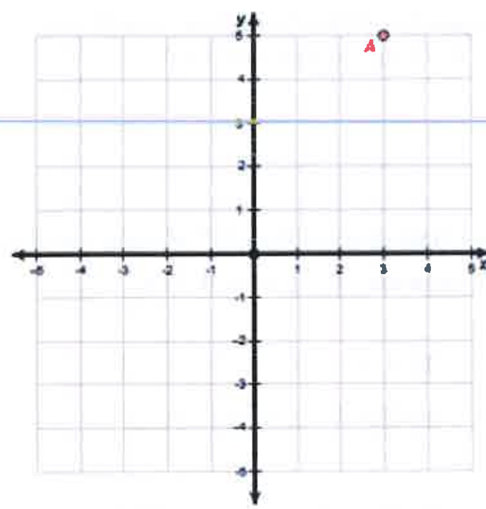
b. First, translate the point C to the left 1 and up 3. Then, dilate that point by a scale factor of $\frac{1}{2}$ to the origin.

c. First, translate the point C to the right 1 and down 3. Then, dilate that point by a scale factor of 2 from the origin.

d. First, translate the point C to the left 1 and up 3. Then, dilate that point by a scale factor of 2 from the origin.

____ 3.

Consider the point A located at the coordinates (3, 5). First, reflect the point over the x-axis. Then, translate that reflected point up 4 units and label this point A". What are the coordinates of A"?



a. A"(-3, 9)

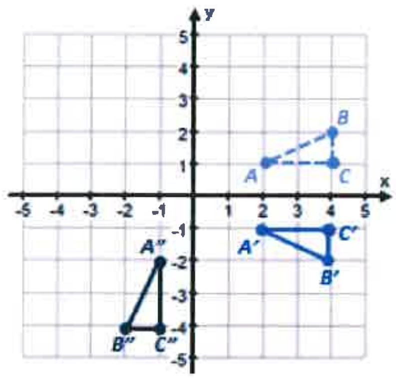
b. A"(7, -5)

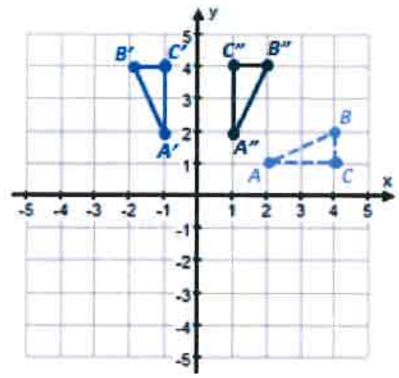
c. A"(1, 5)

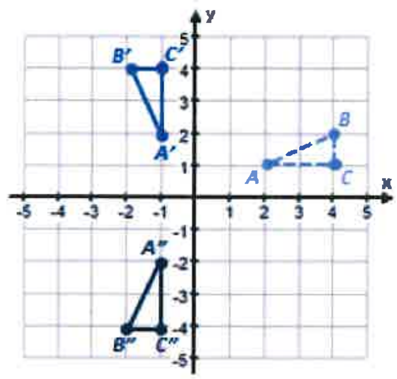
d. A"(3, -1)

Name: ________________________ **ID: A**

____ 4. ΔABC has vertices $A(2, 1)$, $B(4, 2)$, and $C(4, 1)$. Rotate ΔABC 90° counterclockwise about the origin and then reflect it across the x-axis. Which graph below shows this sequence of transformations?

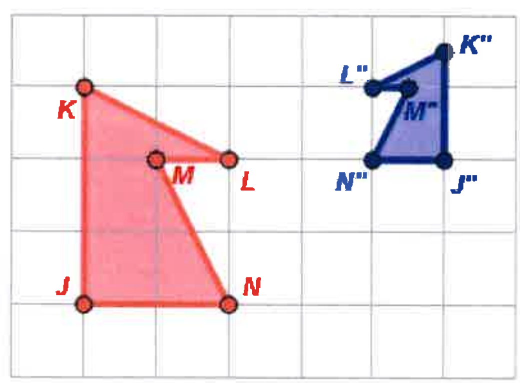
a. 

c. 

b. 

d. None of these show the correct sequence of transformations.

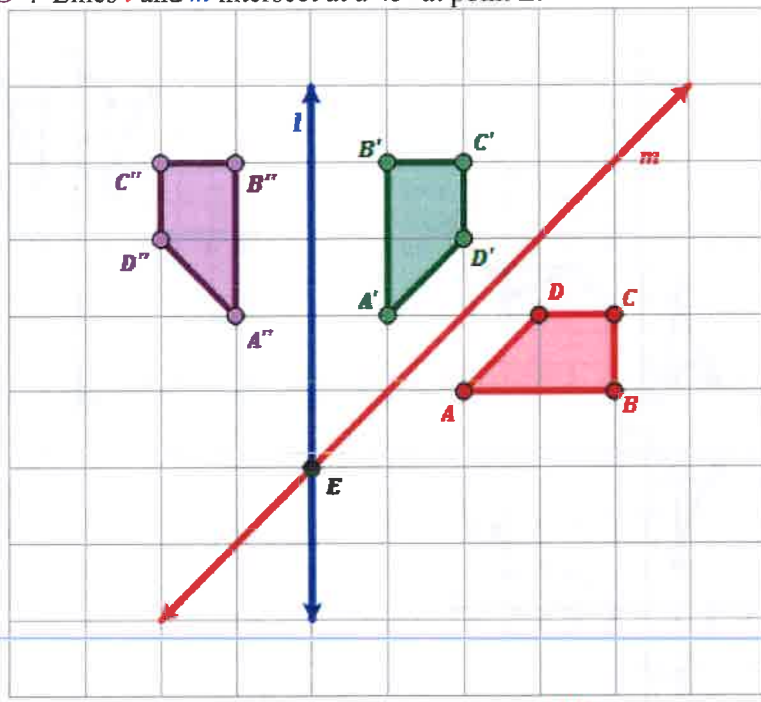
____ 5. Which two types of transformations must have been used to map the pentagon JKLMN onto the pentagon J"K"L"M"N"?



a. Translation and Rotation
b. Reflection and Dilation
c. Rotation and Dilation
d. Dilation and Translation

Name: ________________________ ID: A

____ 6. First, quadrilateral ABCD is first reflected over line m. Then, the image is reflected over line l to create the quadrilateral A”B”C”D”. Lines l and m intersect at a 45° at point E.



What is another type of transformation that would map quadrilateral ABCD onto quadrilateral A”B”C”D”?

a. Rotate quadrilateral ABCD 90° about point E where lines l and m intersect
b. Reflect quadrilateral ABCD over a line that is half way between lines l and m.
c. Translate quadrilateral ABCD left 3 units and up 1 unit.
d. Dilate quadrilateral ABCD by a scale factor of 45 from point E.

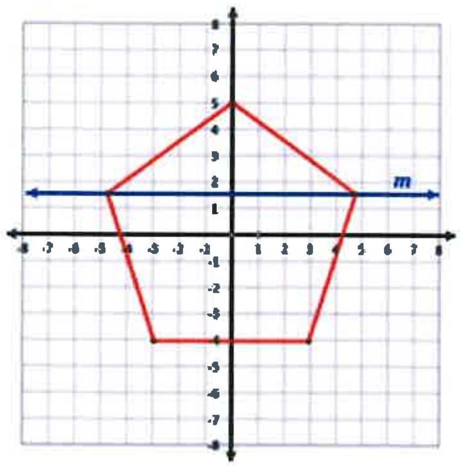
Name: ________________________ Class: ___________________ Date: __________ ID: A

Section-01-05-Symmetries

Multiple Choice

Identify the choice that best completes the statement or answers the question.

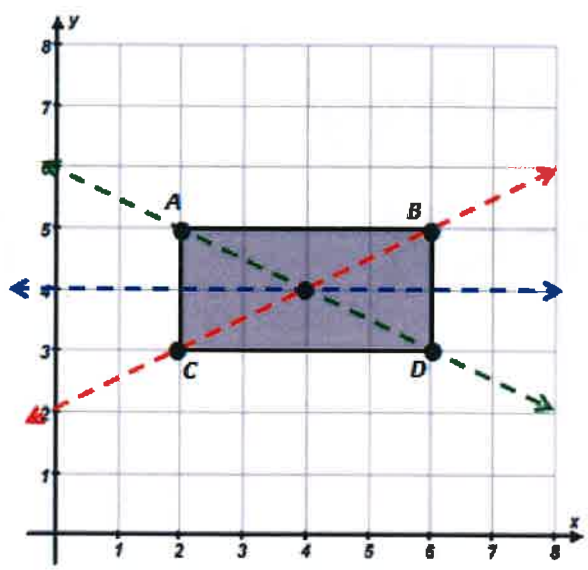
____ 1. A regular pentagon is centered about the origin and has a vertex at (0,5).



Which transformation maps the pentagon to itself?

a. a reflection across line *m*
b. a reflection across the *x*-axis
c. a clockwise rotation of 100° about the origin
d. a clockwise rotation of 144° about the origin

____ 2. A rectangle has vertices at (2,5), (6,5), (6,3), and (2, 3).

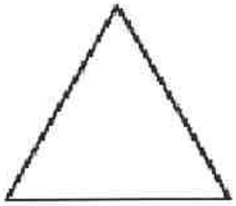


Which transformation maps the rectangle to itself?

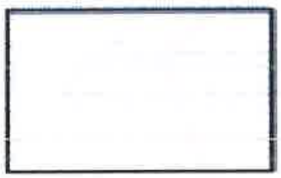
a. a reflection across the line $y = \frac{1}{2}x + 2$
b. a reflection across the line $y = -\frac{1}{2}x + 6$
c. a reflection across the line $y = 4$
d. a rotation of 90° about the point (4,4)

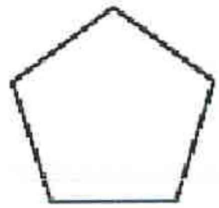
Name: ______________________ ID: A

____ 3. Which figure has exactly two lines of symmetry?

a. 

c. 

b. 

d. 

____ 4. Which letter shown below has more than one line of symmetry?

a. **C**

c. **A**

b. **H**

d. **T**

____ 5. Which word shown below appears to have point symmetry (i.e. rotational symmetry of 180º)?

a. **WOW**

c. **MOW**

b. **HAH**

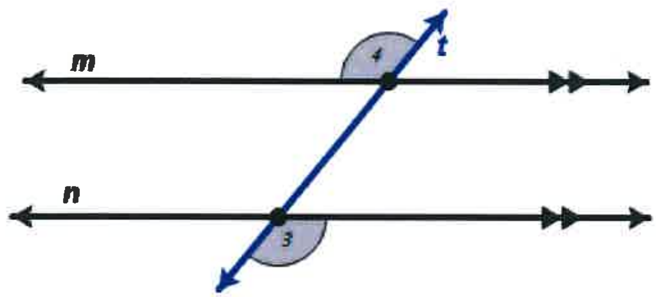
d. **PAP**

Name: ________________________ Class: ___________________ Date: __________ ID: A

02-01-Sample Quiz-Parallel Lines and Angles

Multiple Choice
Identify the choice that best completes the statement or answers the question.

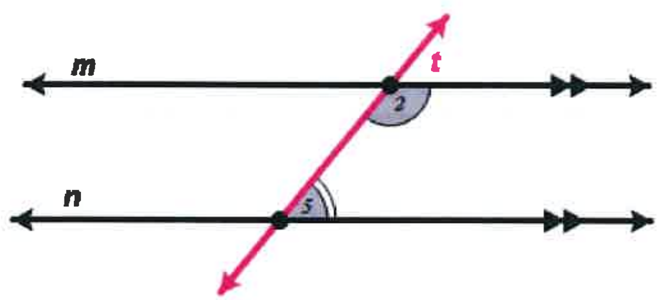
____ 1. *In the diagram line **m** is parallel to line **n** with a transversal line t.*



Which of the below terms best describe the relationship between ∠3 and ∠4 ?

a. Alternate Exterior Angles
b. Alternate Interior Angles
c. Corresponding Angles
d. Vertical Angles

____ 2. *In the diagram line **m** is parallel to line **n** with a transversal line t.*

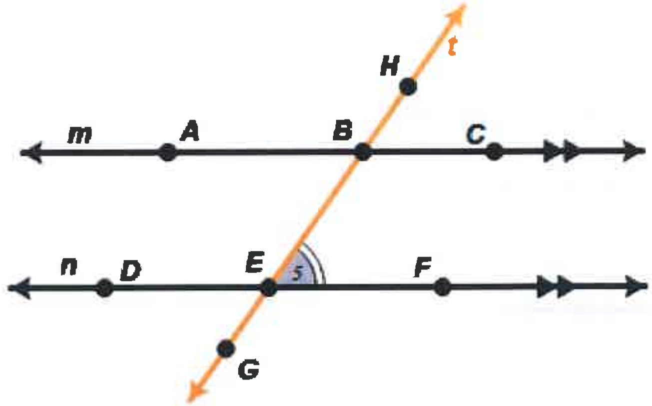


Which of the below terms best describe the relationship between ∠5 and ∠2 ?

a. Consecutive Exterior Angles
b. Alternate Exterior Angles
c. Consecutive Interior Angles
d. Alternate Interior Angles
e. Vertical Angles

Name: ______________________ **ID: A**

____ 3. *In the diagram line **m** is parallel to line **n** with a transversal line t.*

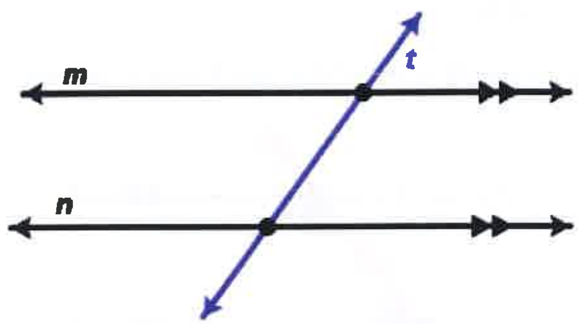


Which other angle would be the **alternate interior angle** to $\angle BEF$?

a. $\angle DEG$
b. $\angle EBC$
c. $\angle ABE$
d. $\angle CBH$

____ 4.

Given 2 **parallel lines** and a transversal that intersects both parallel lines, what must be true about pairs of **Corresponding Angles**?

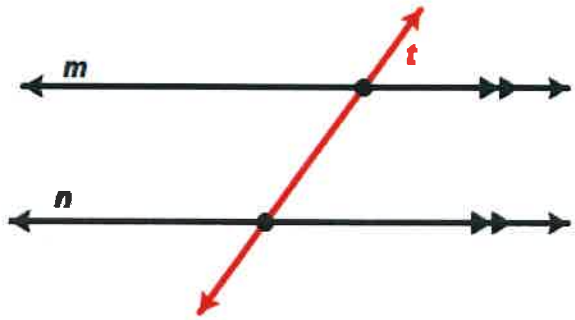


*In the diagram line **m** is parallel to line **n** with a transversal line t.*

a. The angles must be **congruent**.
b. The angles must be **complementary**.
c. The angles must be **supplementary**.
d. The angles must be **obtuse.**.

____ 5.

Given 2 **parallel lines** and a transversal that intersects both parallel lines, how many pairs of **Corresponding Angles** exist in the diagram at the right?

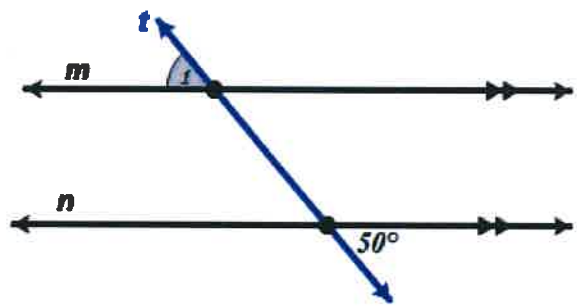


*In the diagram line **m** is parallel to line **n** with a transversal line t.*

a. 1
b. 2
c. 3
d. 4

____ 6.

Use the diagram to determine the measure of angle $m\angle 1$.



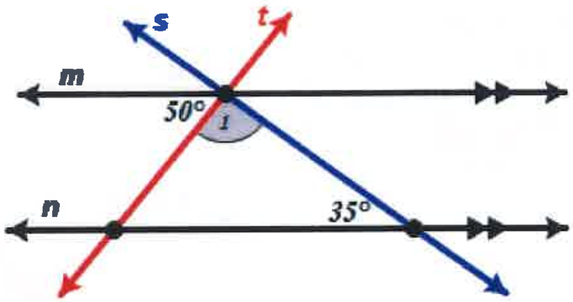
*In the diagram line **m** is parallel to line **n** with a transversal line **t**.*

a. 40°
b. 50°
c. 80°
d. 130°

Name: ______________________ ID: A

____ 7.

Use the diagram to determine the measure of angle $m\angle 1$.

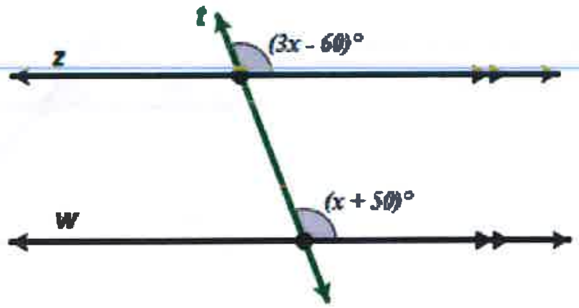


*In the diagram line **m** is parallel to line **n** with a transversal line **t** and transversal line **s**.*

a. 50°
b. 85°
c. 95°
d. 100°

____ 8.

Use the diagram to determine the measure of the value of x.



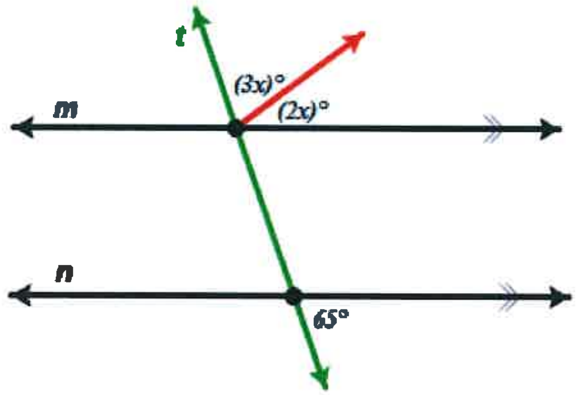
*In the diagram line z is parallel to line w with a transversal line **t**.*

a. $x = 42$
b. $x = 47.5$
c. $x = 55$
d. $x = 60$

Name: ______________________ ID: A

____ 9.

Use the diagram to determine the measure of the value of ***x***.



*In the diagram line **m** is parallel to line **n** with a transversal line **t**.*

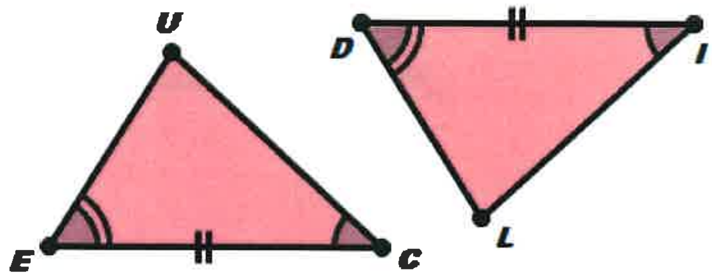
a. x = 5
b. x = 13
c. x = 20
d. x = 23

Name: ________________________ Class: ___________________ Date: __________ ID: A

02-05-Sample Quiz-Congruence Statements

Multiple Choice
Identify the choice that best completes the statement or answers the question.

____ 1. The triangles shown are congruent. Which of the below statements is a correct congruence statement?



a. $\triangle EUC \cong \triangle LID$
b. $\triangle EUC \cong \triangle DLI$
c. $\triangle EUC \cong \triangle LDI$
d. $\triangle EUC \cong \triangle IDL$
e. None of these is a correct congruence statement.

____ 2. Consider two congruent triangles such that $\triangle ABC \cong \triangle XYZ$ and the following measures:

- $AB = 5$
- $BC = 7$
- $CA = 8$

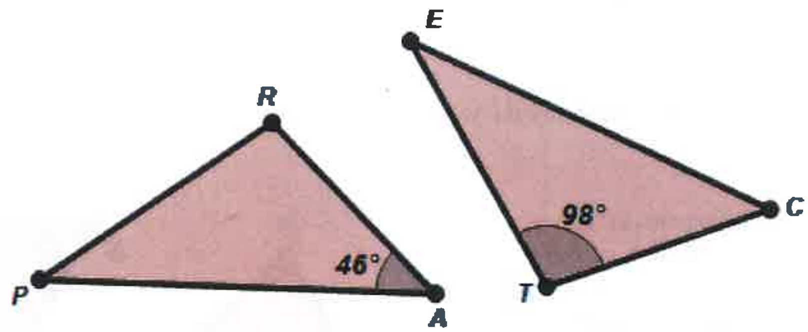
What is the length of the segment $\overline{ZY}$?

a. 5
b. 7
c. 8
d. Not enough information was provided.

Name: ________________________ ID: A

____ 3.

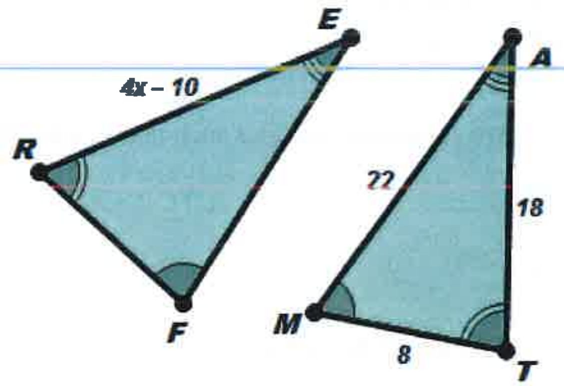
Given the statement, $\triangle PRA \cong \triangle ETC$ and the diagrams shown, determine the measure of $m\angle P$.



a. $m\angle P = 36°$
b. $m\angle P = 46°$
c. $m\angle P = 52°$
d. $m\angle P = 98°$

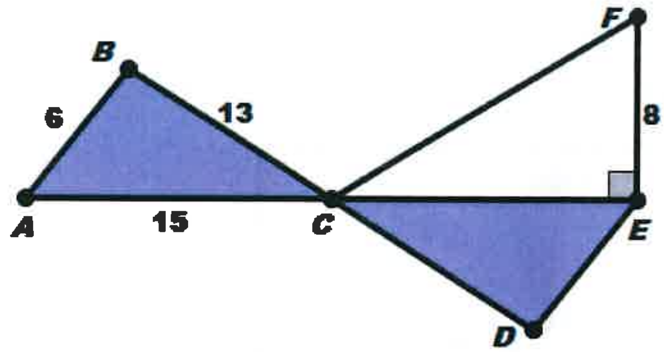
____ 4.

Given the statement, $\triangle FER \cong \triangle MAT$ and the diagrams shown, determine the value of x.



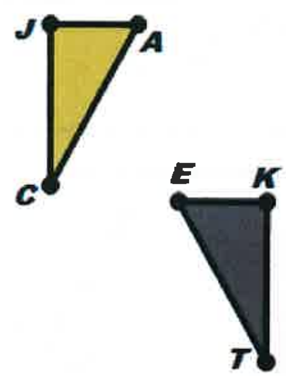
a. 4.5
b. 7
c. 8
d. Not enough information was provided.

____ 5. Given the statement $\triangle ABC \cong \triangle EDC$ and the diagrams shown, determine the length of segment $\overline{FC}$.



a. 15
b. 17
c. 19
d. Not enough information was provided.

____ 6. Given $\triangle JAC \cong \triangle KET$, which series of transformations would most directly map $\triangle JAC$ onto $\triangle KET$.

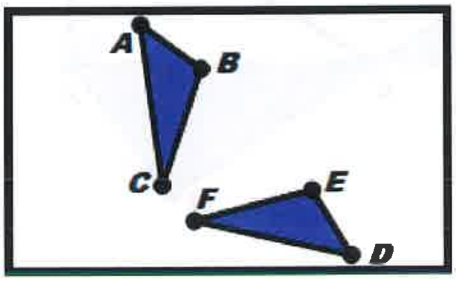


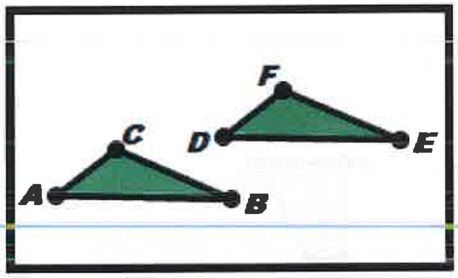
a. Reflection and then Dilation
b. Rotation and then Translation
c. Dilation and then Rotation
d. Reflection and then Translation

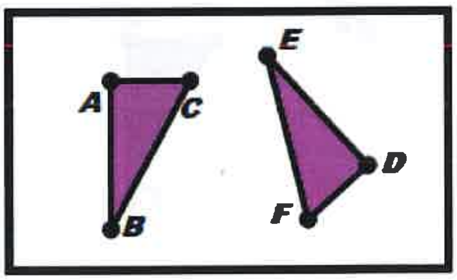
Name: ________________________ ID: A

____ 7. Each of the following pairs of triangles shown are congruent *(In each diagram $\triangle ABC \cong \triangle DEF$)*.

Which is the only pair that requires a **REFLECTION** at some point to map one triangle onto the other corresponding congruent triangle using transformations?

a. 

b. 

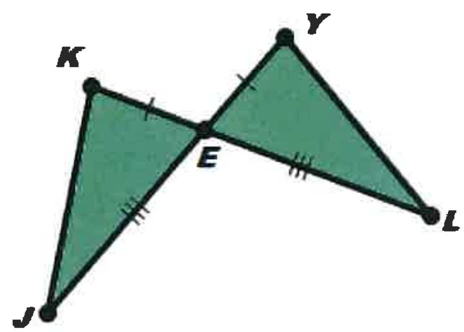
c. 

Name: ________________________ Class: ___________________ Date: __________ ID: A

02-06-Sample Quiz-Triangle Proofs

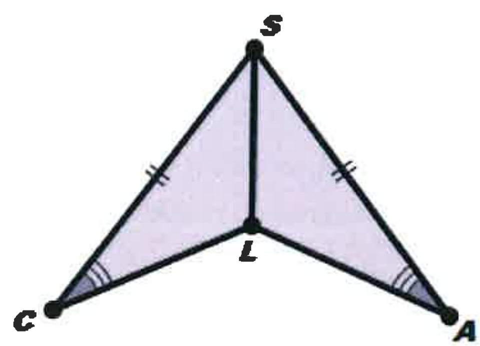
Multiple Choice
Identify the choice that best completes the statement or answers the question.

____ 1. Based on the diagram and congruent marks, what would be a correct congruence statement and the most likely reason the triangles might be congruent?



a. $\triangle JKE \cong \triangle LYE$ because **SSS**
b. $\triangle JKE \cong \triangle LYE$ because **SAS**
c. $\triangle JKE \cong \triangle LYE$ because **ASA**
d. $\triangle JKE \cong \triangle LYE$ because **AAS**
e. There is not enough information to guarantee congruent triangles.

____ 2. Based on the diagram and congruent marks, what would be a correct congruence statement and the most likely reason the triangles might be congruent?



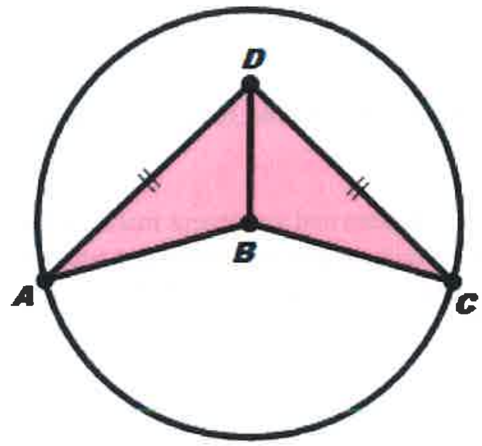
a. $\triangle CSL \cong \triangle ASL$ because **SSS**
b. $\triangle CSL \cong \triangle ASL$ because **SAS**
c. $\triangle CSL \cong \triangle ASL$ because **ASA**
d. $\triangle CSL \cong \triangle ASL$ because **AAS**
e. There is not enough information to guarantee congruent triangles.

Name: ________________________ ID: A

____ 3.

Consider the diagram, congruent marks, and circle with a center at point B.

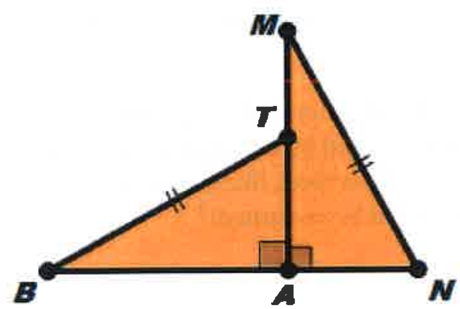
What would be a correct congruence statement and the most likely direct reason the triangles might be congruent?



a. $\triangle ABD \cong \triangle CBD$ because **SSS**
b. $\triangle ABD \cong \triangle CBD$ because **SAS**
c. $\triangle ABD \cong \triangle CBD$ because **ASA**
d. $\triangle ABD \cong \triangle CBD$ because **AAS**
e. There is not enough information to guarantee congruent triangles.

____ 4.

Based on the diagram and congruent marks, what would be a correct congruence statement and the most likely reason the triangles might be congruent?

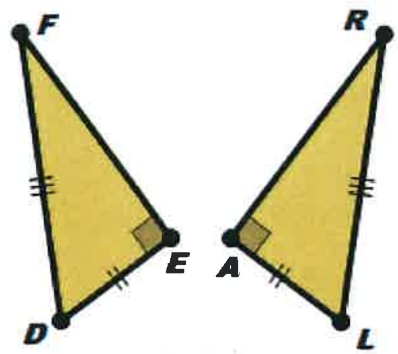


a. $\triangle BAT \cong \triangle MAN$ because **HL**
b. $\triangle BAT \cong \triangle MAN$ because **SAS**
c. $\triangle BAT \cong \triangle MAN$ because **ASA**
d. $\triangle BAT \cong \triangle MAN$ because **AAS**
e. There is not enough information to guarantee congruent triangles.

Name: ______________________ **ID: A**

____ 5.

Based on the diagram and congruent marks, what would be a correct congruence statement and the most likely reason the triangles might be congruent?

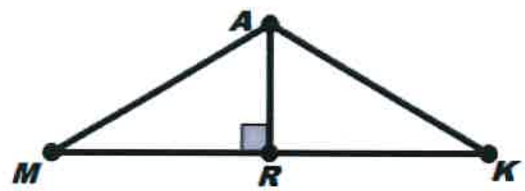


a. $\triangle FED \cong \triangle RAL$ because **SSS**
b. $\triangle FED \cong \triangle RAL$ because **SAS**
c. $\triangle FED \cong \triangle RAL$ because **ASA**
d. $\triangle FED \cong \triangle RAL$ because **HL**
e. There is not enough information to guarantee congruent triangles.

____ 6.

Given:
• $\overline{AR}$ is the perpendicular bisector of $\overline{MK}$.

Prove: $\triangle MAR \cong \triangle KAR$.



Statements	Reasons
1. $\overline{AR}$ is the perpendicular bisector of $\overline{MK}$.	Given
2. $\overline{MR} \cong \overline{KR}$	Definition of a bisector
3. $\overline{AR} \cong \overline{AR}$	Reflexive Property of Congruence
4. $\measuredangle MRA$ and $\measuredangle KRA$ are right angles.	Definition of perpendicular lines
5. $\measuredangle MRA \cong \measuredangle KRA$	All right angles are congruent
6. $\Delta MAR \cong \Delta KAR$	???

What would be the **reason** on the last step of the proof?

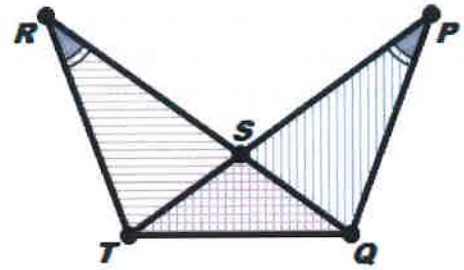
a. **AAS**
b. **ASA**
c. **SAS**
d. **SSS**
e. **HL**

Name: ________________________ **ID: A**

____ 7.

Given:

- $\angle R \cong \angle P$
- $\triangle TSQ$ forms an isosceles triangle with a base of $\overline{TQ}$



Prove: $\triangle TRQ \cong \triangle QPT$.

Statements	Reasons
1. $\measuredangle R \cong \measuredangle P$	Given
2. ΔTSQ is an isosceles triangle with base $\overline{TQ}$	Given
3. $\measuredangle PTQ \cong \measuredangle RQT$	???
4. $\overline{QT} \cong \overline{QT}$	Reflexive Property of Congruence
5. $\Delta TRQ \cong \Delta QPT$	Angle-Angle-Side (AAS) Theorem

What would be the **reason** on the 3rd step of the proof?

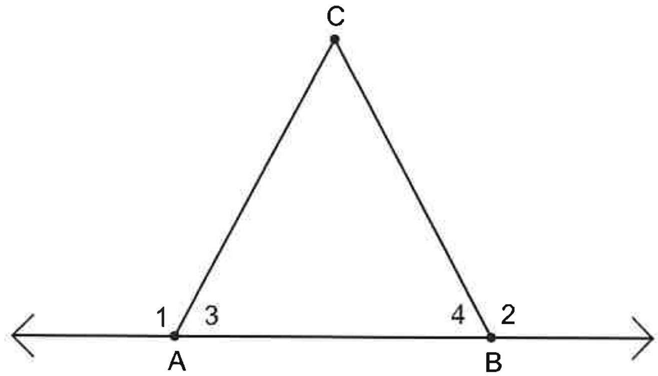
a. Reflexive Property of Congruence
b. Alternate Interior Angles Theorem
c. Definition of Midpoint
d. Isosceles Triangle Theorem
e. Vertical Angles are Congruent

Geometry
Proof Worksheet (3)

Name:
Date:

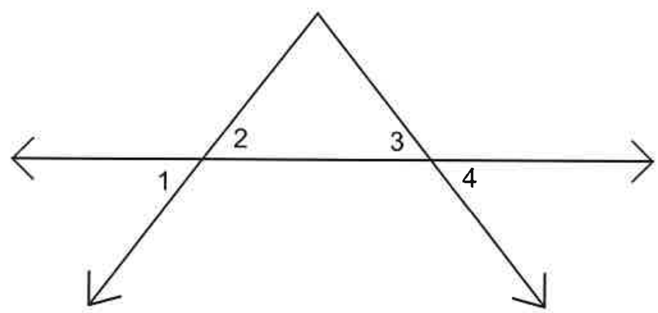
1. Given: $\angle 3 \cong \angle 4$

Prove: $\angle 1 \cong \angle 2$



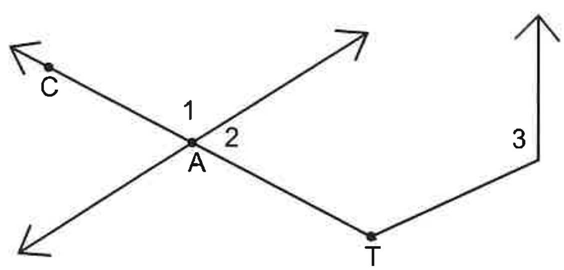
2. Given: $\angle 1 \cong \angle 4$

Prove: $\angle 2 \cong \angle 3$



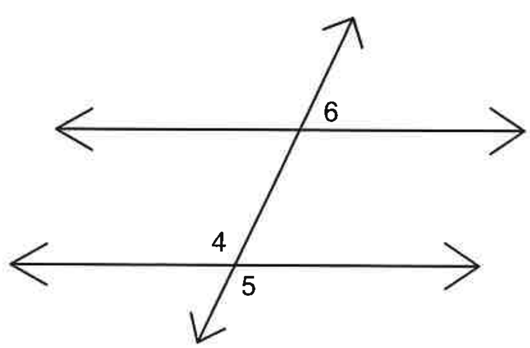
3. Given: $\angle 1 \cong \angle 3$

Prove: $\angle 2$ is supplementary to $\angle 3$



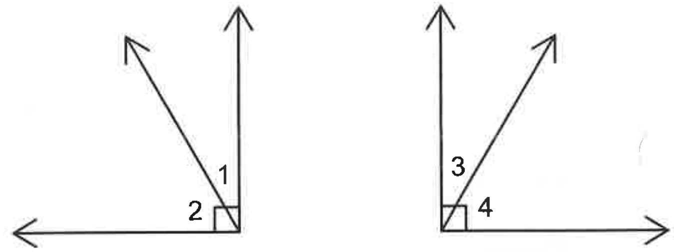
4. Given: $\angle 4 \cong \angle 6$

Prove: $\angle 5 \cong \angle 6$



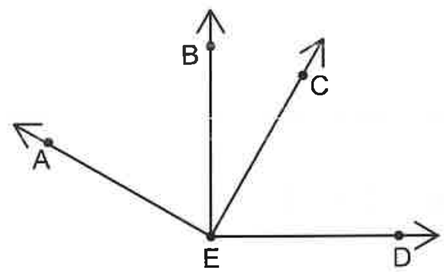
5. Given: $\angle 1 \cong \angle 3$

Prove: $\angle 2 \cong \angle 4$



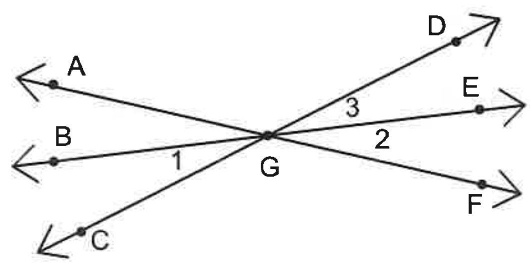
6. Given: $\angle AEC$ is a right angle
$\angle BED$ is a right angle

Prove: $\angle AEB \cong \angle DEC$



7. Given: $\overrightarrow{GE}$ bisects $\angle DGF$

Prove: $\angle 1 \cong \angle 2$



8. If a pair of vertical angles are supplementary, what can we conclude about the angles? Sketch a diagram that supports your reasoning?

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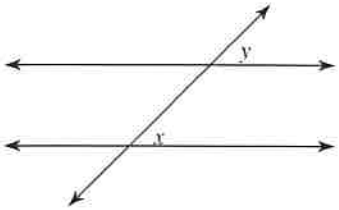
Name________________________________

Parallel Lines and Transversals

Date________________ Period____

Identify each pair of angles as corresponding, alternate interior, alternate exterior, or consecutive interior.

1)



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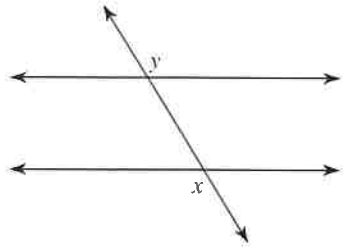

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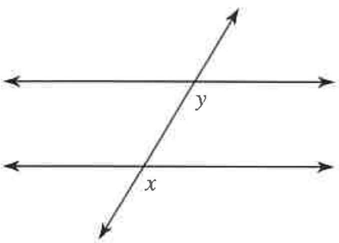

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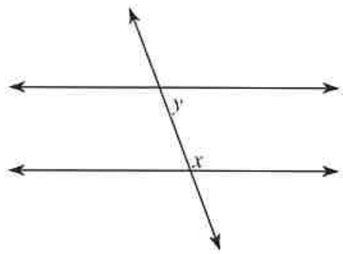

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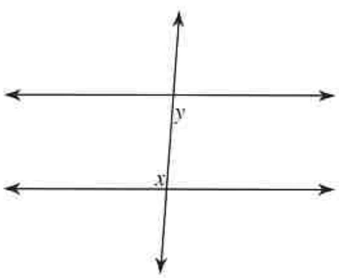

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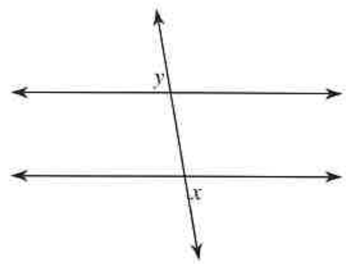

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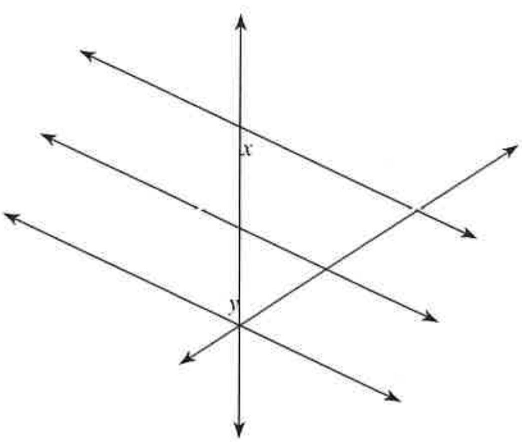

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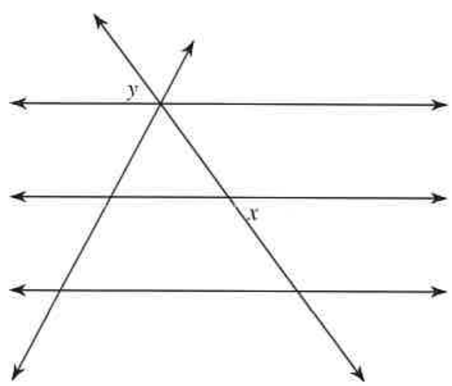

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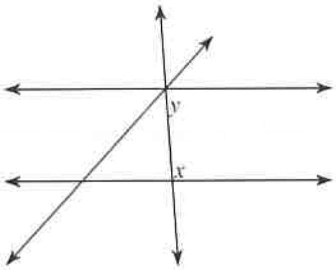

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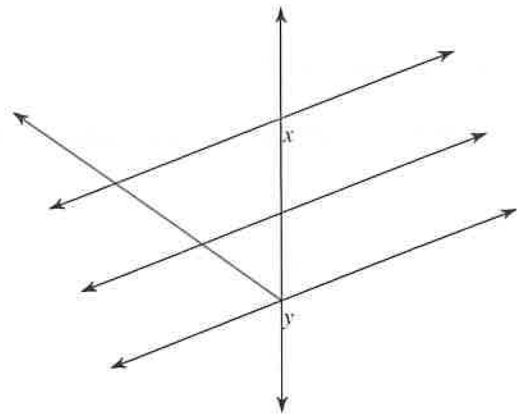
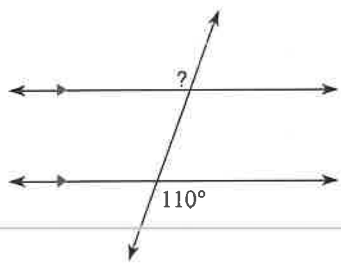


Find the measure of each angle indicated.

11)



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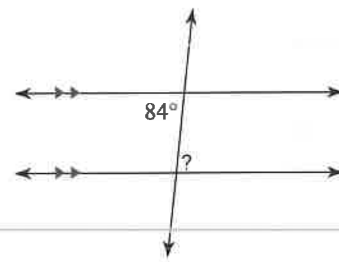

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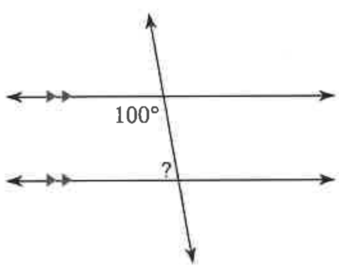

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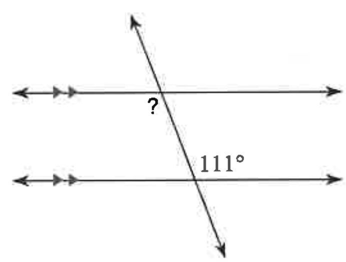

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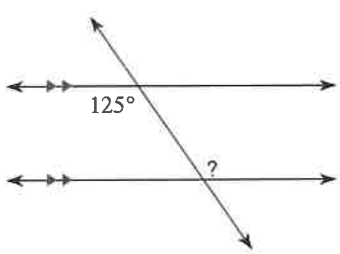

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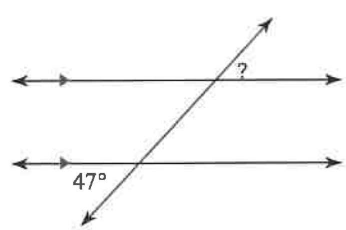

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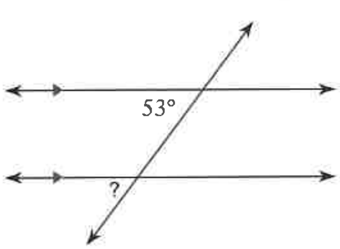

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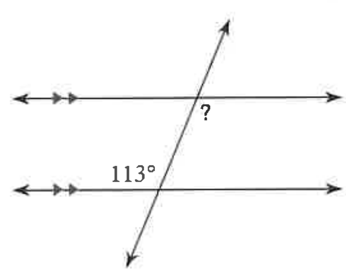
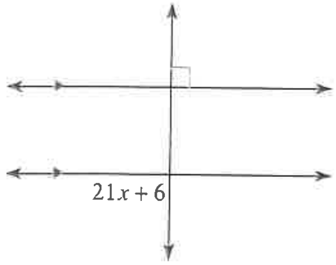
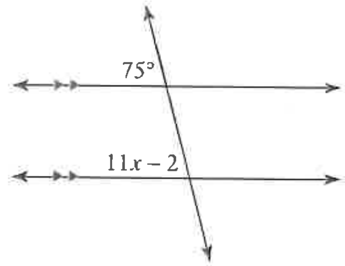


Solve for x.

19)



20)



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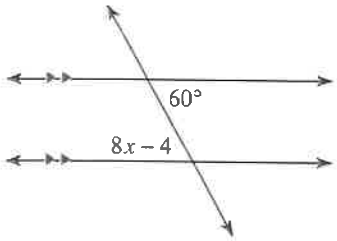

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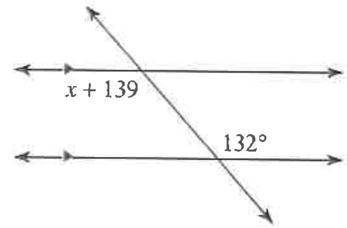

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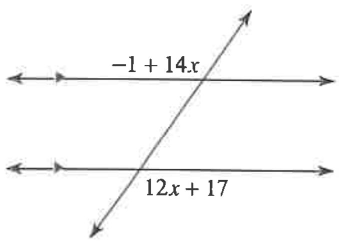

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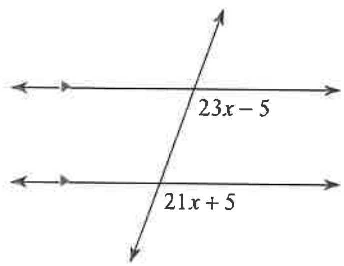
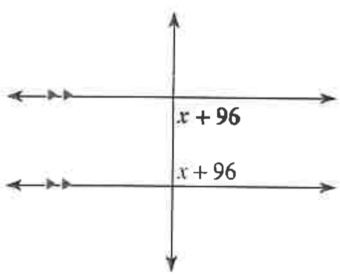


Find the measure of the angle indicated in bold.

25)



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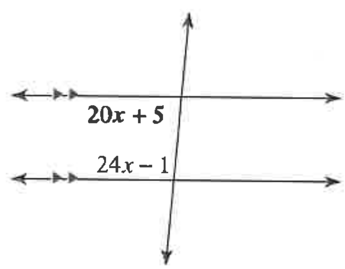

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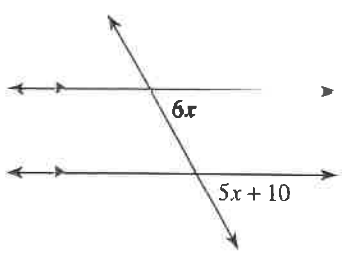

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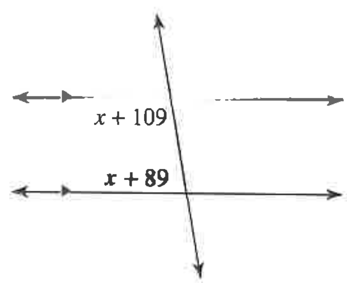

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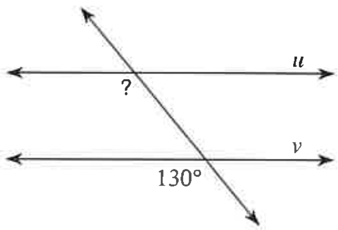
Name_______________________________

Proving Lines Parallel

Date________________ Period____

Find the measure of the indicated angle that makes lines u and v parallel.

1)



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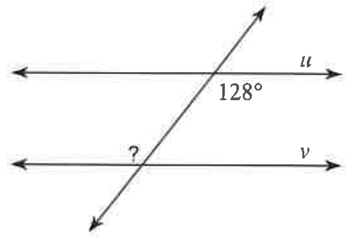

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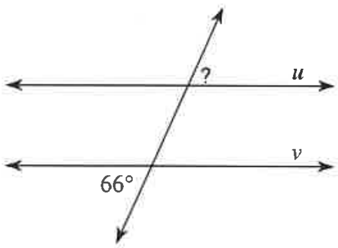

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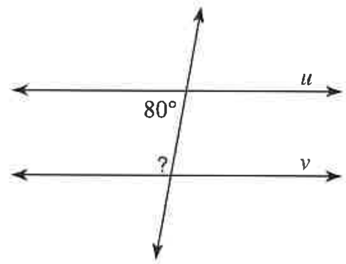

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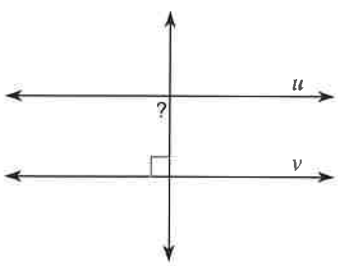

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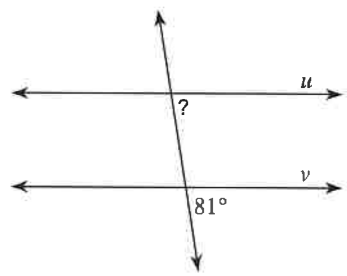

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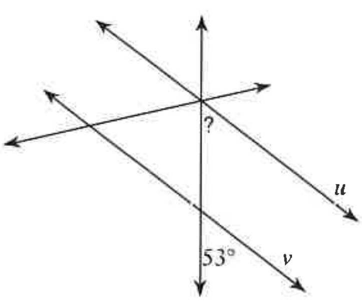

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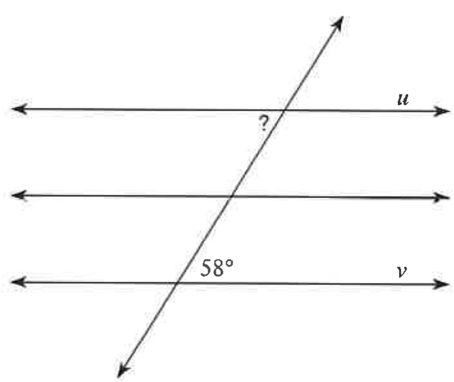

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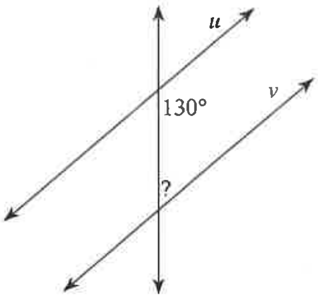

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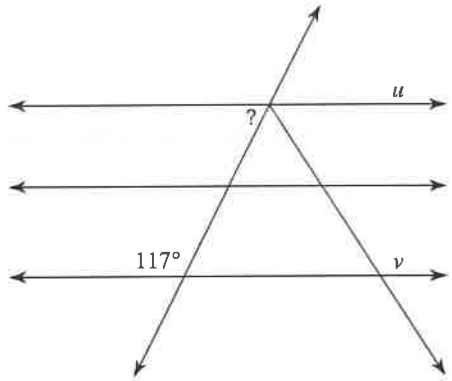
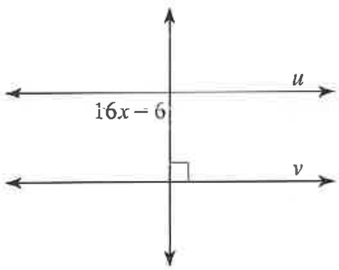


Find the value of x that makes lines u and v parallel.

11)



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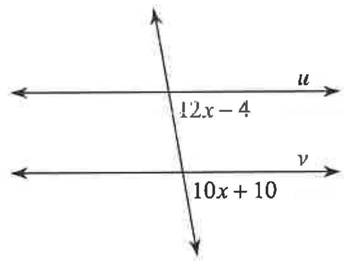

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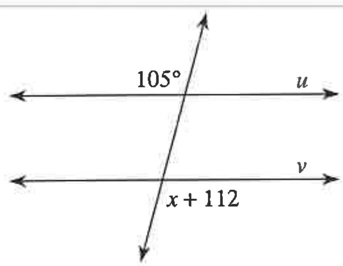

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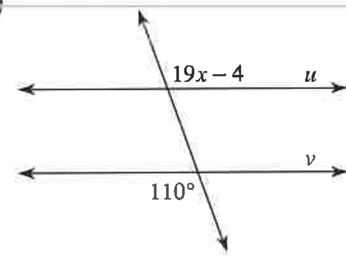

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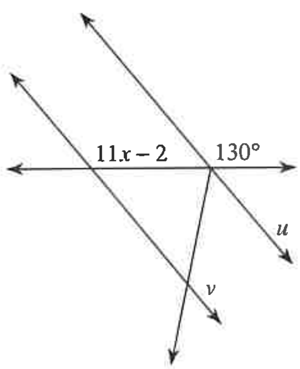

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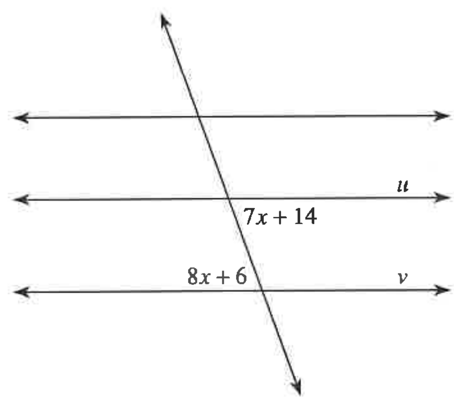


Critical thinking questions:

17) For question #16, find a value of x that makes lines u and v intersect.

18) Even if the lines in question #16 were not parallel, could $x = 25$? Why or why not?

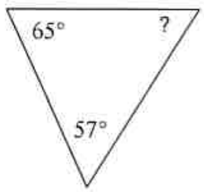
Kuta Software - Infinite Geometry

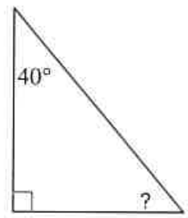
Name________________________________

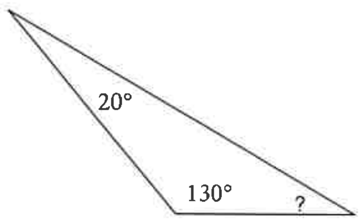
Angles in a Triangle

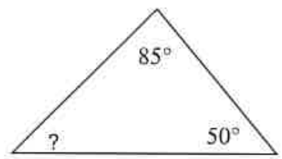
Date________________ Period____

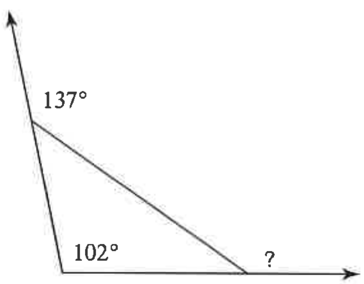
Find the measure of each angle indicated.

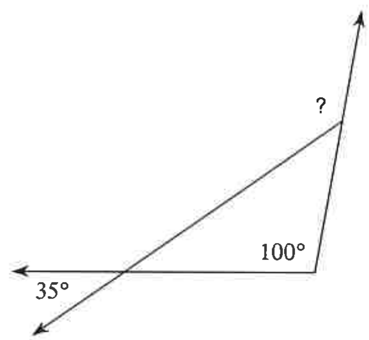
1) 

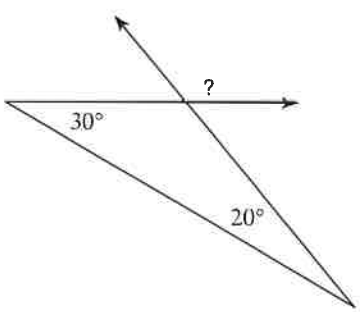
2) 

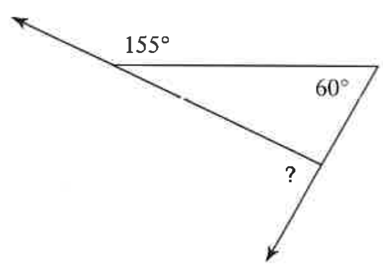
3) 

4) 

5) 

6) 

7) 

8) 

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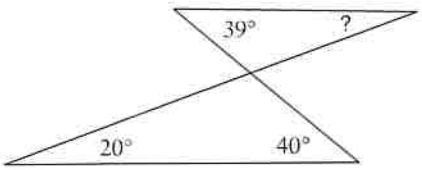

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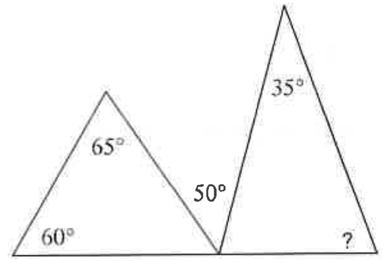

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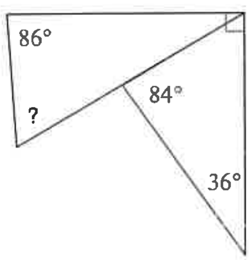

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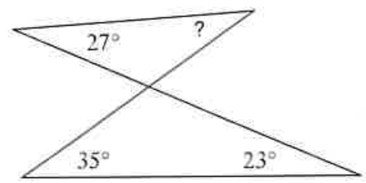

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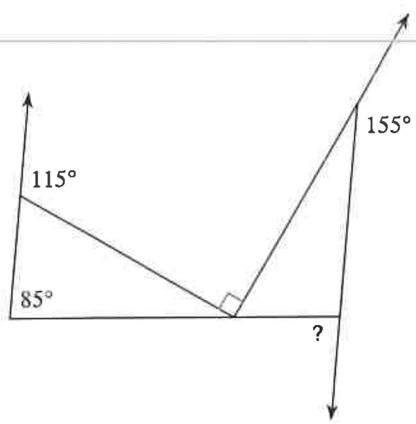

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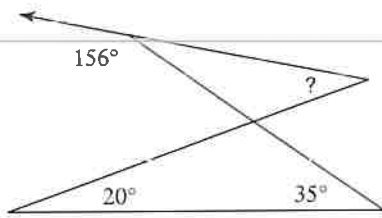

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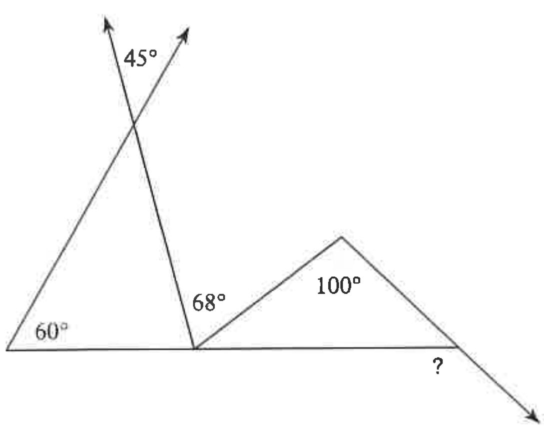

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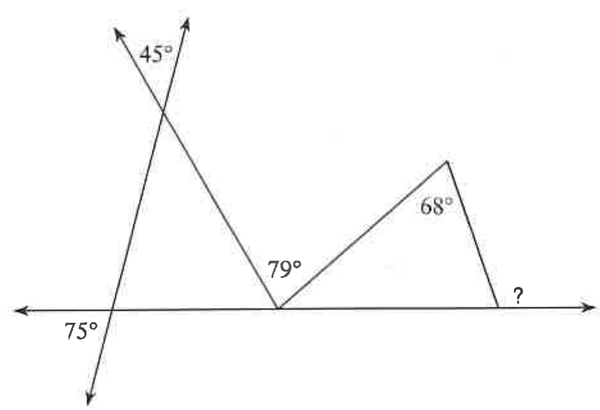
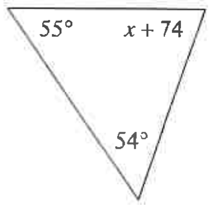


Solve for x.

17)



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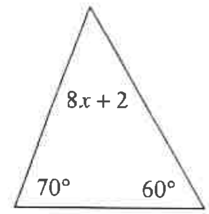

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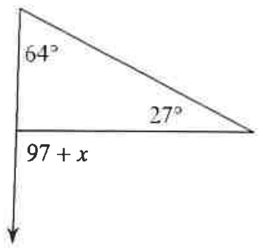
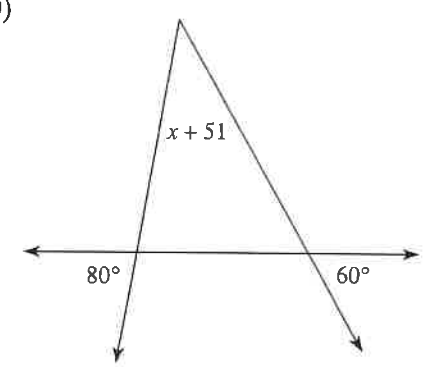
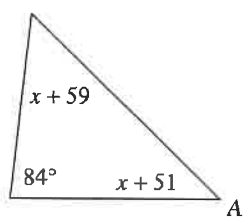


20)



Find the measure of angle A.

21)



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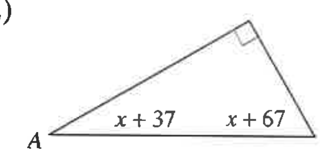

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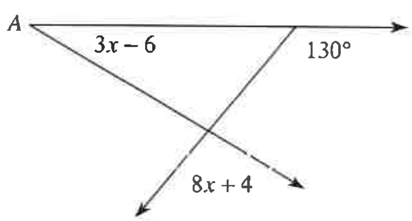

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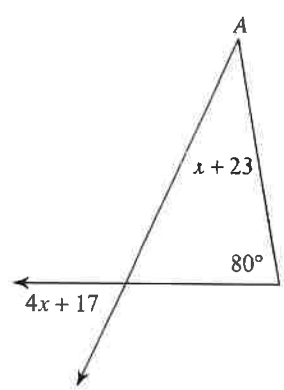


Kuta Software - Infinite Geometry

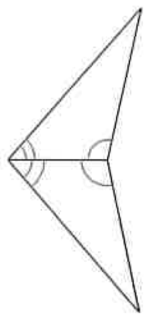
Name________________________

ASA and AAS Congruence

Date________________ Period____

State if the two triangles are congruent. If they are, state how you know.

1)



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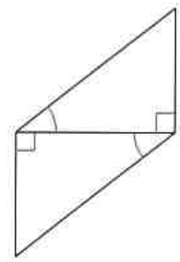
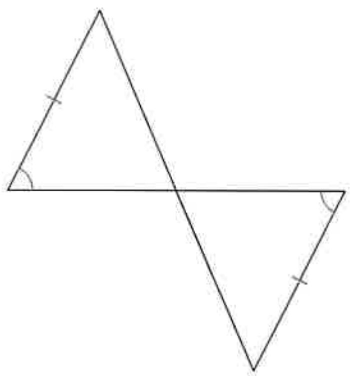


3)



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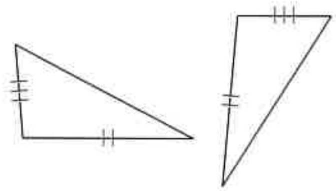

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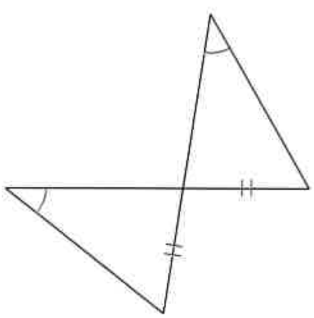

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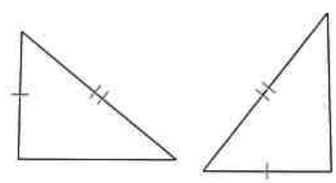

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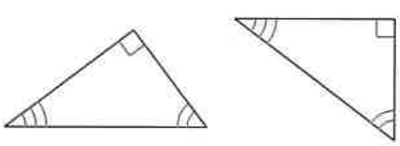

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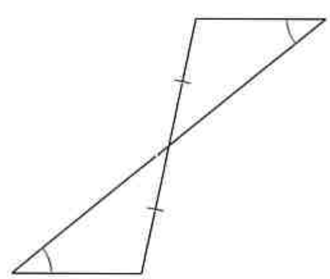

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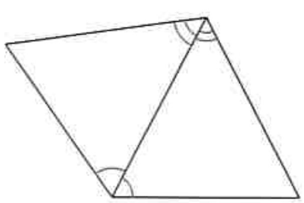

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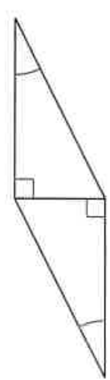
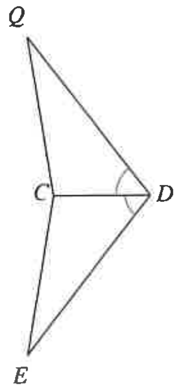
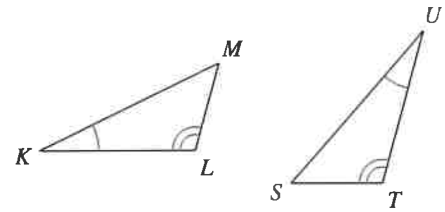


State what additional information is required in order to know that the triangles are congruent for the reason given.

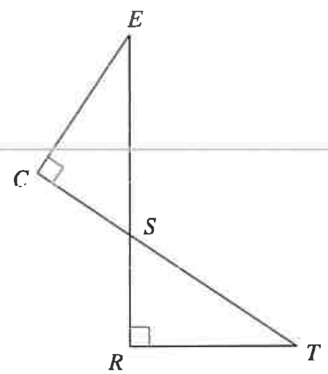
11) ASA



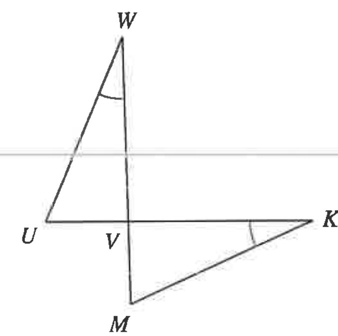
12) ASA



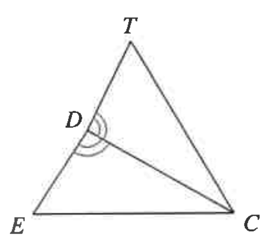
13) ASA



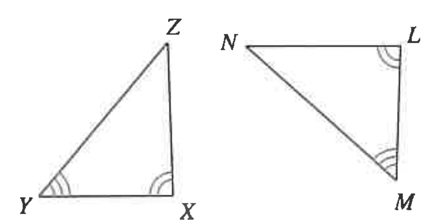
14) ASA



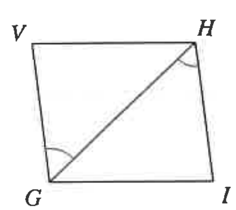
15) AAS



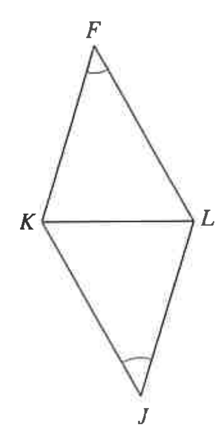
16) AAS



17) ASA



18) AAS



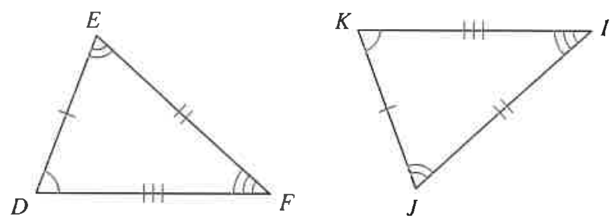
Name________________________________

Congruence and Triangles

Date________________ Period____

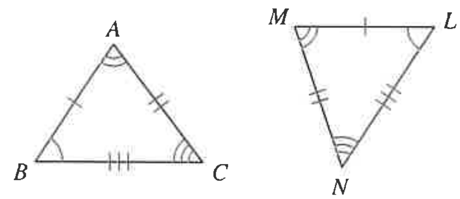
Complete each congruence statement by naming the corresponding angle or side.

1) $\Delta DEF \cong \Delta KJI$



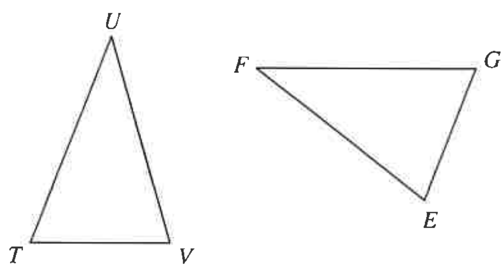
$\overline{FD} \cong ?$

2) $\Delta BAC \cong \Delta LMN$



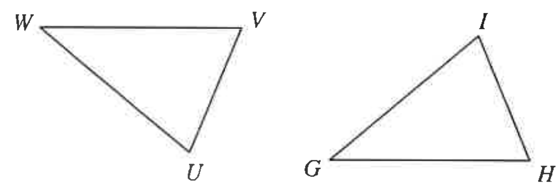
$\angle A \cong ?$

3) $\Delta TUV \cong \Delta GFE$



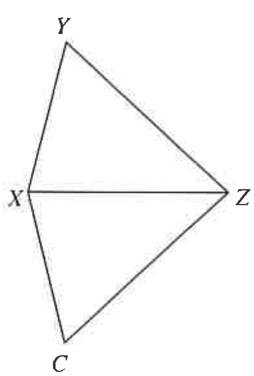
$\angle U \cong ?$

4) $\Delta WVU \cong \Delta GHI$



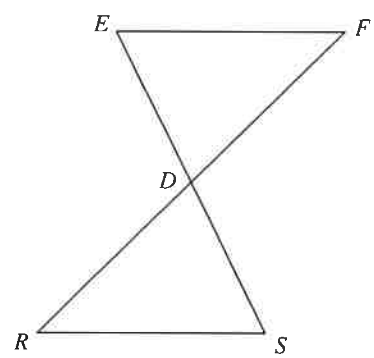
$\angle W \cong ?$

5) $\Delta ZXY \cong \Delta ZXC$



$\angle Y \cong ?$

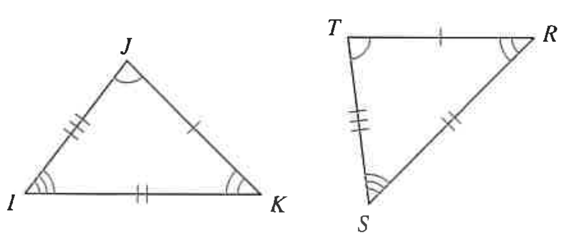
6) $\Delta DEF \cong \Delta DSR$



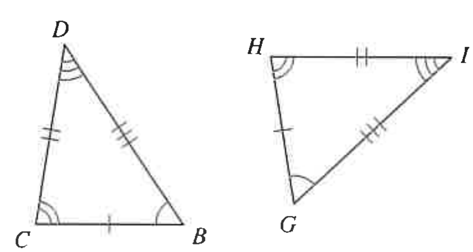
$\angle F \cong ?$

Write a statement that indicates that the triangles in each pair are congruent.

7)



8)



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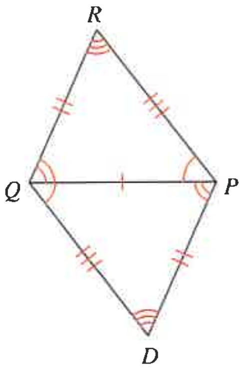

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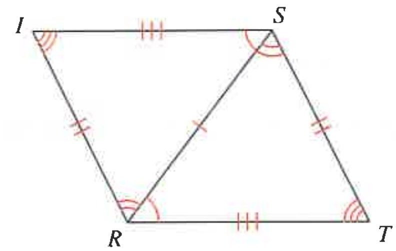

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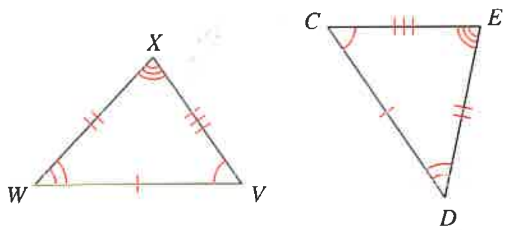

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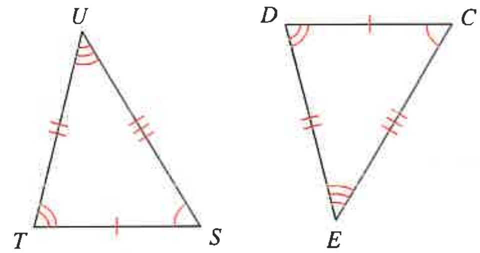
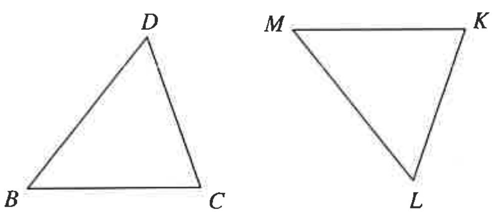
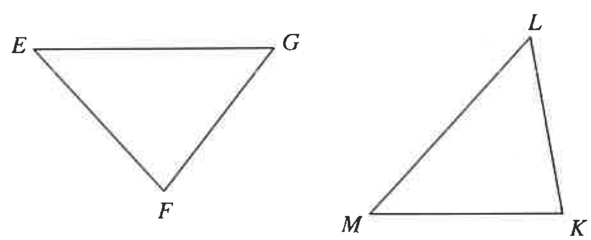


Mark the angles and sides of each pair of triangles to indicate that they are congruent.

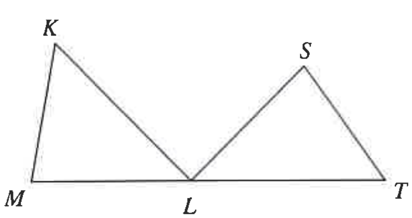
13) $\Delta BDC \cong \Delta MLK$



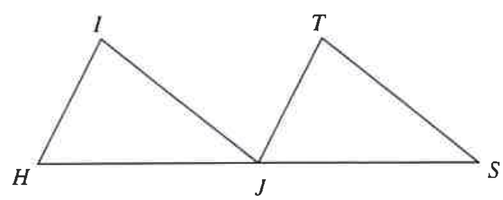
14) $\Delta GFE \cong \Delta LKM$



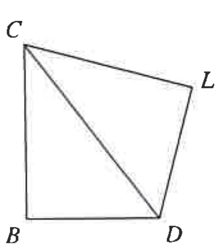
15) $\Delta MKL \cong \Delta STL$



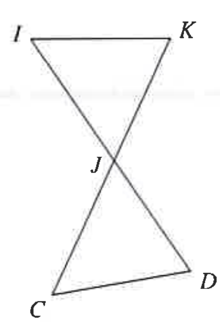
16) $\Delta HIJ \cong \Delta JTS$



17) $\Delta CDB \cong \Delta CDL$



18) $\Delta JIK \cong \Delta JCD$



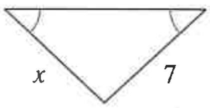
Kuta Software - Infinite Geometry

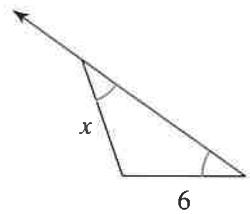
Name________________________________

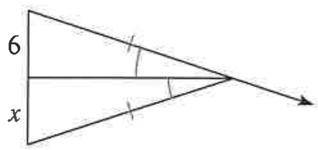
Isosceles and Equilateral Triangles

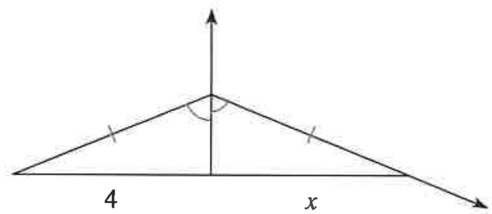
Date________________ Period____

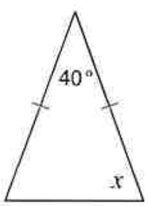
Find the value of *x*.

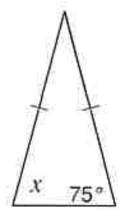
1) 

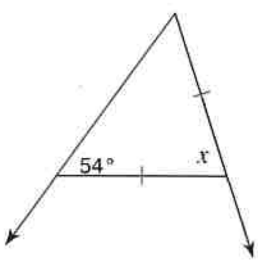
2) 

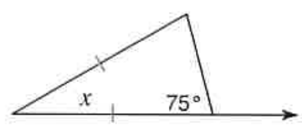
3) 

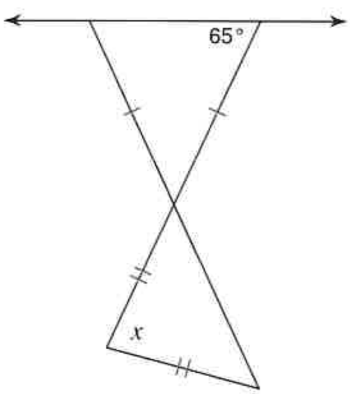
4) 

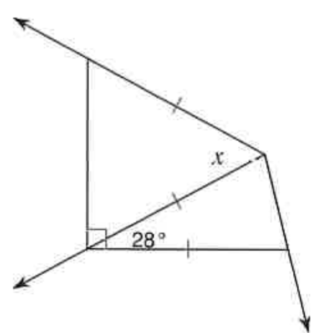
5) 

6) 

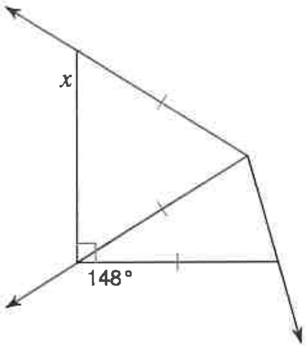
7) 

8) 

9) 

10) 

11)



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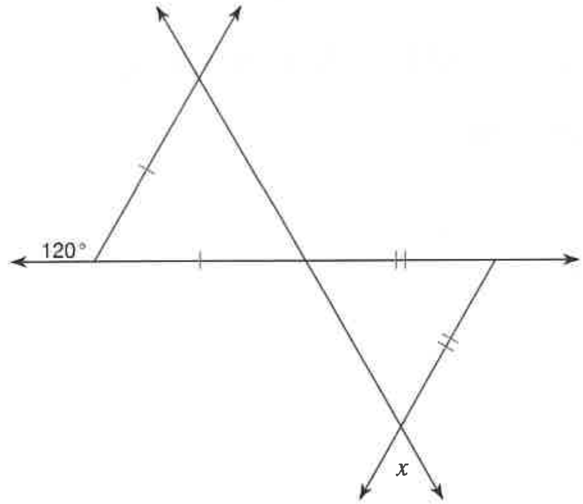
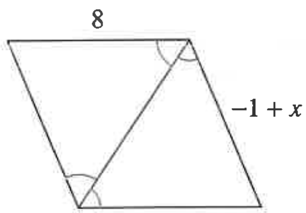


13)



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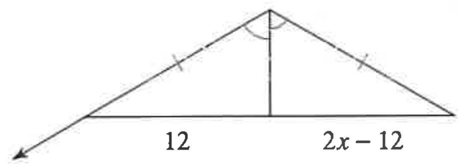
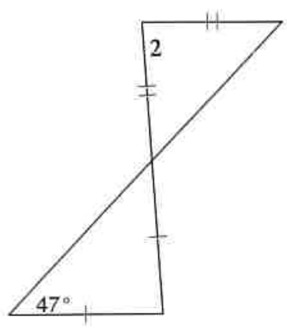
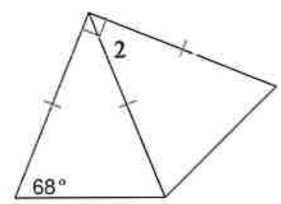


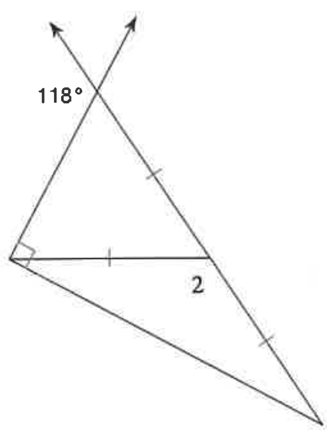
15) $m\angle 2 = x + 94$



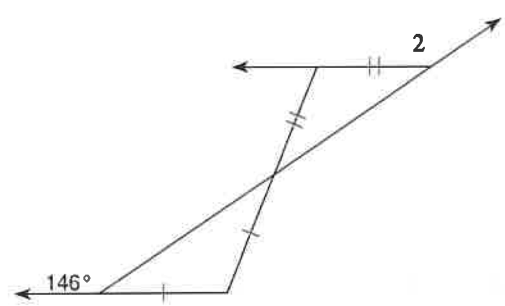
16) $m\angle 2 = 4x - 2$



17) $m\angle 2 = 12x + 4$



18) $m\angle 2 = 13x + 3$



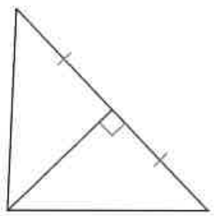
Kuta Software - Infinite Geometry

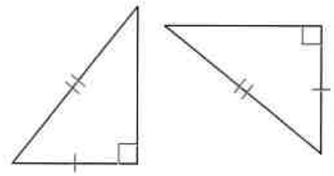
Name________________________________

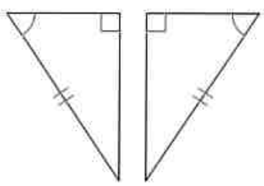
Right Triangle Congruence

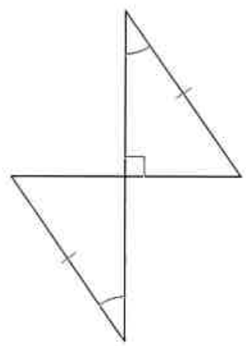
Date________________ Period____

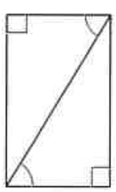
State if the two triangles are congruent. If they are, state how you know.

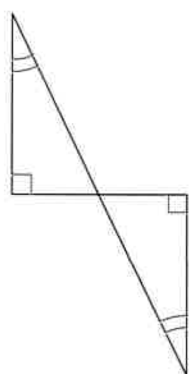
1) 

2) 

3) 

4) 

5) 

6) 

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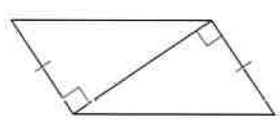
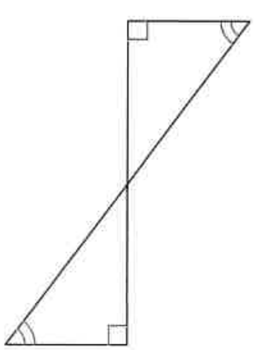
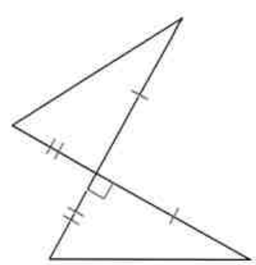
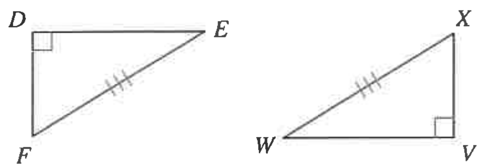
8) 

9) 

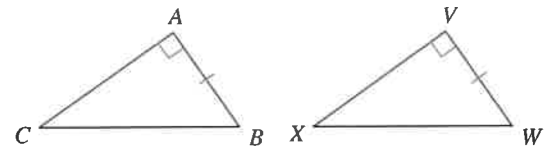
10) 

State what additional information is required in order to know that the triangles are congruent for the reason given.

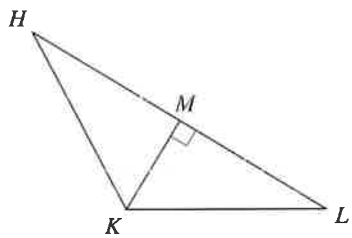
11) HL



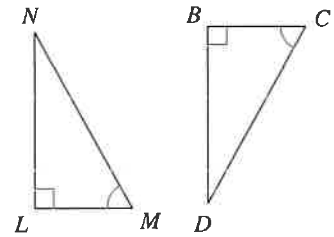
12) LL



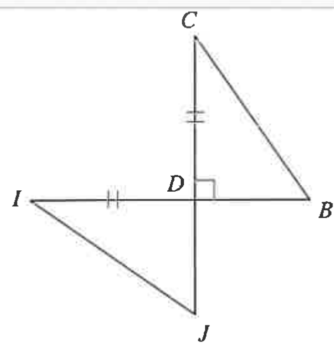
13) LL



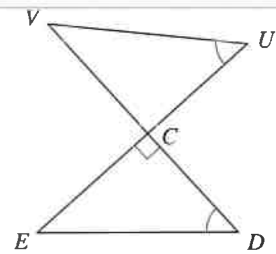
14) HA



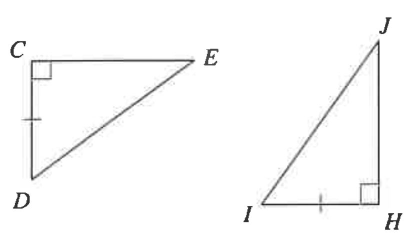
15) LA



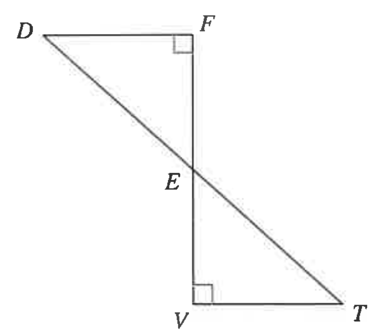
16) HA



17) HL



18) LA



Kuta Software - Infinite Geometry

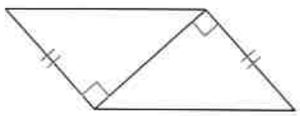
Name______________________________

SSS and SAS Congruence

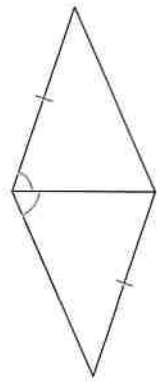
Date_________________ Period____

State if the two triangles are congruent. If they are, state how you know.

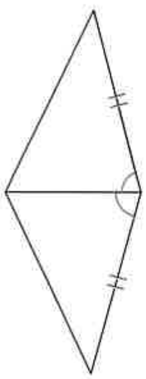
1)



2)



3)



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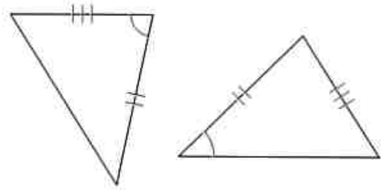
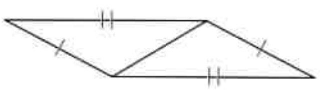
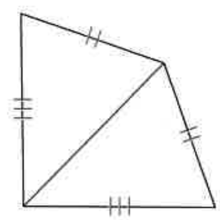


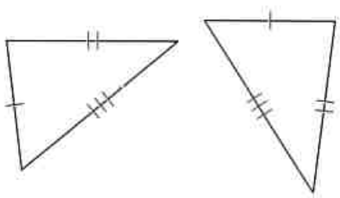
5)



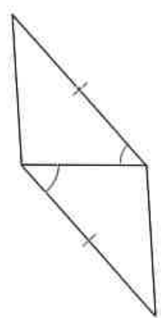
6)



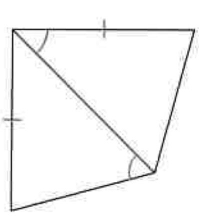
7)



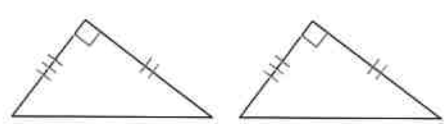
8)



9)

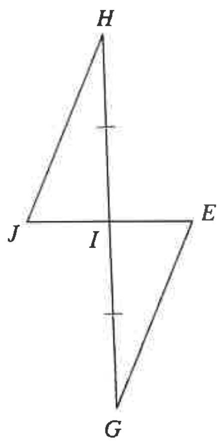


10)

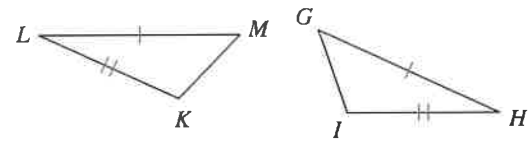


State what additional information is required in order to know that the triangles are congruent for the reason given.

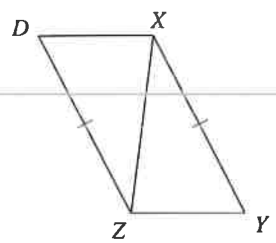
11) SAS



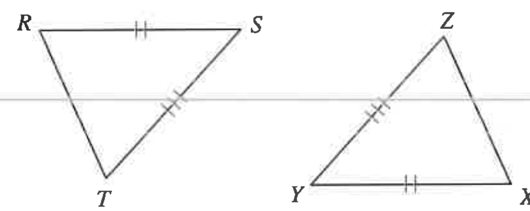
12) SAS



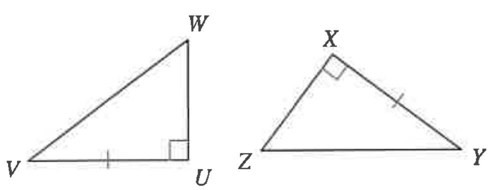
13) SSS



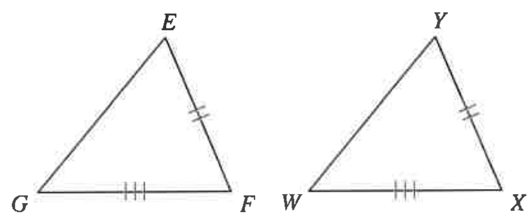
14) SSS



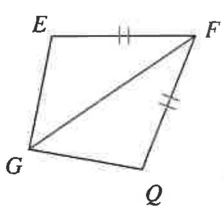
15) SAS



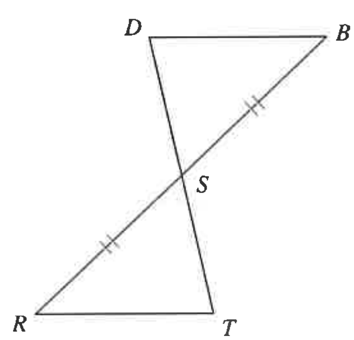
16) SSS



17) SAS



18) SAS



Kuta Software - Infinite Geometry

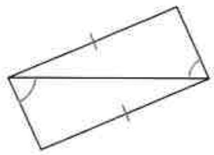
Name________________________________

SSS, SAS, ASA, and AAS Congruence

Date________________ Period____

State if the two triangles are congruent. If they are, state how you know.

1)



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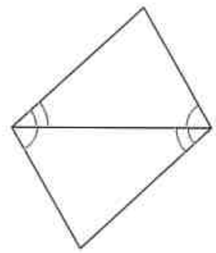

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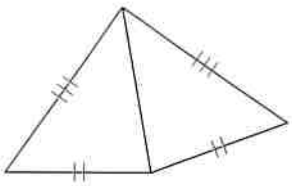

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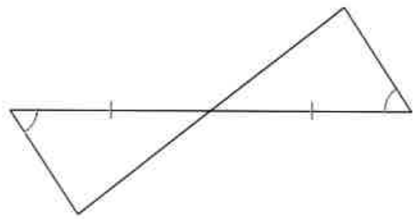

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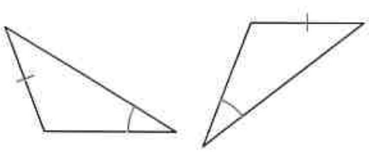
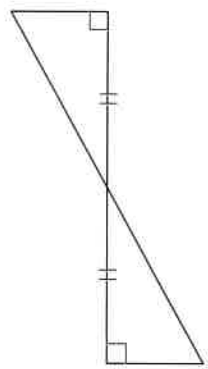
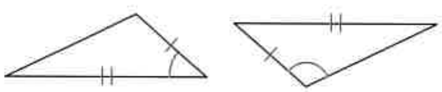


6)



7)



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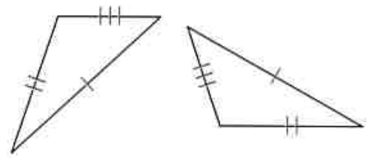
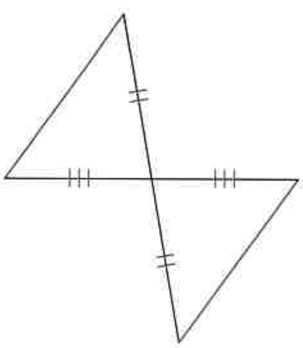
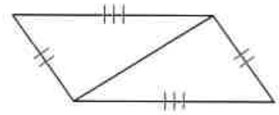


9)

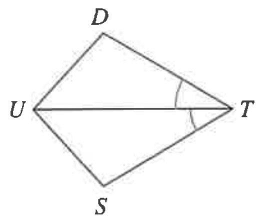


10)

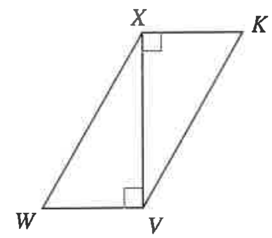


State what additional information is required in order to know that the triangles are congruent for the reason given.

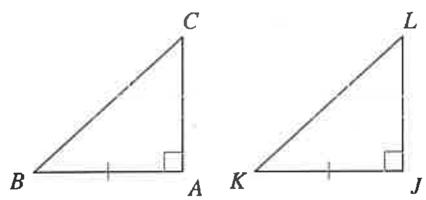
11) ASA



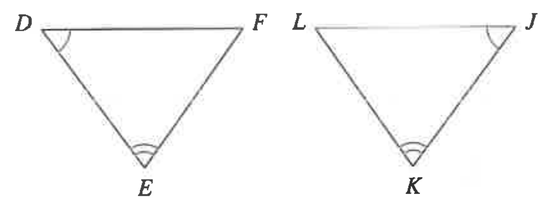
12) SAS



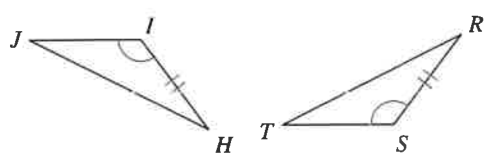
13) SAS



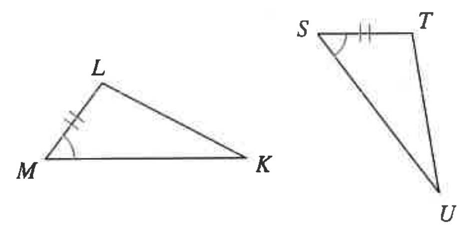
14) ASA



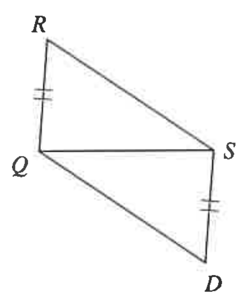
15) SAS



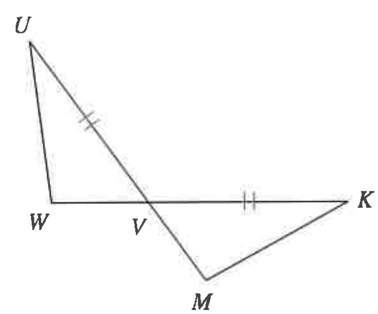
16) ASA



17) SSS



18) SAS



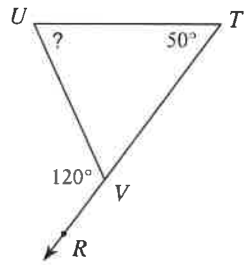
Kuta Software - Infinite Geometry

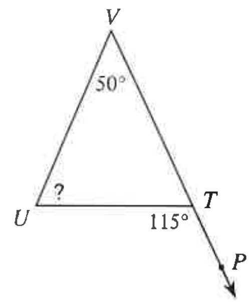
Name________________________

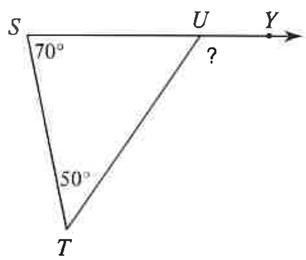
The Exterior Angle Theorem

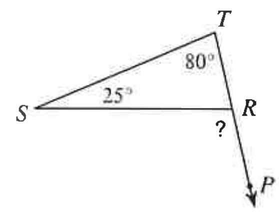
Date________________ Period____

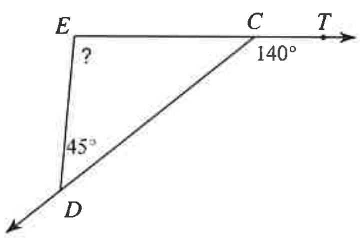
Find the measure of each angle indicated.

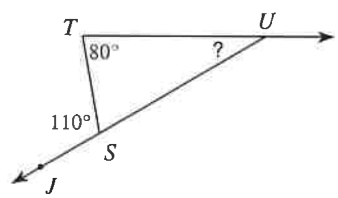
1) 

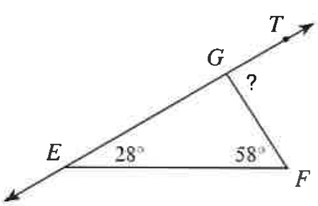
2) 

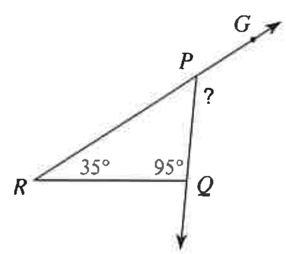
3) 

4) 

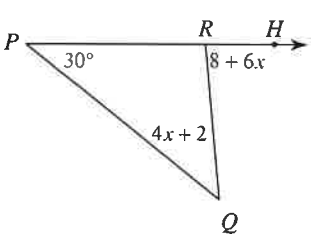
5) 

6) 

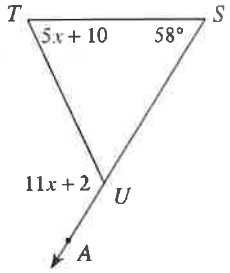
7) 

8) 

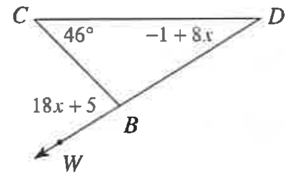
Solve for *x*.

9) 

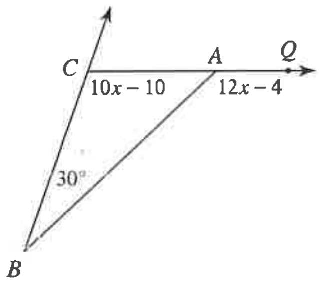
10)



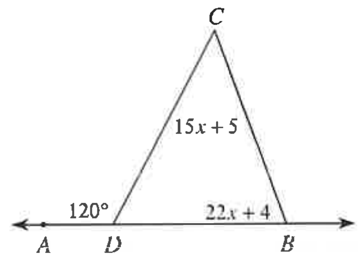
11)



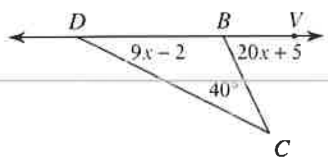
12)



13)

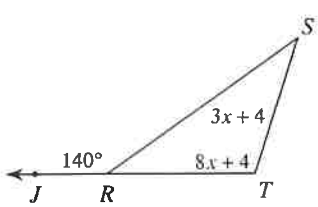


14)

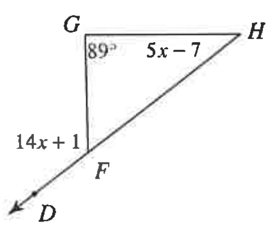


Find the measure of the angle indicated.

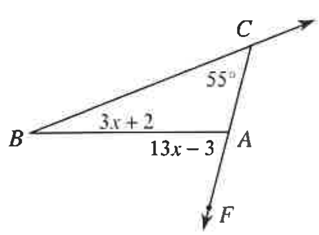
15) Find $m\angle S$.



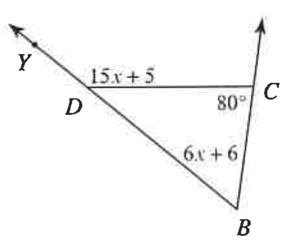
16) Find $m\angle H$.



17) Find $m\angle FAB$.



18) Find $m\angle YDC$.



Kuta Software - Infinite Geometry

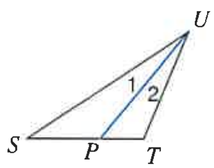
Name______________________

Angle Bisectors of Triangles

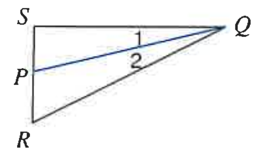
Date________________ Period____

Each figure shows a triangle with one of its angle bisectors.

1) $m\angle SUT = 34°$. Find $m\angle 1$.

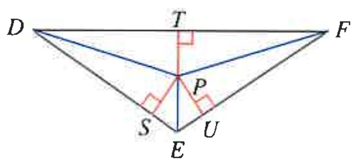


2) Find $m\angle SQR$ if $m\angle 2 = 13°$.

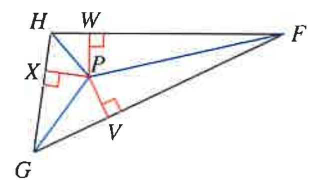


Each figure shows a triangle with its three angle bisectors intersecting at point P.

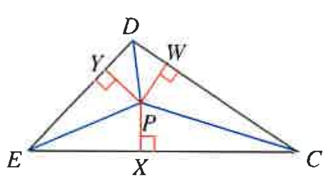
3) $PT = 3$. Find PU.



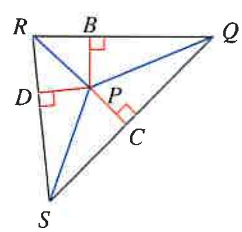
4) Find PV if $PW = 7$.



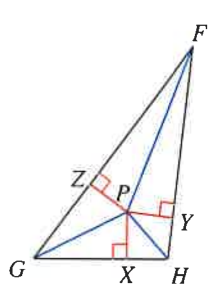
5) Find PW if $PX = 5$.



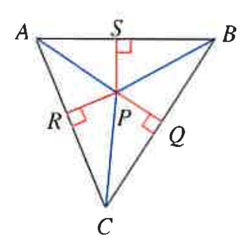
6) Find PD if $PC = 8$.



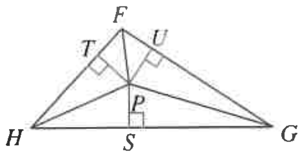
7) $PY = 2$ and $HP = 3$.
Find HY.



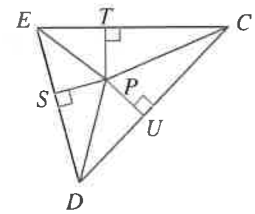
8) Find AP if $PQ = 1$
and $AR = 2$.



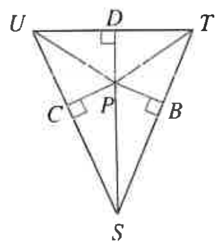
9) $PT = 5$ and $FP = 7$.
Find FT.



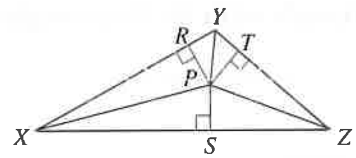
10) $PT = 3$ and $CP = 8$.
Find CT.



11) Find PB if $UC = 2$ and $UP = 3$.

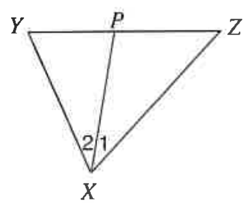


12) $PS = 3$ and $XP = 5$.
Find XS.

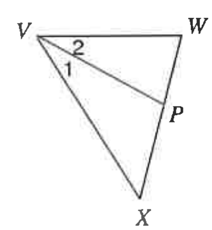


Each figure shows a triangle with one of its angle bisectors.

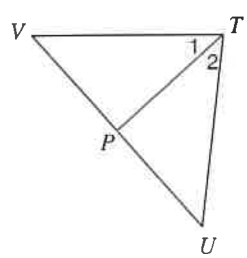
13) Find x if $m\angle 2 = 4x + 5$ and $m\angle 1 = 5x - 2$.



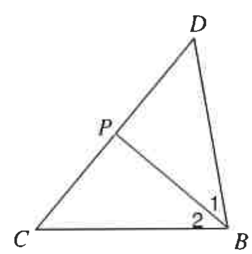
14) Find x if $m\angle 2 = 1 + 28x$ and $m\angle XVW = 59x - 1$.



15) $m\angle 1 = 7x + 7$ and $m\angle VTU = 16x + 4$.
Find $m\angle 1$.



16) Find $m\angle 2$ if $m\angle 2 = 7x + 5$ and $m\angle 1 = 9x - 5$.



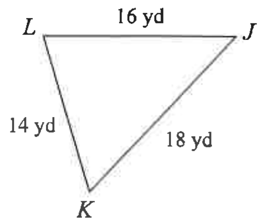
Kuta Software - Infinite Geometry

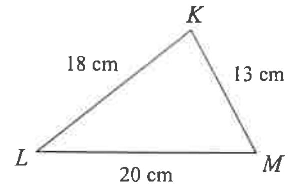
Name________________________

Inequalities in One Triangle

Date________________ Period____

Order the angles in each triangle from smallest to largest.

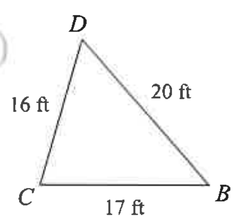


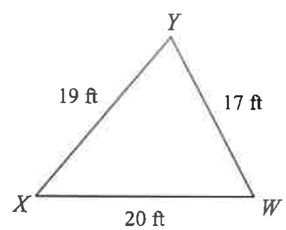
2) 

3) In ΔRQP
$QP = 15$ ft
$RP = 25$ ft
$RQ = 13$ ft

4) In ΔTUV
$UV = 17$ yd
$TV = 14$ yd
$TU = 9$ yd

Name the largest and smallest angle in each triangle.

5) 

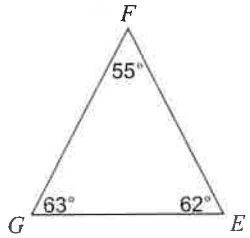
6) 

7) In ΔUVW
$VW = 13$ m
$UW = 11.7$ m
$UV = 5.8$ m

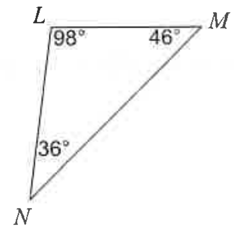
8) In ΔEFG
$FG = 10.9$ in
$EG = 17$ in
$EF = 10.9$ in

Order the sides of each triangle from shortest to longest.

9)



10)

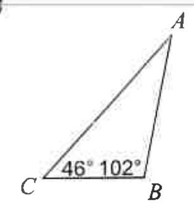


11) In ΔVWX
$m\angle V = 50°$
$m\angle W = 48°$
$m\angle X = 82°$

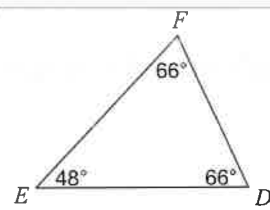
12) In ΔSTU
$m\angle S = 50°$
$m\angle T = 70°$
$m\angle U = 60°$

Name the longest and shortest side in each triangle.

13)



14)



15) In ΔDEF
$m\angle D = 35°$
$m\angle F = 95°$

16) In ΔKLM
$m\angle K = 50°$
$m\angle L = 100°$
$m\angle M = 30°$

Critical thinking questions:

17) In triangle ABC:
AB is the longest side.
70° is the measure of angle B.

Find the range of possible measures for angle A.

18) In triangle XYZ:
XY is the shortest side.
30° is the measure of angle Y.

Find the range of possible measures for angle X.

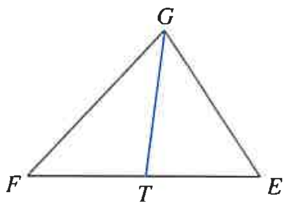
Name___________________________________

Medians

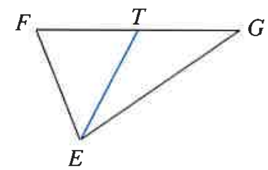
Date__________________ Period____

Each figure shows a triangle with one or more of its medians.

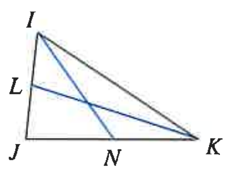
1) Find FE if $TE = 8$



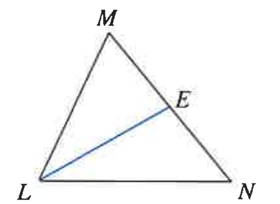
2) Find GF if $TF = 6.3$



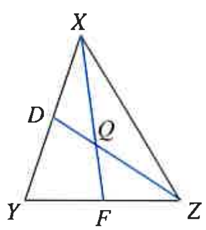
3) Find LJ if $IJ = 6$



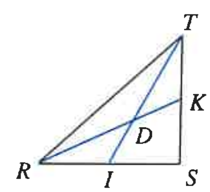
4) Find NM if $EM = 10$



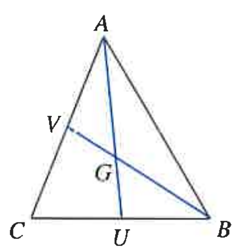
5) Find ZQ if $ZD = 6$



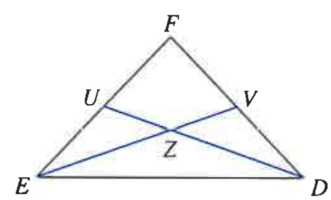
6) Find RK if $DK = 3.4$



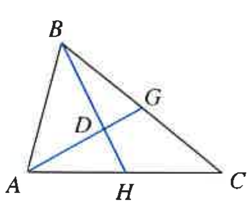
7) Find BG if $BV = 3.9$



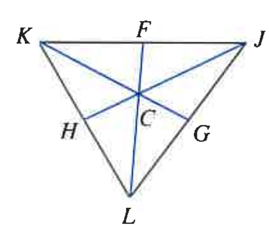
8) Find EZ if $ZV = 12$



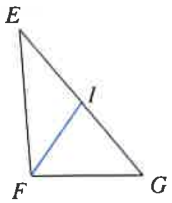
9) Find DH if $BH = 4.5$



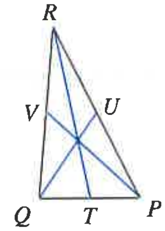
10) Find CG if $KG = 41.4$



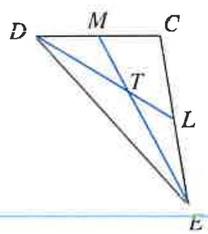
11) Find x if $GE = 3x + 5$ and $IE = x + 6$



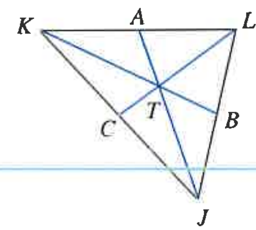
12) Find x if $TP = 2x + 1$ and $TQ = 3x - 5$



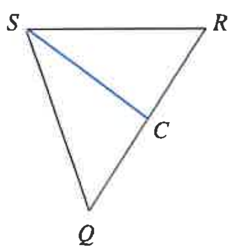
13) Find x if $ET = 3x + 2$ and $EM = 5x$



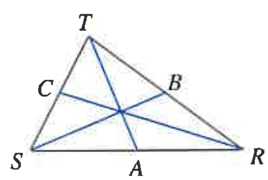
14) Find x if $KT = \dfrac{6x+6}{5}$ and $KB = \dfrac{11}{5}x - \dfrac{6}{5}$



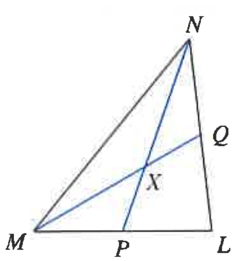
15) Find CQ if $CR = x$ and $CQ = 2x - 6$



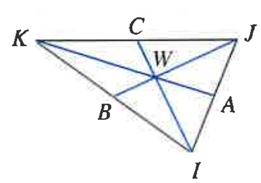
16) Find AS if $AR = x - \dfrac{1}{2}$ and $AS = \dfrac{x+5}{2}$



17) Find XQ if $MQ = 3x - 3$ and $XQ = 2x - 6$



18) Find JW if $JW = 6x + 2$ and $JB = 10x + 1$



Kuta Software - Infinite Geometry

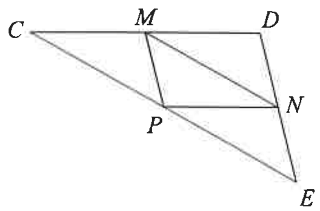
Name________________________

Midsegment of a Triangle

Date________________ Period____

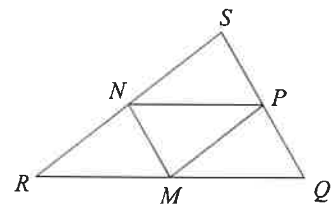
In each triangle, M, N, and P are the midpoints of the sides. Name a segment parallel to the one given.

1)



$\overline{CD} \parallel$ ____

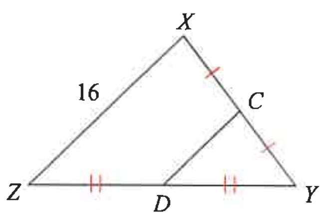
2)



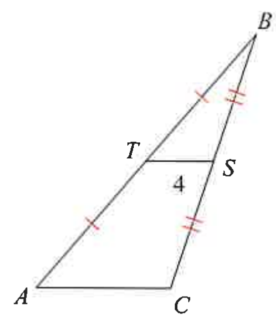
____ $\parallel \overline{QS}$

Find the missing length indicated.

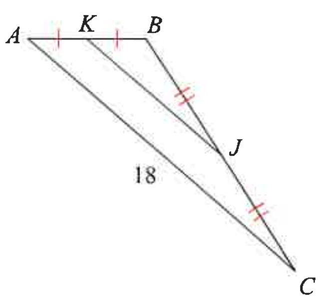
3) Find *CD*



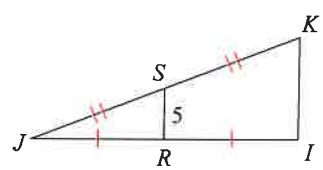
4) Find *AC*



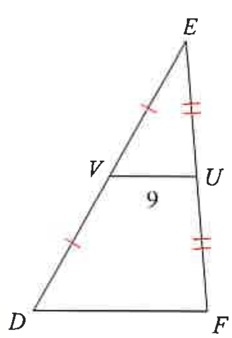
5) Find *KJ*



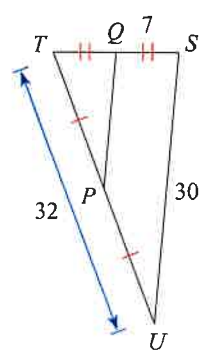
6) Find *IK*



7) Find *DF*

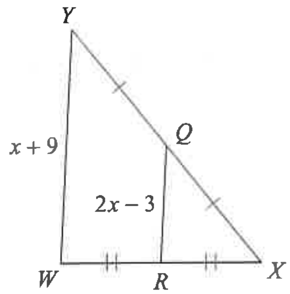


8) Find *PQ*

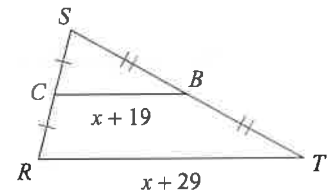


Solve for *x*.

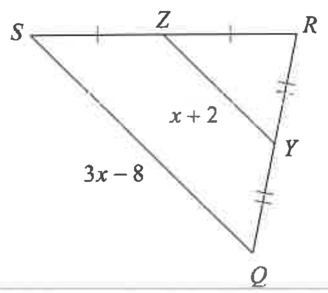
9)



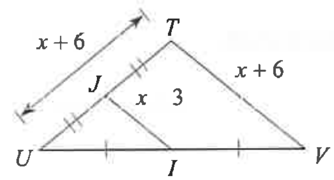
10)



11)

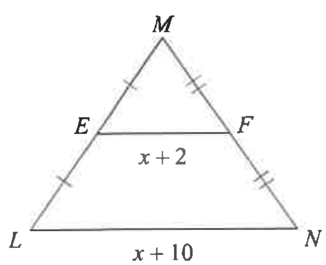


12)

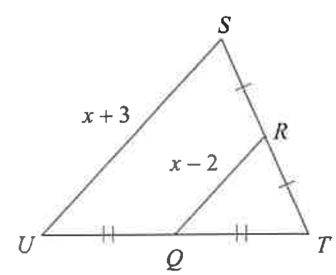


Find the missing length indicated.

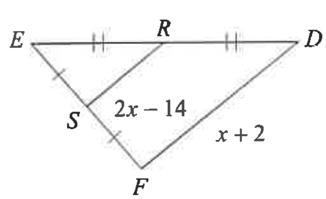
13) Find LN



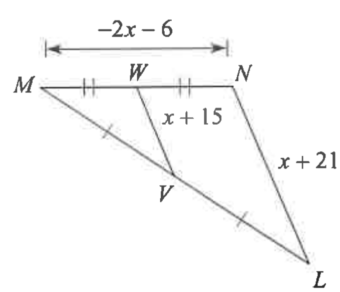
14) Find RQ



15) Find SR



16) Find VW



Kuta Software - Infinite Geometry

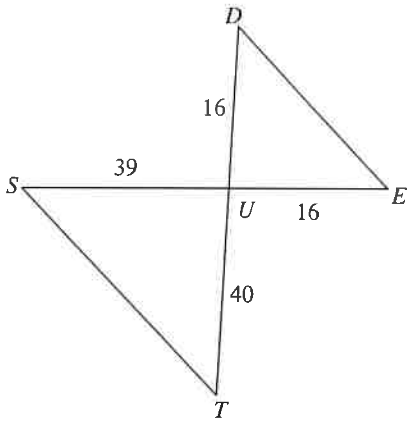
Name_______________________________

Similar Triangles

Date________________ Period____

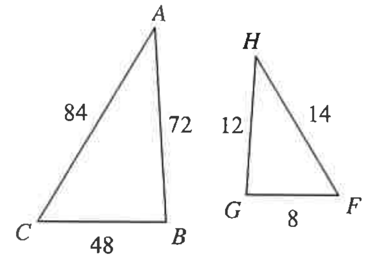
State if the triangles in each pair are similar. If so, state how you know they are similar and complete the similarity statement.

1)



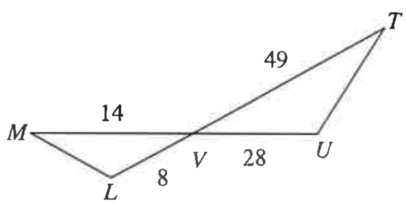
$\Delta UTS \sim$ ______

2)



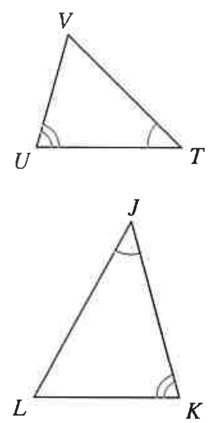
$\Delta CBA \sim$ ______

3)



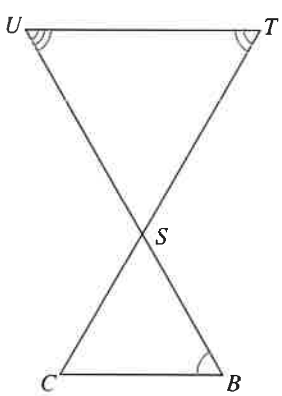
$\Delta VUT \sim$ ______

4)



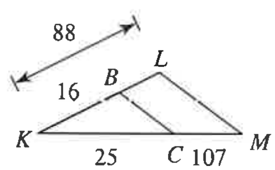
$\Delta JKL \sim$ ______

5)



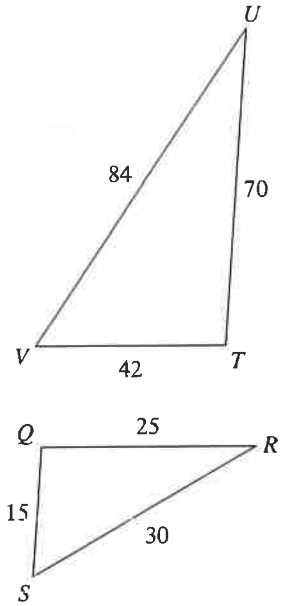
$\Delta STU \sim$ ______

6)



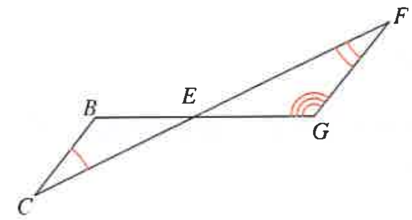
$\Delta KLM \sim$ ______

7)



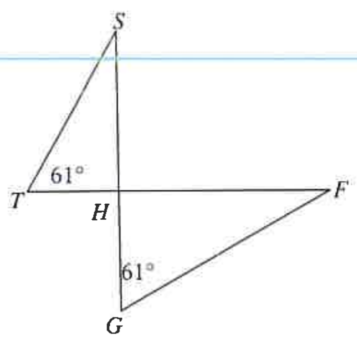
$\Delta TUV \sim$ ______

8)



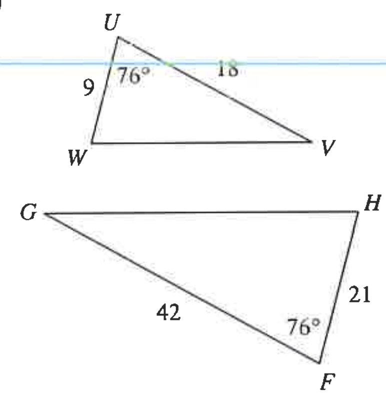
$\Delta EFG \sim$ ______

9)



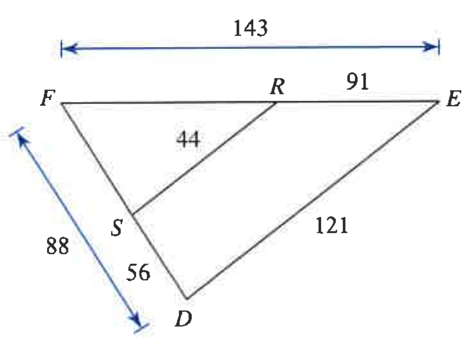
$\Delta HGF \sim$ ______

10)



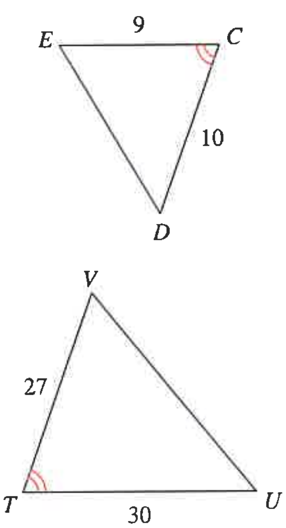
$\Delta FGH \sim$ ______

11)



$\Delta FED \sim$ ______

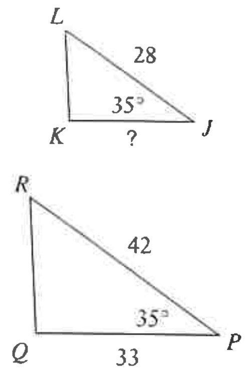
12)



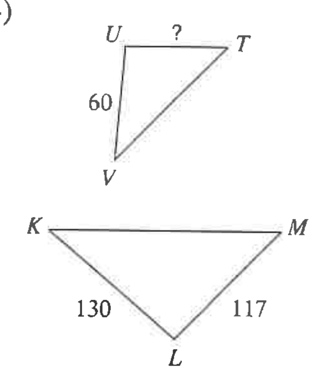
$\Delta TUV \sim$ ______

Find the missing length. The triangles in each pair are similar.

13)



14)



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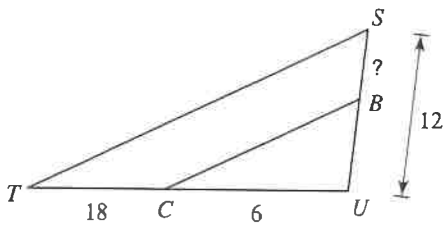
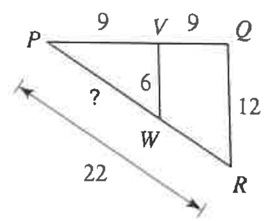
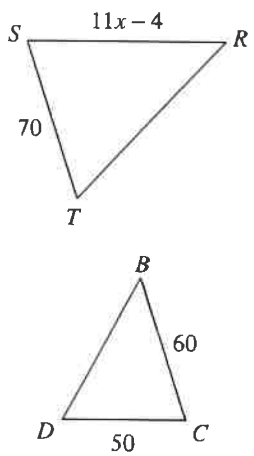


16)

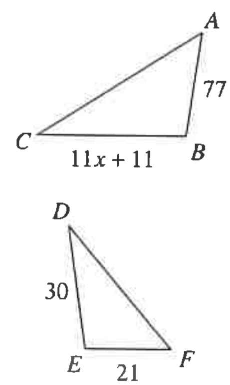


Solve for x. The triangles in each pair are similar.

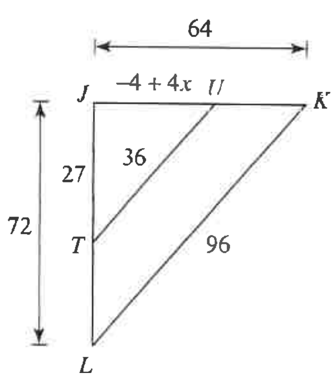
17)



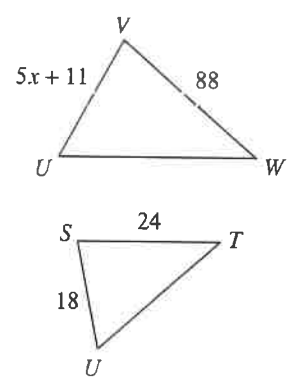
18)



19)



20)



Kuta Software - Infinite Geometry

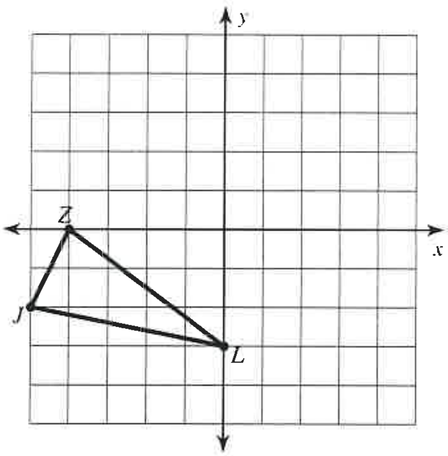
Name________________________________

All Transformations

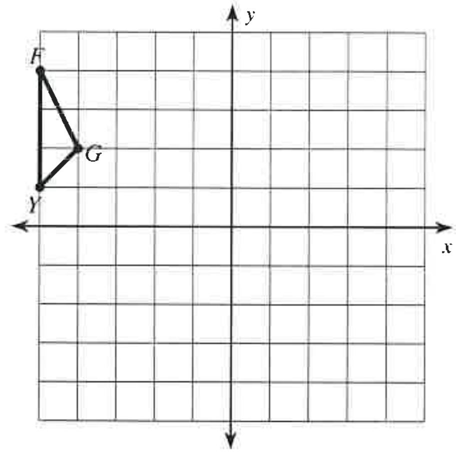
Date________________ Period____

Graph the image of the figure using the transformation given.

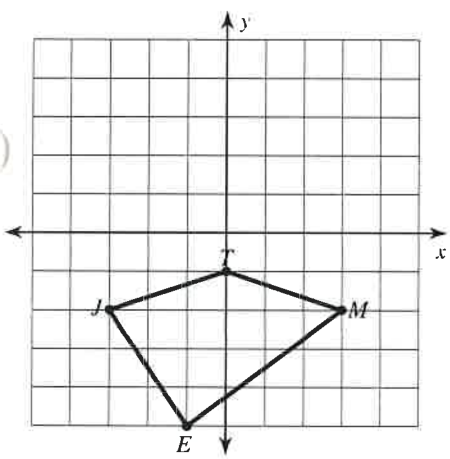
1) rotation 90° counterclockwise about the origin



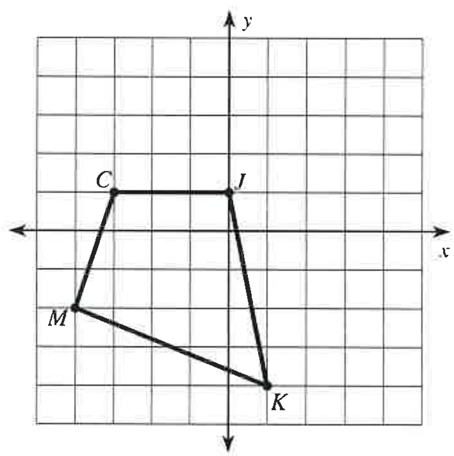
2) translation: 4 units right and 1 unit down



3) translation: 1 unit right and 1 unit up

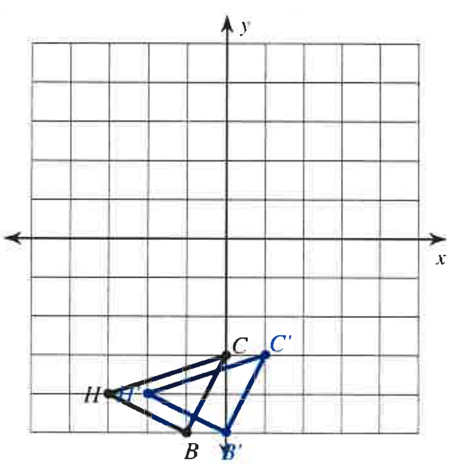


4) reflection across the x-axis

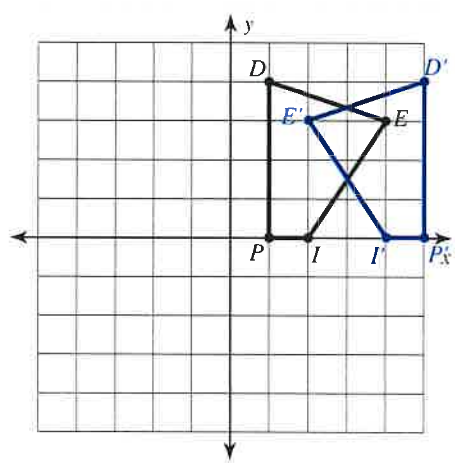


Write a rule to describe each transformation.

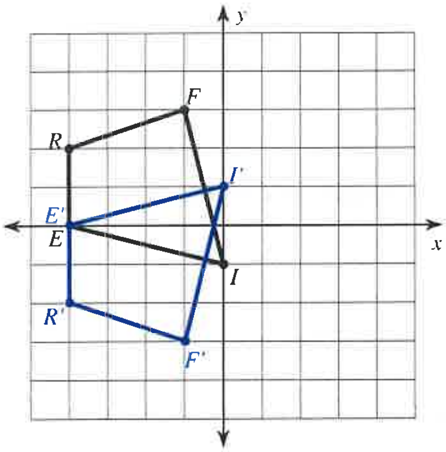
5)



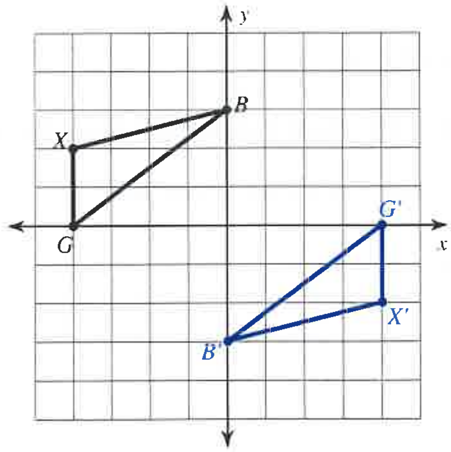
6)



7)

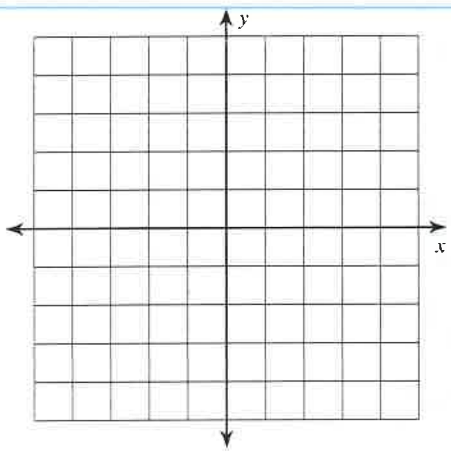


8)

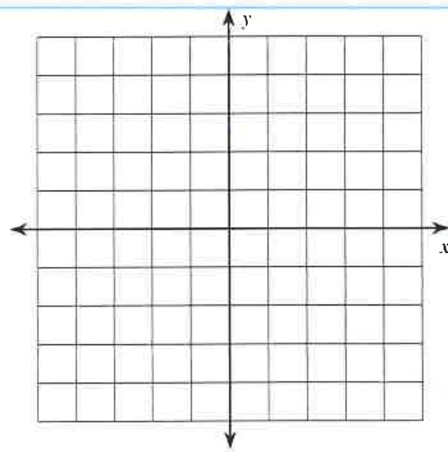


Graph the image of the figure using the transformation given.

9) rotation 90° clockwise about the origin
$B(-2, 0)$, $C(-4, 3)$, $Z(-3, 4)$, $X(-1, 4)$



10) reflection across $y = x$
$K(-5, -2)$, $A(-4, 1)$, $I(0, -1)$, $J(-2, -4)$



Find the coordinates of the vertices of each figure after the given transformation.

11) rotation 180° about the origin
$E(2, -2)$, $J(1, 2)$, $R(3, 3)$, $S(5, 2)$

12) reflection across $y = 2$
$J(1, 3)$, $U(0, 5)$, $R(1, 5)$, $C(3, 2)$

13) translation: 7 units right and 1 unit down
$J(-3, 1)$, $F(-2, 3)$, $N(-2, 0)$

14) translation: 6 units right and 3 units down
$S(-3, 3)$, $C(-1, 4)$, $W(-2, -1)$

Kuta Software - Infinite Geometry

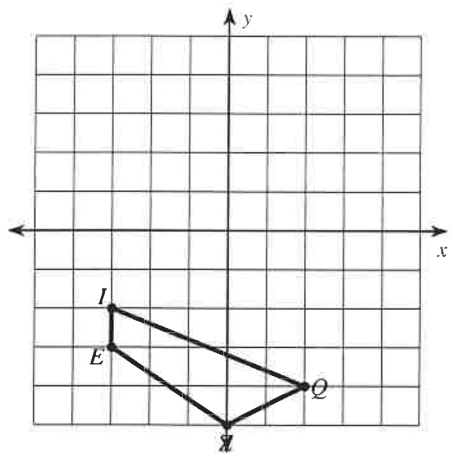
Name________________________

Reflections

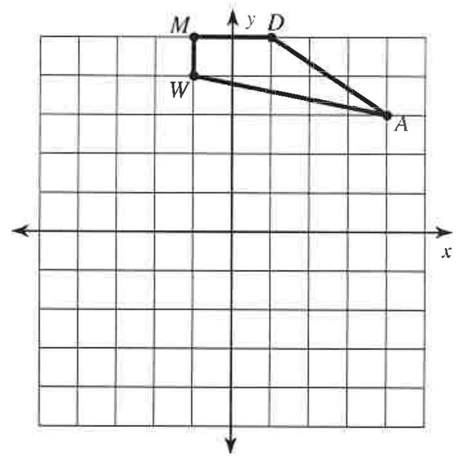
Date________________ Period____

Graph the image of the figure using the transformation given.

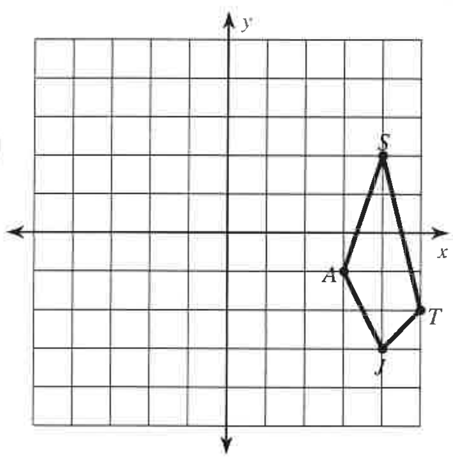
1) reflection across $y = -2$



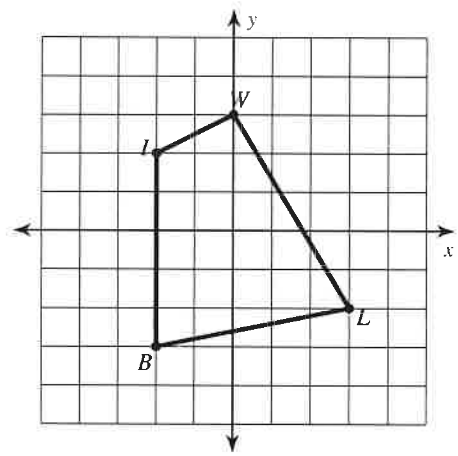
2) reflection across the x-axis



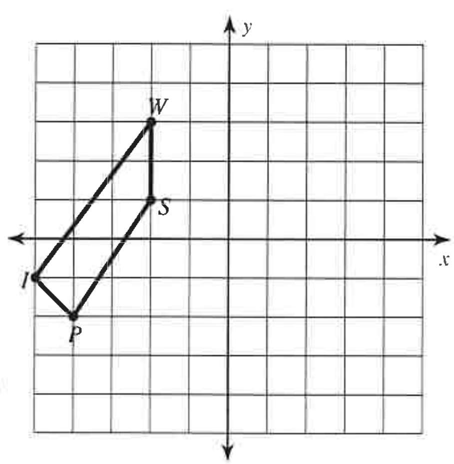
3) reflection across $y = -x$



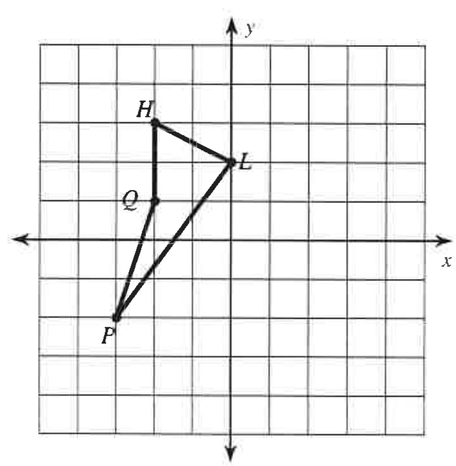
4) reflection across $y = -1$



5) reflection across $x = -3$

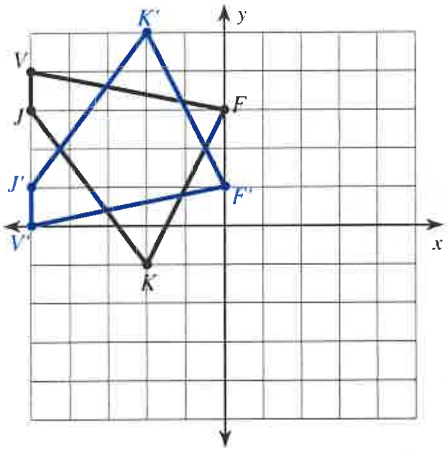


6) reflection across $y = x$

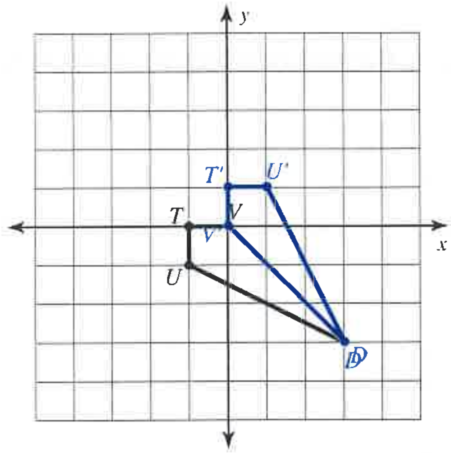


Write a rule to describe each transformation.

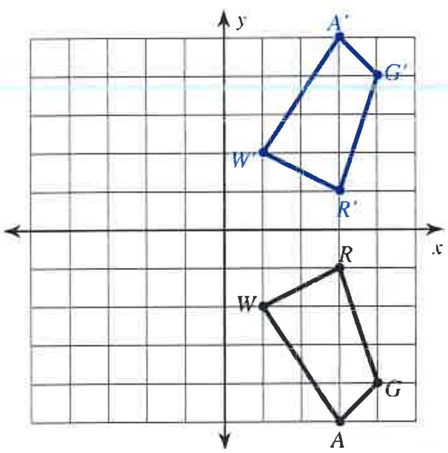
7)



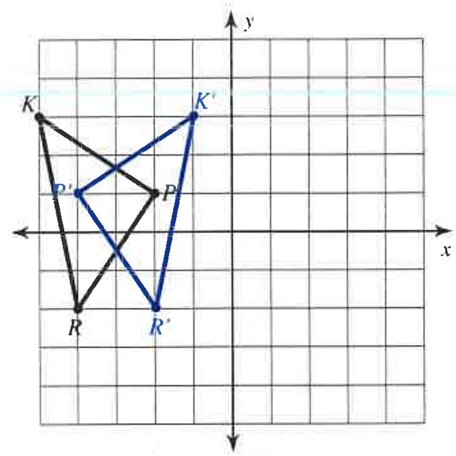
8)



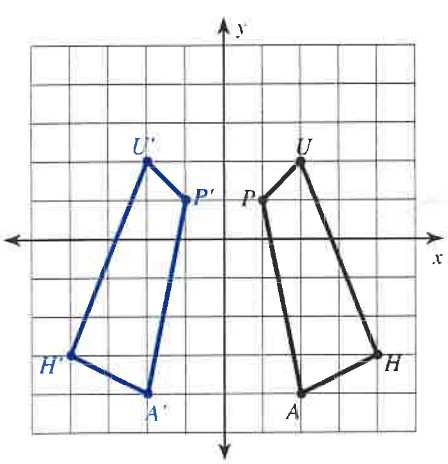
9)



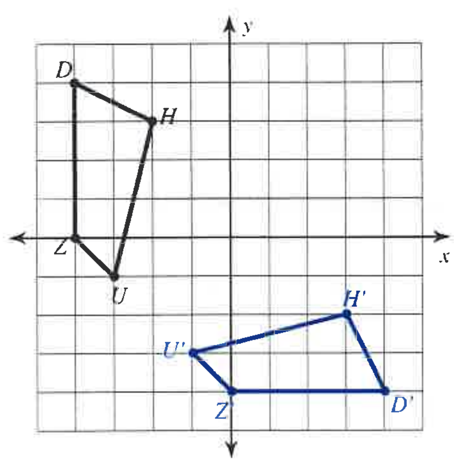
10)



11)



12)



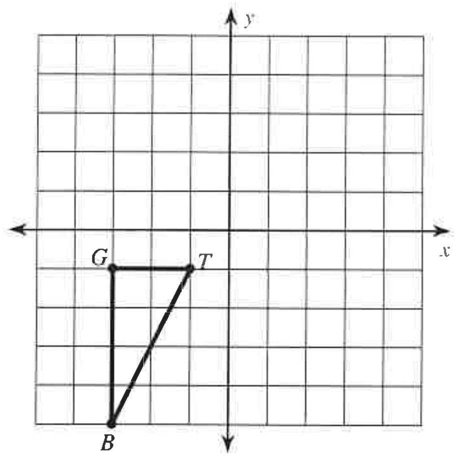
Name________________________________

Translations

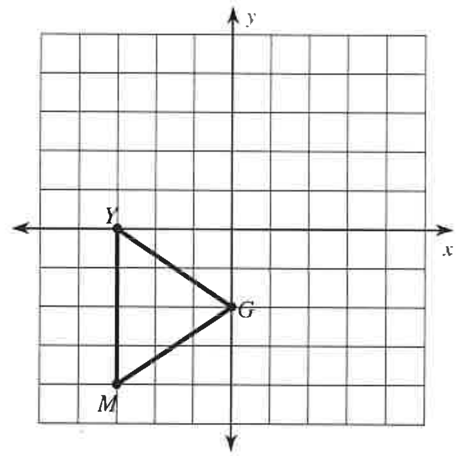
Date________________ Period____

Graph the image of the figure using the transformation given.

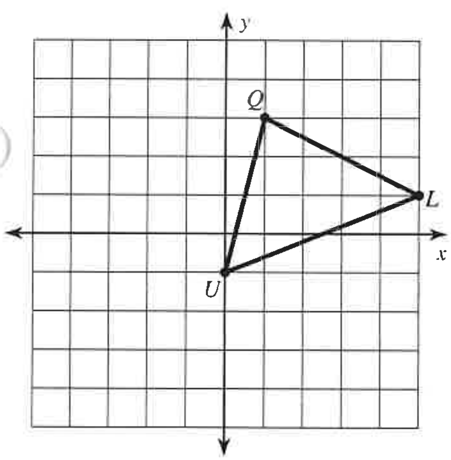
1) translation: 5 units right and 1 unit up



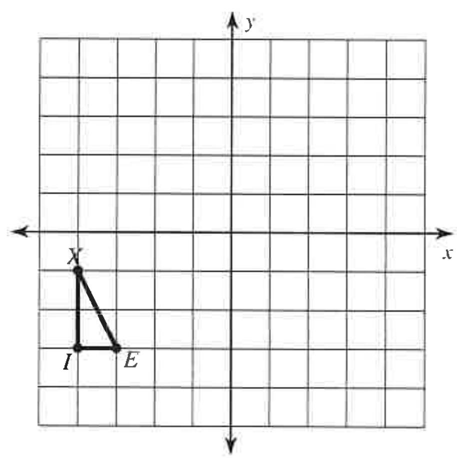
2) translation: 1 unit left and 2 units up



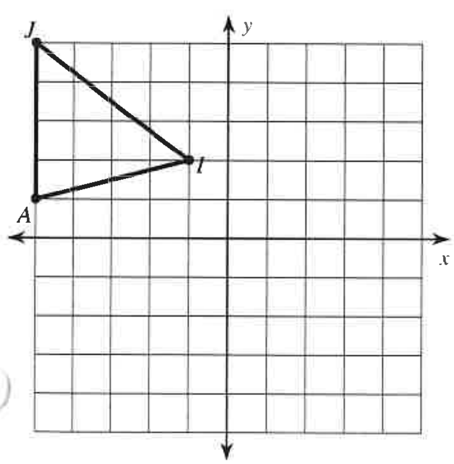
3) translation: 3 units down



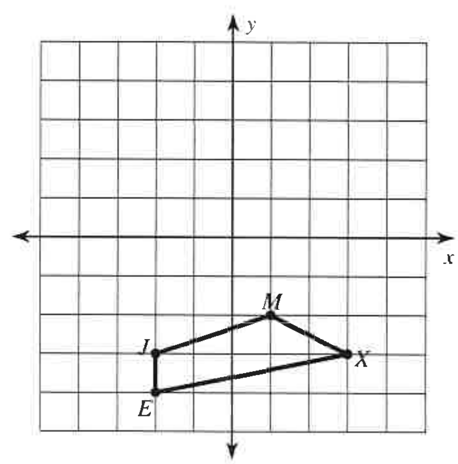
4) translation: 5 units right and 2 units up



5) translation: 4 units right and 4 units down

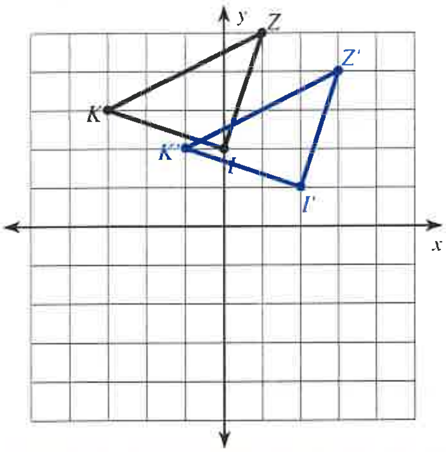


6) translation: 2 units right and 3 units up



Write a rule to describe each transformation.

7)



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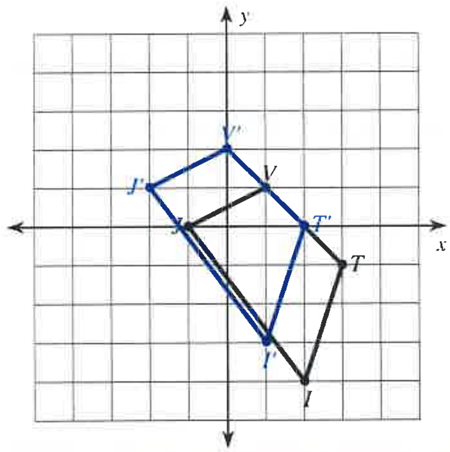
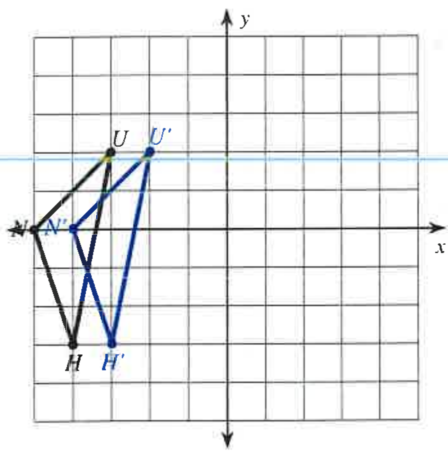


9)



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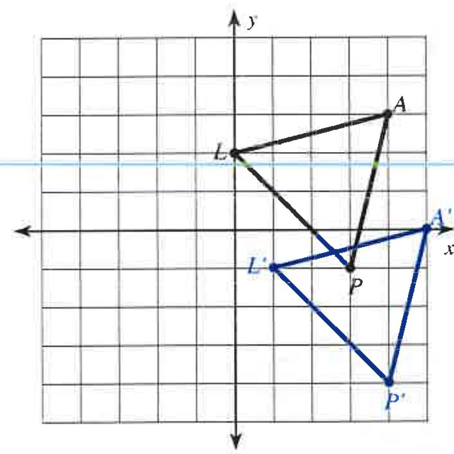

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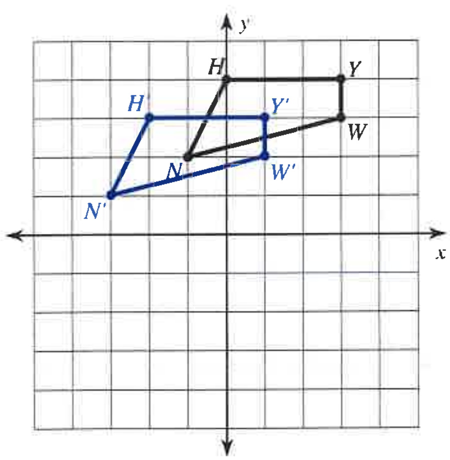

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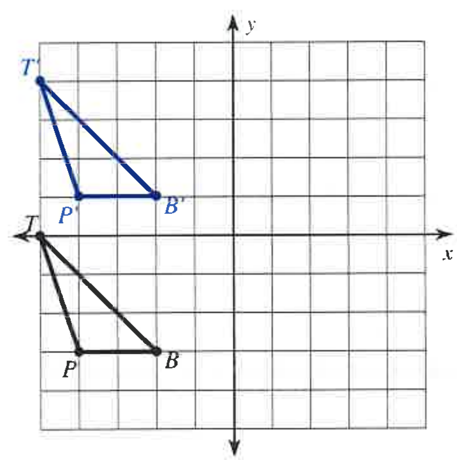


Geometry **Proofs Practice**

Name ______________________

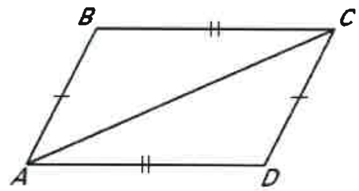
Period ____ Date ___________

4.3-4.5 – Triangle Congruence Proofs (Part 2)

1. Complete the proof by providing the correct statement or reason.

GIVEN: $\overline{AB} \cong \overline{CD}$, $\overline{BC} \cong \overline{AD}$

PROVE: $\Delta ABC \cong \Delta CDA$

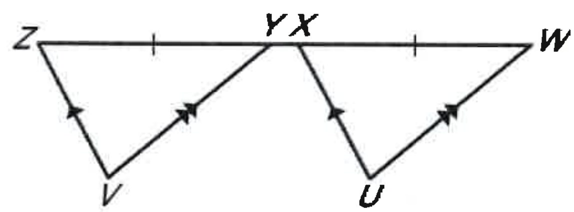


Statements	**Reason**
1.	**1.** Given
2.	**2.**
3. $\Delta ABC \cong \Delta CDA$	**3.**

Complete the proof.

2. GIVEN: $\overline{WU} \parallel \overline{YV}, \overline{XU} \parallel \overline{ZV}, \overline{WX} \cong \overline{YZ}$

PROVE: $\Delta WXU \cong \Delta YZV$

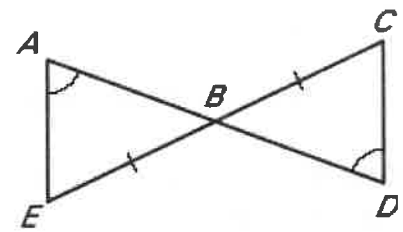


Statements	**Reasons**
1. $\overline{WU} \parallel \overline{YV}$	1.
2. $\angle UWX \cong \angle VYZ$	2.
3. $\overline{XU} \parallel \overline{ZV}$	3.
4. $\angle UXW \cong \angle VZY$	4.
5. $\overline{WX} \cong \overline{YZ}$	5.
6. $\Delta WXU \cong \Delta YZV$	6.

3. Complete the proof by providing the correct reason for each statement.

GIVEN: $\overline{BE} \cong \overline{BC}$ and $\angle A \cong \angle D$

PROVE: $\Delta ABE \cong \Delta DBC$

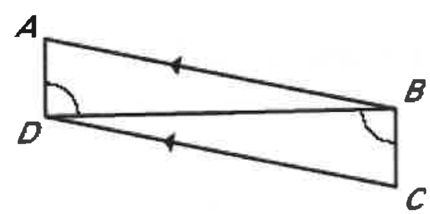


Statements	Reason
1.	1.
2.	2.
3.	3.

4. Complete the proof by providing the missing information.

GIVEN: $\overline{AB} \parallel \overline{DC}$, $\angle ADB \cong \angle CBD$

PROVE: $\Delta ABD \cong \Delta CDB$

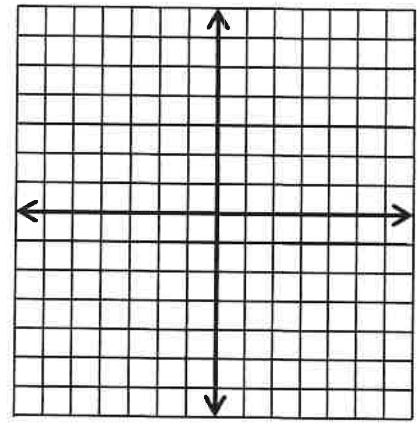


Statements	Reasons

5. Given: The coordinates of two triangles are ΔPVH : P(-6, 2), V(-1, 4), H(-2, -1)
and ΔNQR : N(2, -2), Q(4, -7), R(-1, -6).

(You will need to calculate the length of the sides by using the distance formula/Pythagorean Theorem.)

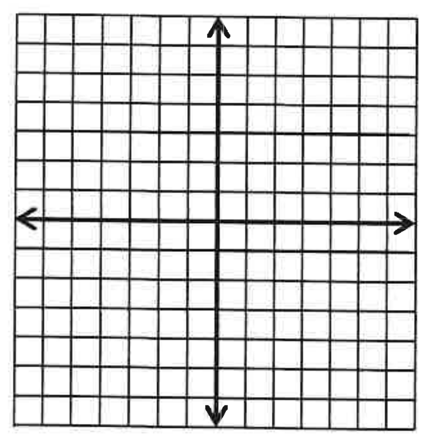
Prove: $\Delta PVH \cong \Delta NQR$ **by using SSS**



6. Given: The coordinates of two triangles are ΔPVH : P(-1, 6), V(-4, 4), H(0, -2)
and ΔNQR : N(5, -3), Q(3, -6), R(-3, -2).

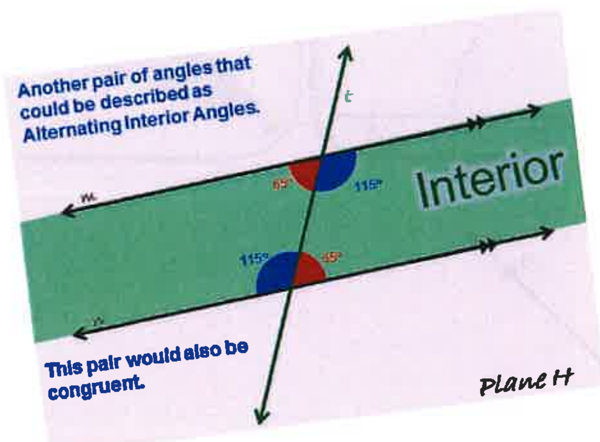
(You will need to calculate length like in #5. You might need to calculate slope as well.)

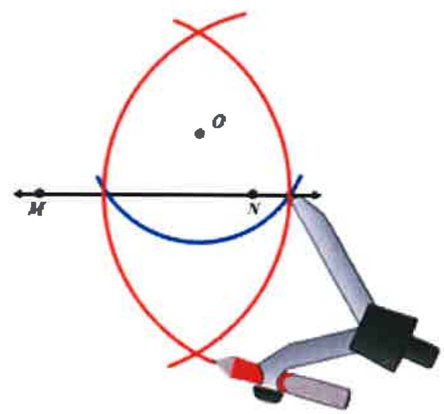
Prove: $\Delta PVH \cong \Delta NQR$ **by using HL**

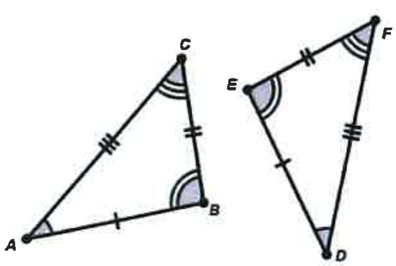


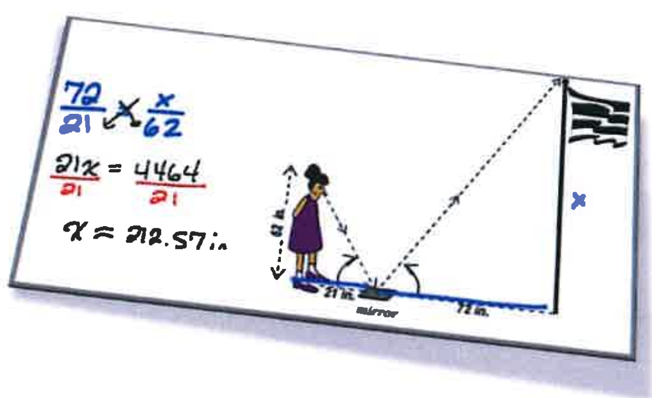
Geometry

Unit 2: Constructions & Similarity









MAtTHew M. Winking

Sec 2.1 Geometry - Constructions

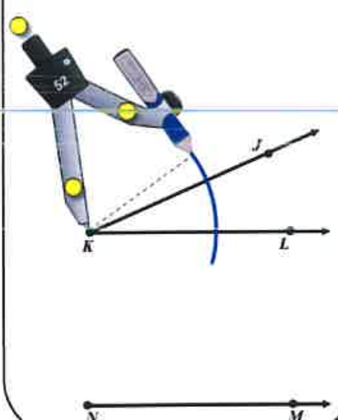
Name:

1. **[COPY SEGMENT] Construct a segment with an endpoint of C and congruent to the segment AB.**

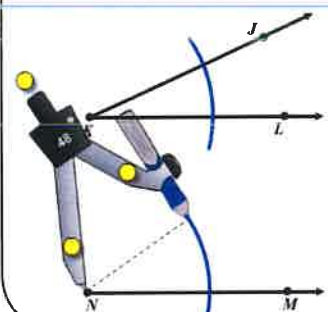
***Using a ruler measure the two lengths to make sure they have the same measure.**

2. **[COPY ANGLE] Construct an angle with ray $\overrightarrow{HI}$ and congruent to the angle $\measuredangle DEF$**

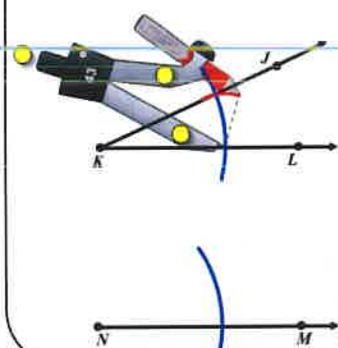
Step I: Open the compass to any length and create an arc on the original angle with the needle at the vertex



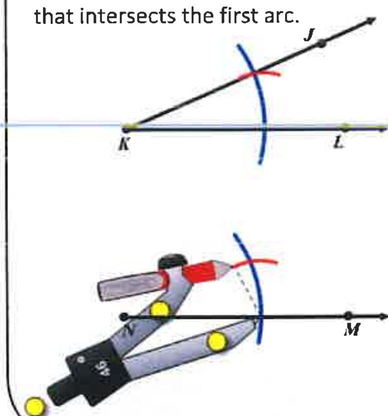
Step 2: Leave the compass set to the exact same length and create a new arc on the new ray with the needle at the end point of the ray.



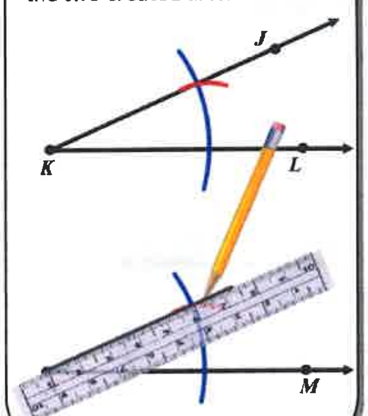
Step 3: Open the compass to the length of the intercepted arc in the original angle and draw a small arc to be sure the created arcs intersect on the ray of the angle.

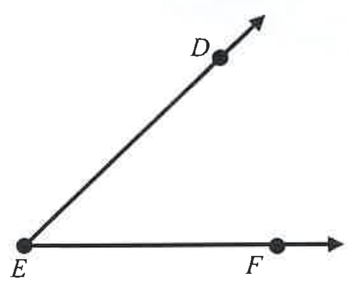


Step 4: Leave the compass open to the exact same length as it was from step 3. Then with the needle at the intersection of the second arc and the copied ray, create an arc that intersects the first arc.



Step 5: Finally, use a straight edge to draw a ray from the endpoint of the original bottom ray through the intersection of the two created arcs.

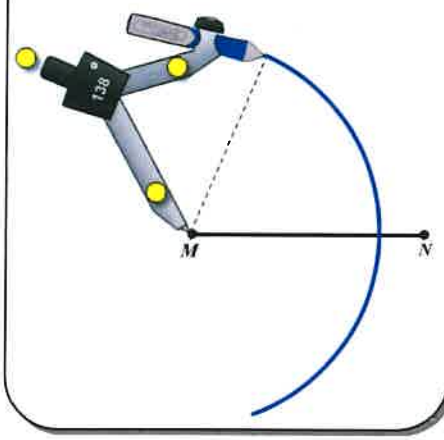




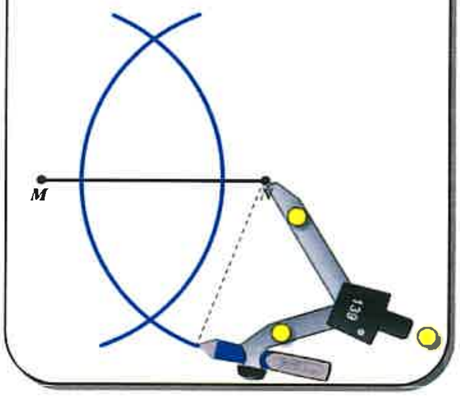
***Using a protractor measure the two angles to make sure they have the same measure.**

3. [Perpendicular Bisector] Construct a perpendicular bisector to the segment $\overline{AB}$.

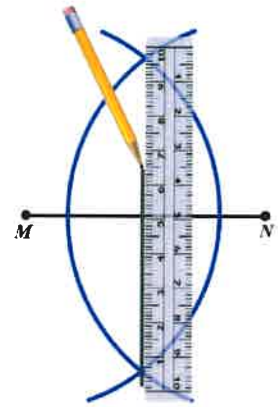
Step I: Open the compass to a length that is **certainly larger than half** of the segment length. Draw an arc with the needle at the left endpoint of the segment.



Step 2: Leave the compass open to the same length from step 1 and create another arc only this time starting with the needle at the right endpoint.



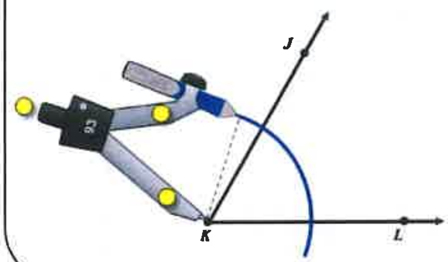
Step 3: Using a straight edge and a pencil, draw a line that passes through the intersection point of the two arcs.



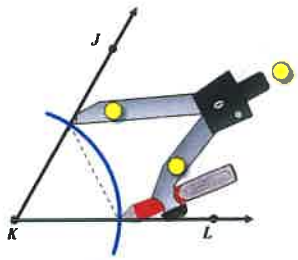
Using a ruler measure the two halves of the segment to make sure they have the same measure.

4. [Angle Bisector] Construct an angle bisector of the angle $\measuredangle DEF$

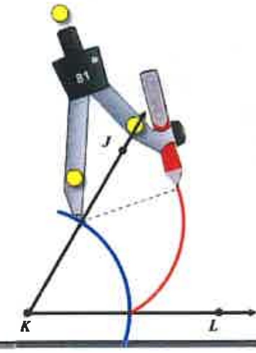
Step I: Open the compass to any length and create an arc with the needle of the compass at the vertex of the angle.



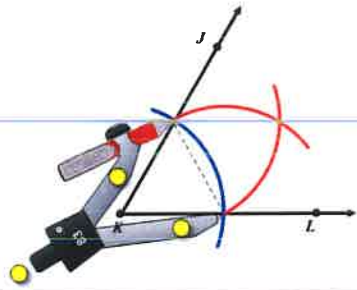
Step 2: Open the compass so that the needle is on one intersection of the arc and the angle and the pencil is at the other intersection of the arc and the angle.



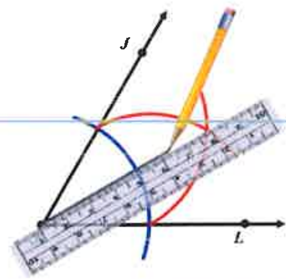
Step 3: Create an arc with the opened length determined in step 2 towards the interior of the angle.

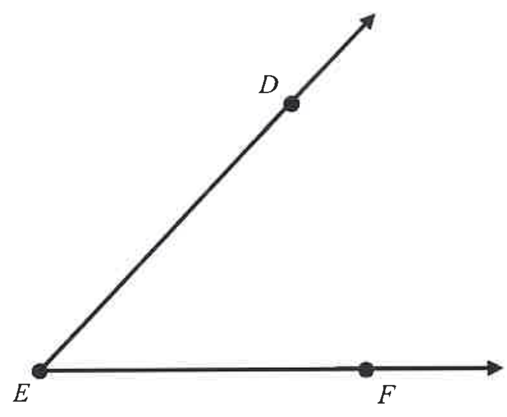


Step 4: Move the compass so that the needle and pencil are on the opposite intersection points that they were on in step 2 and create another arc to the exterior of the angle.



Step 5: Using a straight edge, draw ray from the vertex of the angle through were the two outer arcs intersect.

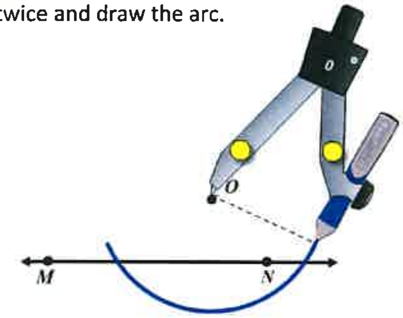




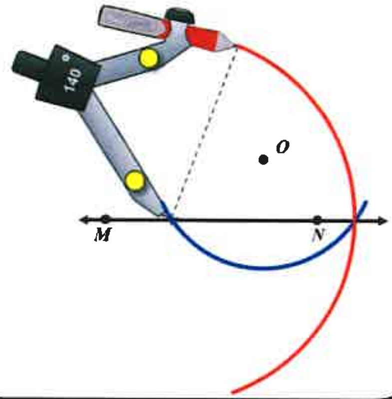
***Using a ruler measure the two halves of the segment to make sure they have the same measure.**

5. [Perpendicular to a Line Through a Point] Construct a perpendicular line to $\overleftrightarrow{AB}$ through point C.

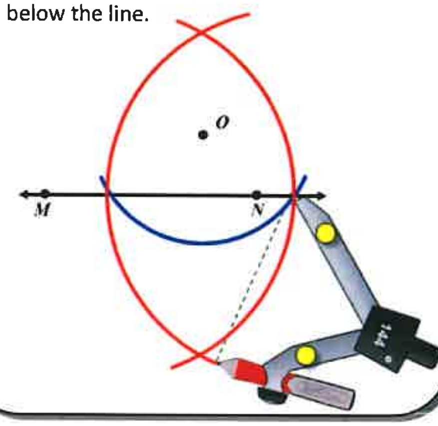
Step I: Start by placing the needle on the point not on the line. Then, open your compass to a length such that when you draw an arc it will pass through the line twice and draw the arc.



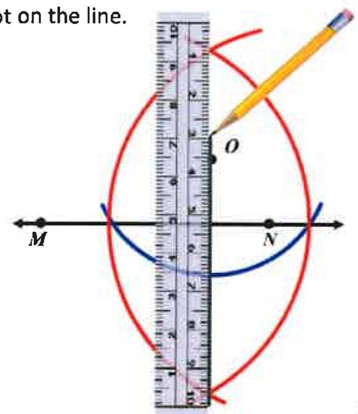
Step 2: Open the compass so that the needle is on one intersection of the arc and the pencil is on the other intersection of the arc. Then, create a arc above and below the line.

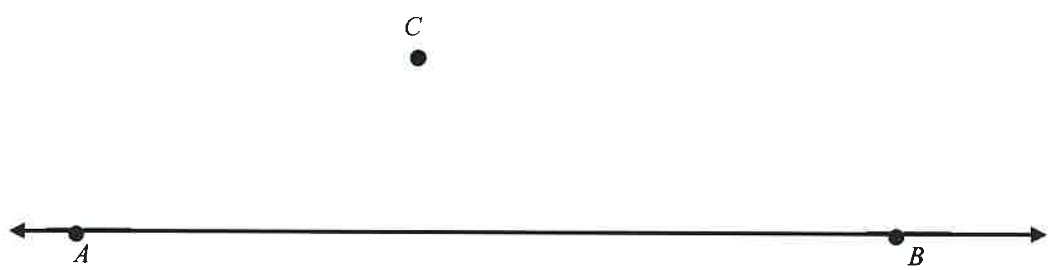


Step 3: Opening the compass to the same length as the previous step and switching the needle and pencil to the opposite intersections of arcs that they were on in step 2, create another arc above and below the line.



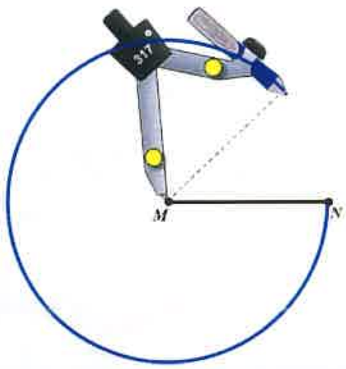
Step 3: Using a straight edge, draw a line passing through the intersection of the two most recent arc intersections. The line should incidentally also pass through the point not on the line.



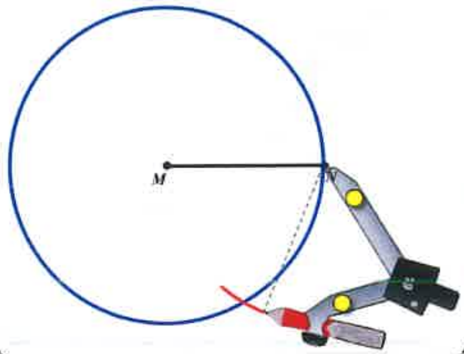


6. [Hexagon inscribed in a Circle] Construct a circle with radius $\overline{AB}$ and an inscribed regular hexagon.

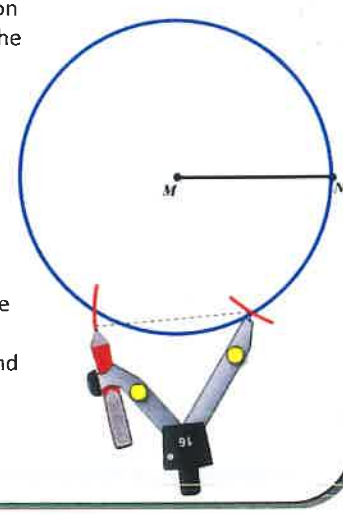
Step I: Start by placing the needle on the point that is to be the center of the circle and the pencil on the other endpoint of the radius. Create the entire circle with the compass.



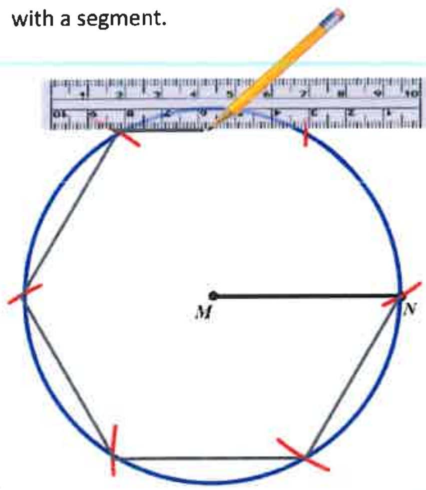
Step 2: With the compass still open to the exact length of the radius place the needle on the endpoint of the radius that is on the circle and with the pencil mark an intersection on the circle with a new arc.



Step 3: Leaving the compass still open to the same length move the needle to the new intersection point and mark another intersection point on the circle again. Continue repeating this process until you have made it all the way around the circle.

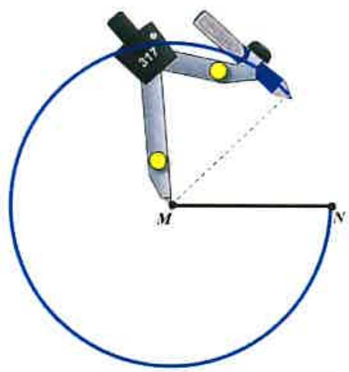


Step 4: The last mark should end right were you started with the needle. Then, connect each consecutive arc intersection with a segment.

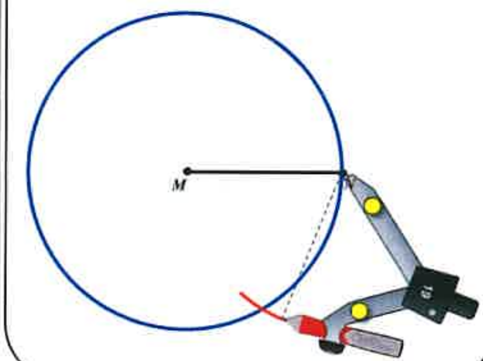


7. [Triangle inscribed in a Circle] Construct a circle with radius $\overline{AB}$ and an inscribed regular triangle.

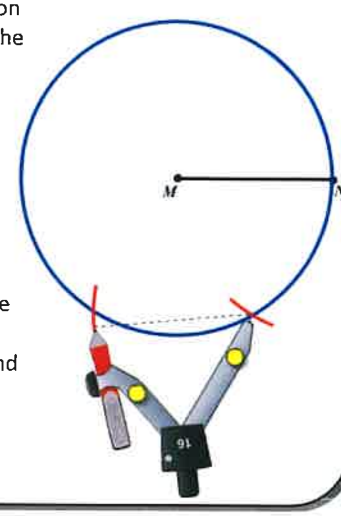
Step I: Start by placing the needle on the point that is to be the center of the circle and the pencil on the other endpoint of the radius. Create the entire circle with the compass.



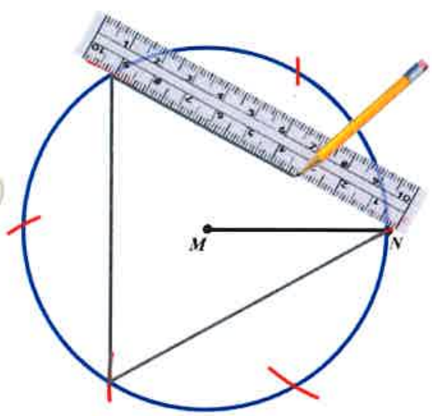
Step 2: With the compass still open to the exact length of the radius place the needle on the endpoint of the radius that is on the circle and with the pencil mark an intersection on the circle with a new arc.



Step 3: Leaving the compass still open to the same length move the needle to the new intersection point and mark another intersection point on the circle again. Continue repeating this process until you have made it all the way around the circle.

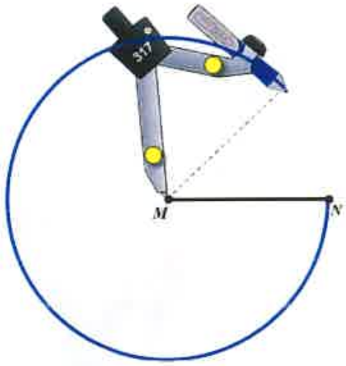


Step 4: The last mark should end right were you started with the needle. Then, connect every other consecutive arc intersection with a segment.

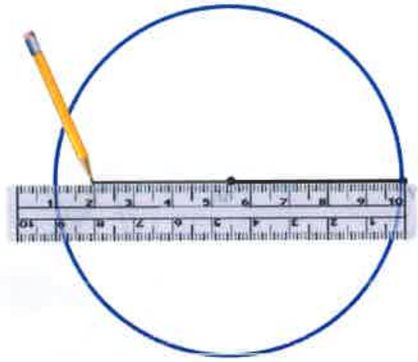


8. [Square inscribed in a Circle] Construct a circle with radius $\overline{AB}$ and an inscribed square.

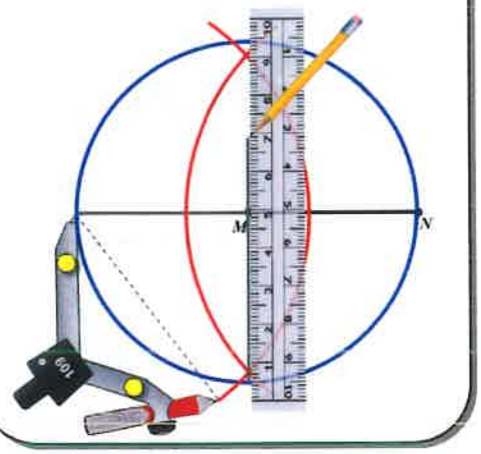
Step I: Start by placing the needle on the point that is to be the center of the circle and the pencil on the other endpoint of the radius. Create the entire circle with the compass.



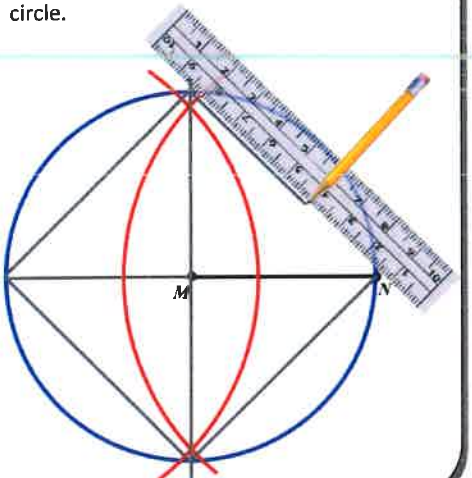
Step 2: Line your straight edge up with the radius and extend the radius segment to create a diameter.



Step 3: Create a perpendicular bisector of the newly created diameter (see previous construction #3 if needed)

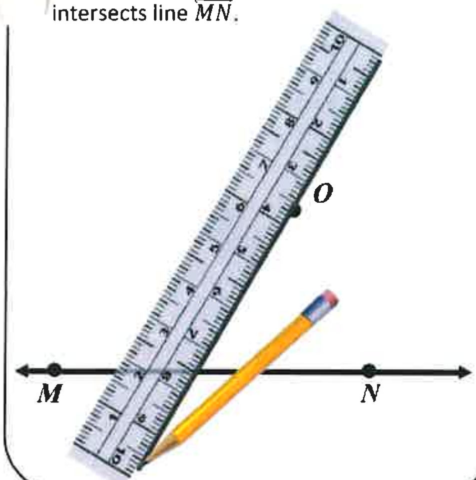


Step 4: Connect the each endpoint of the diameter with each endpoint of where the perpendicular bisector intersects the circle.

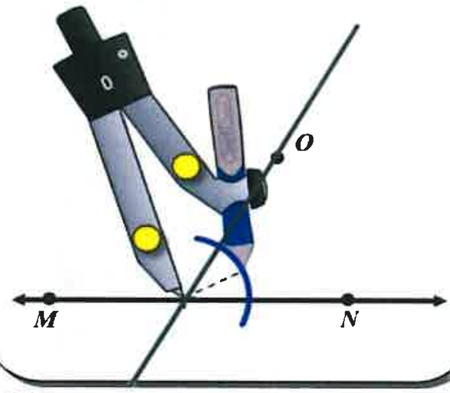


9. [Construct a Parallel Line given a point and a line] Construct a parallel line to $\overleftrightarrow{AB}$ through point C.

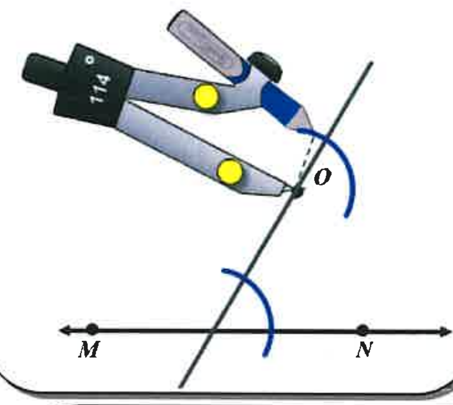
Step I: Start drawing a general line that passes through point O and intersects line $\overleftrightarrow{MN}$.



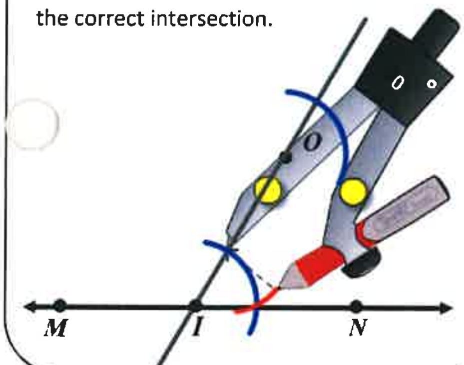
Step 2: Open the compass so that the needle is on the intersection of the new line just created and line $\overleftrightarrow{MN}$. Then, open the compass a little (it needs to be less than the distance to reach point O). Create an arc as shown below.



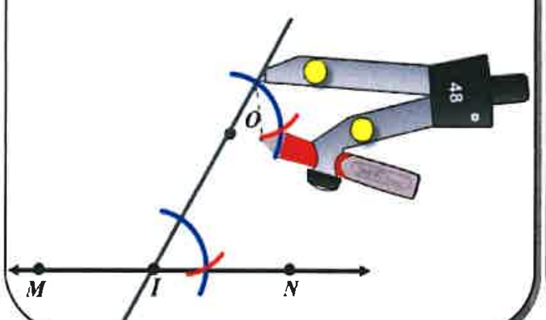
Step 3: Leave the compass set to the same opening and recreate another arc of the same radius but with the needle at point O.



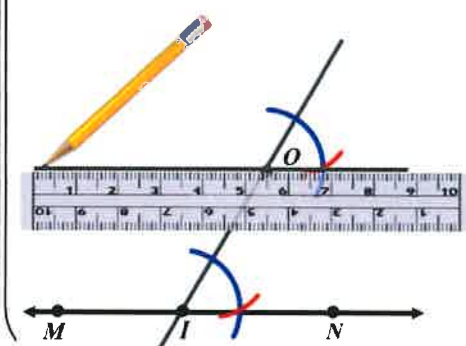
Step 4: Put the compass needle on the intersection of the transversal line and first arc that you created as shown below and open the pen to the intersection of the first arc and line $\overleftrightarrow{MN}$. Create a small arc to verify the intersection of the arcs mark the correct intersection.

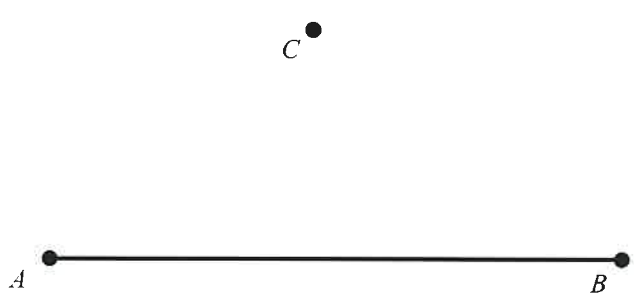


Step 5: Leave the compass open to the same length as the previous step and put the compass needle on the intersection of the transversal line and the second arc that you created. Then, create an arc of the same radius to intersect the second arc created as shown below.



Step 6: D

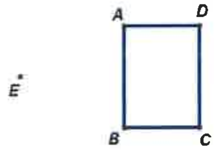




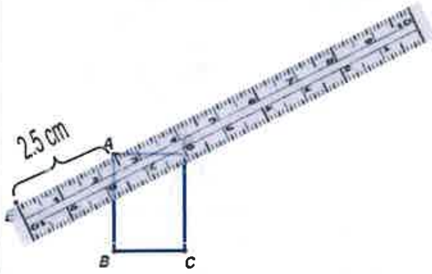
Sec 2.2 Geometry – Dilations

Name:

[Example Dilation]: Dilate the ☐ABCD by a factor of 2.0 from point E.



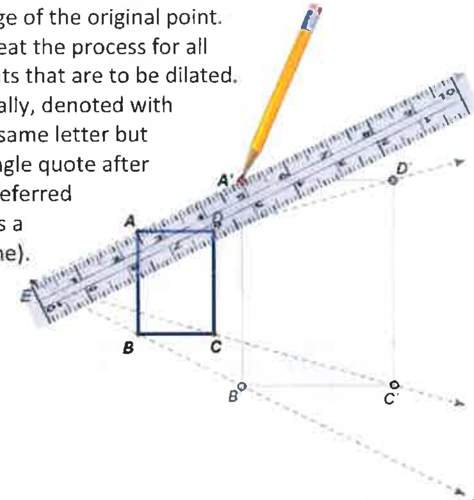
Step I: Measure the distance from the point of dilation to a point to be dilated (preferably using centimeters).



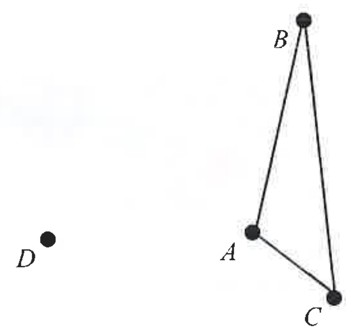
Step 2: Multiply the measured distance by the scale factor.

$2.5cm \times 2.0 = 5\ cm$

Step I: With the ruler in the same place as it was in step #1, mark a point at the measured distance determined in step #2 as the image of the original point. Repeat the process for all points that are to be dilated. Usually, denoted with the same letter but a single quote after it (referred to as a prime).



1. **Dilate the $\triangle$ ABC by a factor of $\frac{3}{2}$ from point D.**



a. **Measure the length of $\overline{AB}$ in centimeters to the nearest tenth.**

AB = ______________

b. **Measure the length of $\overline{A'B'}$ in centimeters to the nearest tenth.**

A'B' = ____________

c. **Determine the value of A'B' divided by AB.**

$\frac{A'B'}{AB}$ = ________________

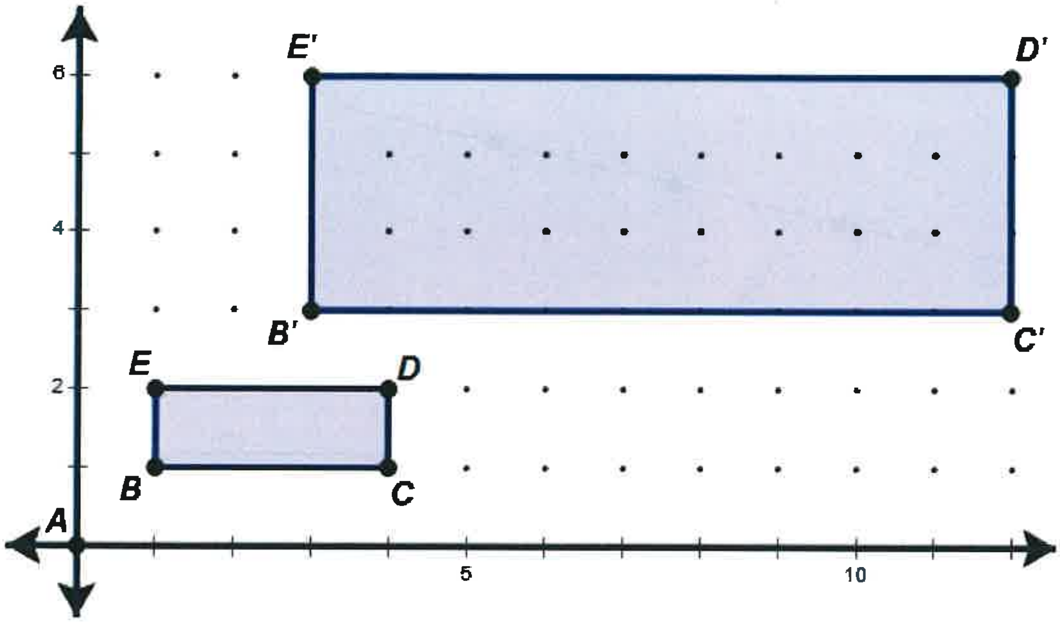
d. **What might you conclude about the scale factor and the ratio of dilated segment measure to its pre-image?**___________________________________

e. **Measure angle $\measuredangle BAC$ and the angle $\measuredangle B'A'C'$ using a protractor.**

$m\measuredangle BAC$ =_______________ $m\measuredangle B'A'C'$ =_______________

f. **What might you conclude about each pair of corresponding angles?**______________

2. Consider the following picture in which □BCDE has been dilated from point A.



a. What is the scale factor of the dilation based on the sides?

b. What is the area of □BCDE?

c. What is the area of □B'C'D'E'?

________________ ________________ ________________

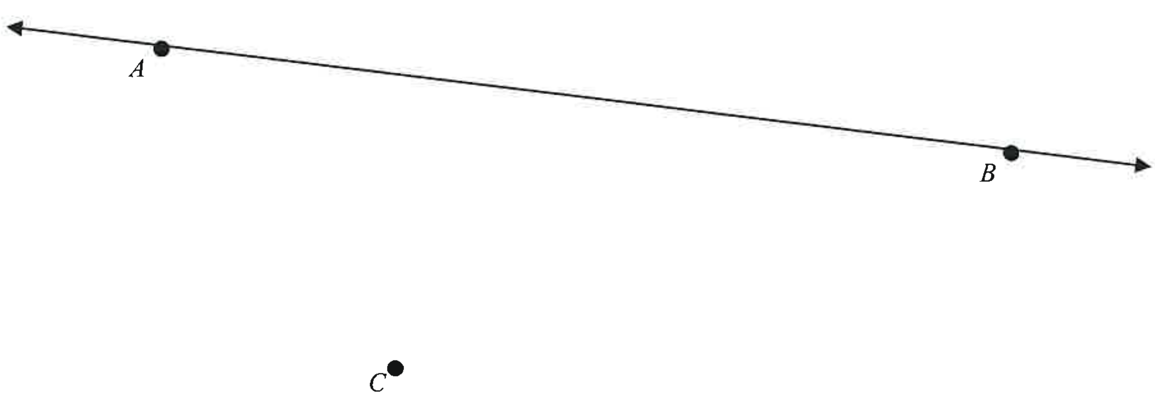
d. What is the value of the area of □B'C'D'E' divided by area of □BCDE? ____________

e. What might you conclude about the ratio of two dilated shapes sides compared to the ratio of their areas?____

____

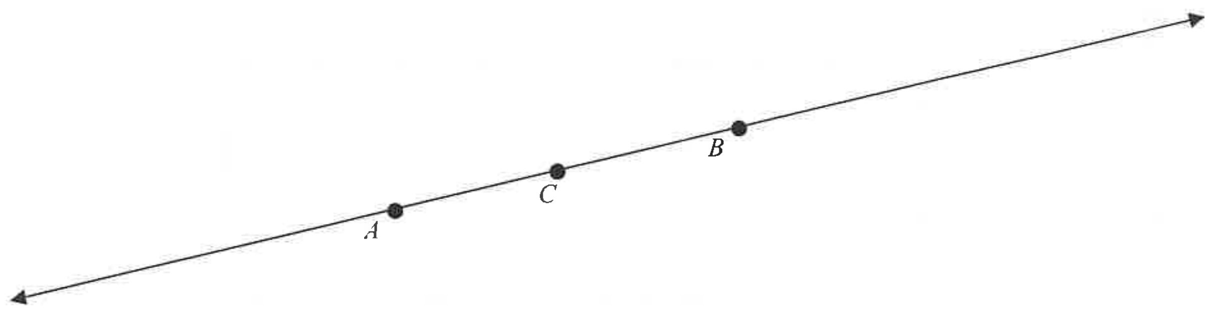
---

3. Dilate the line $\overleftrightarrow{AB}$ by a factor of 0.5 from point C.



How could you characterize the lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{A'B'}$?________________________________

4. Dilate the line $\overleftrightarrow{AB}$ by a factor of 2.2 from point C.



How could you characterize the lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{A'B'}$?______________________________

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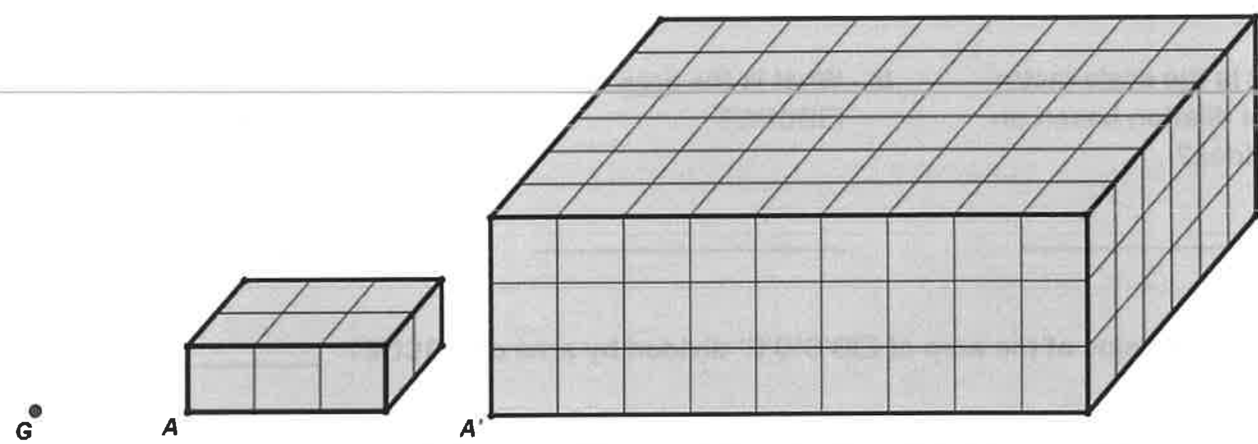
5. Consider the following picture in which rectangular prism A has been dilated from point G.



a. What is the scale factor of the dilation based on the sides?

____

b. What is the volume of rectangular prism A?

____

c. What is the volume of rectangular prism A'?

____

d. What is the value of the volume of prism A' divided by volume of prism A? __________

e. What might you conclude about the ratio of two dilated solids sides compared to the ratio of their volumes?____

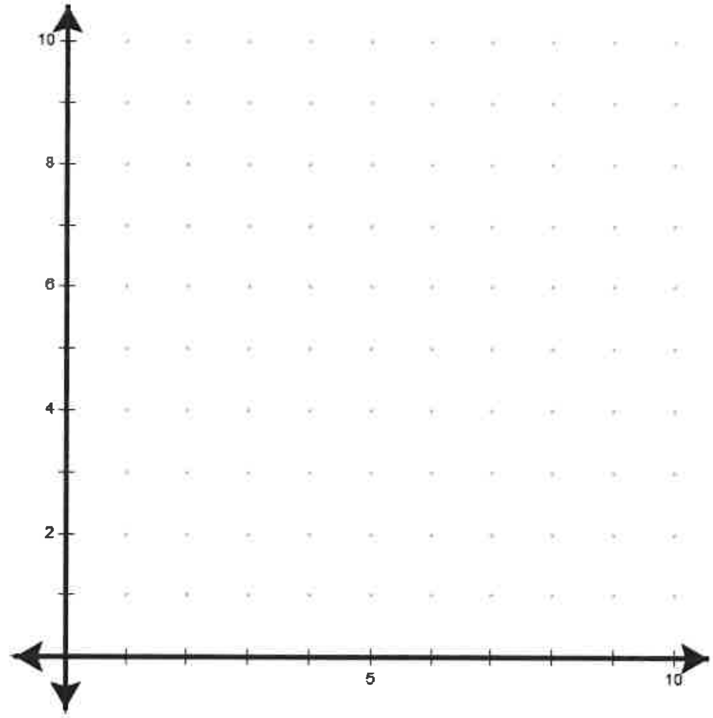
____

Coordinate Dilations

Name:

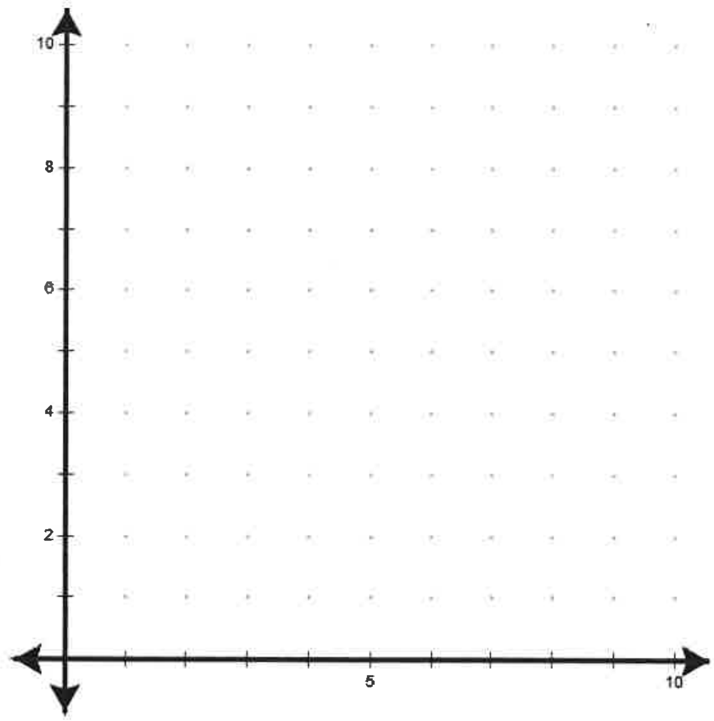
1. Plot the following points and connect the consecutive points.

A(5,1)	H(6,4)	O(2,7)
B(8,2)	I(5,6)	P(3,8)
C(9,3)	J(6,5)	Q(1,7)
D(9,7)	K(5,7)	R(1,3)
E(7,8)	L(4,6)	S(2,2)
F(8,7)	M(4,4)	A(5,1)
G(7,5)	N(3,5)	



2. First use the dilation rule D:(x,y) → (0.5x, 0.5y) using the points from the previous problem, plot the newly created points, and connect the consecutive points.

A'( )	H'( )	O'( )
B'( )	I'( )	P'( )
C'( )	J'( )	Q'( )
D'( )	K'( )	R'( )
E'( )	L'( )	S'( )
F'( )	M'( )	A'( )
G'( )	N'( )	



3. What would happen with the rule: D:(x,y) → (3x, 3y)?______________________________

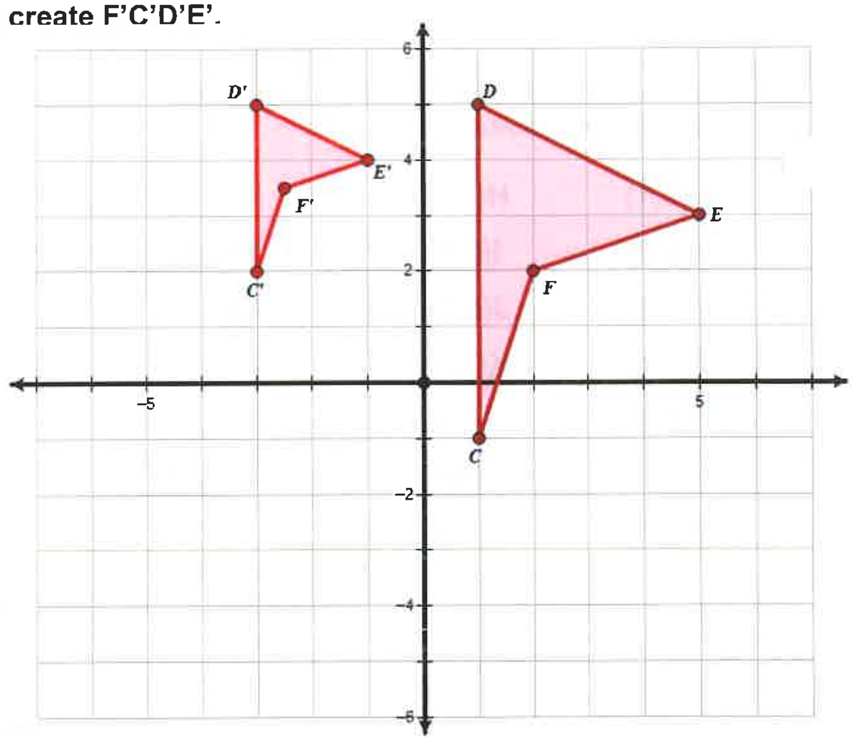
4. What would happen with the rule: D:(x,y) → (0.5x, 1y)?______________________________

5. What would happen with the rule: R:(x,y) → (– 1x, 1y)?______________________________

6. What would happen with the rule: R:(x,y) → (1x, – 1y)?______________________________

7. Figure FCDE has been dilated to create to create F’C’D’E’.

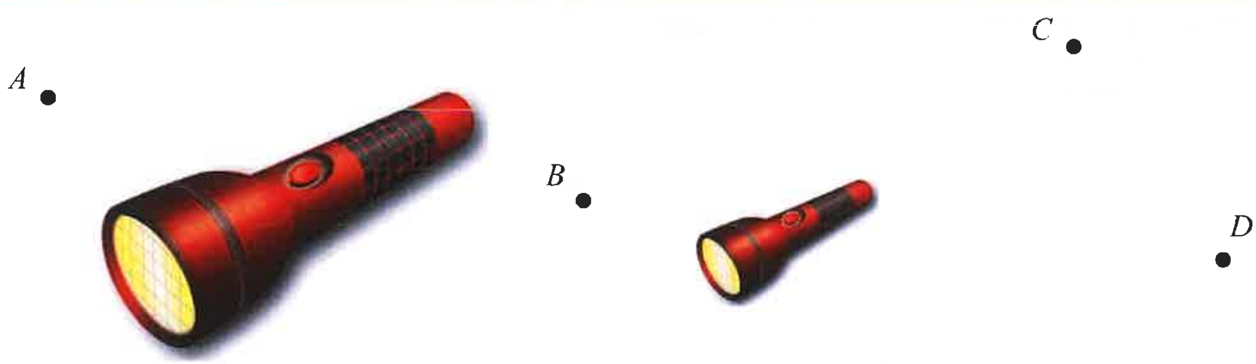
   a. What is the dilation scale factor?



   b. What is the location of the center of dilation?

   c. What is the ratio of the areas?

8. Which point would be the center of dilation?

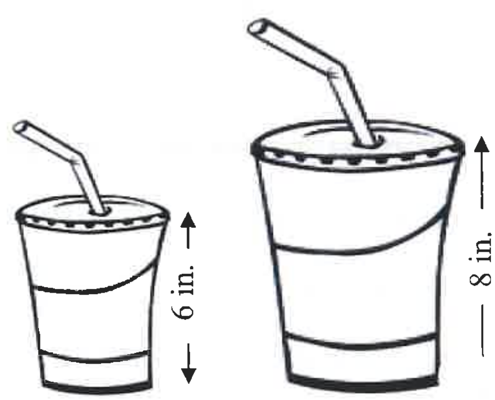


9. The different sizes of soft drink cups at a movie theater are created by using dilations. If the large is 8 inches tall and the medium is 6 inches tall, answer the following.

   a. What is the scale factor of the dilation from a large to a medium drink?

   b. What is the ratio of the volumes of the two drinks?

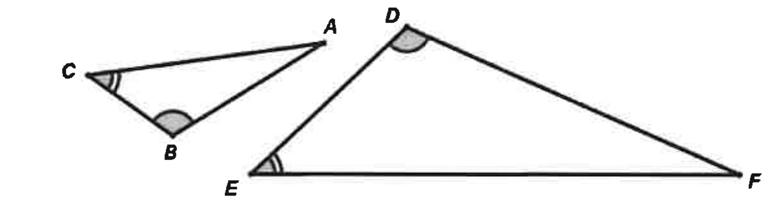
   c. If the large holds 30 ounces, how much does the medium hold?



Two figures are considered to be <u>SIMILAR</u> if the two figures have the same shape but may differ in size. To be similar by definition, all corresponding sides have the same ratio OR all corresponding angles are congruent. Alternately, if one figure can be considered a transformation *(rotating, reflection, translation, or dilation)* of the other then they are also similar.

Two triangles are similar if one of the following is true:

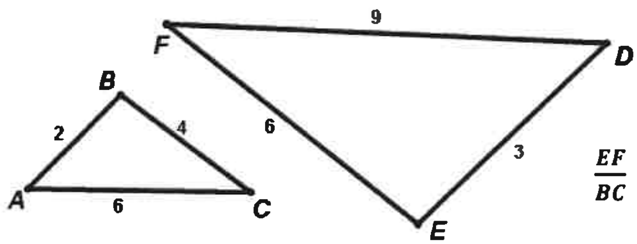
1) (AA) Two corresponding pairs of angles are congruent.



$\Delta ABC \sim \Delta FDE$

Notice that in the similarity statement above that corresponding angles must match up.

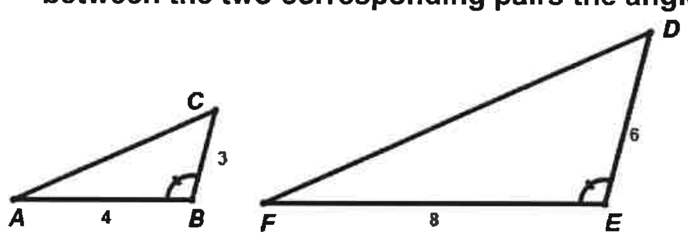
2) (SSS) Each pair of corresponding sides has the same ratio.



$\Delta ABC \sim \Delta DEF$

$$\frac{EF}{BC} = \frac{DE}{AB} = \frac{FD}{CB} = \frac{3}{2} = 1.5$$

3) (SAS) Two pairs of corresponding sides have the same ratio and the angle between the two corresponding pairs the angle is congruent.



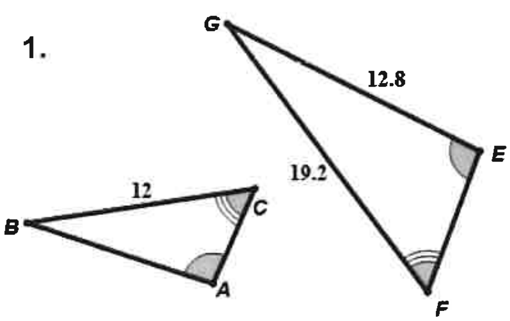
$$\frac{FE}{AB} = \frac{ED}{BC} = 2$$

$$\measuredangle ABC \cong \measuredangle FED$$

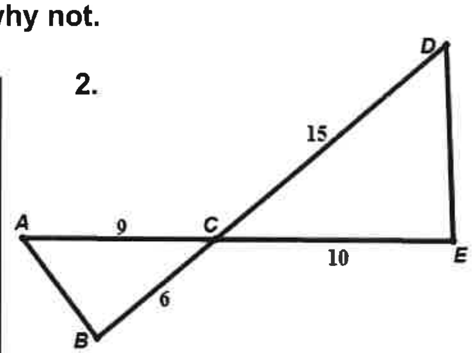
$\Delta ABC \sim \Delta FED$

Determine whether the following figures are similar. If so, write the similarity ratio and a similarity statement. If not, explain why not.

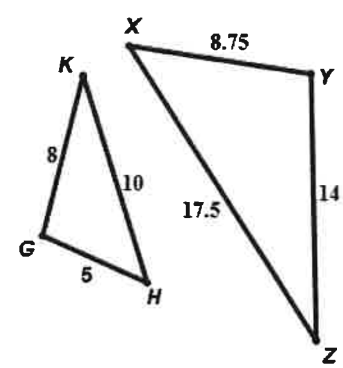
1.



2.



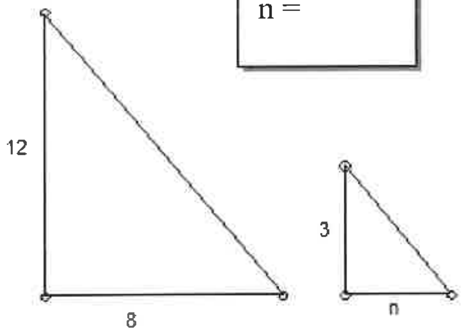
3.



Assuming the following figures are similar use the properties of similar figures to find the unknown.

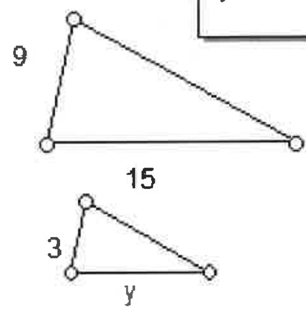
4.

n =



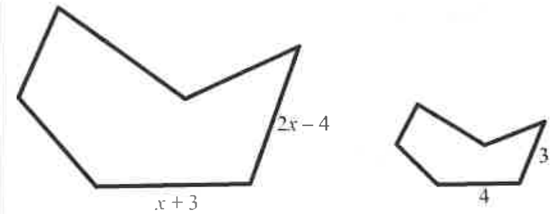
5.

y =



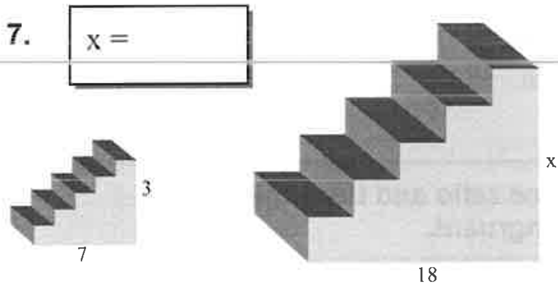
6.

x =



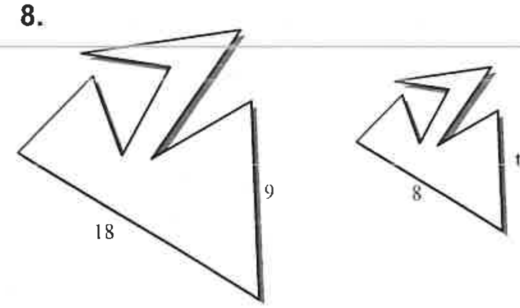
7.

x =



8.

t =



9. **Given the similarity statement $\Delta ABC \sim \Delta DEF$ and the following measures, find the requested measures. It may help to draw a picture.**

- $AB = 8$
- $AC = 10$
- $DE = 20$
- $EF = 30$
- $m\measuredangle ABC = 40°$
- $m\measuredangle EFD = 30°$

a. Find the measure of DF = ________________

b. Find the measure of BC = ________________

c. Find the measure of $m\measuredangle DEF$ = ____________

d. Find the measure of $m\measuredangle BCA$ = ____________

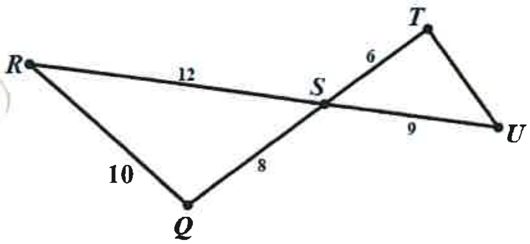
e. Find the measure of $m\measuredangle CAB$ = ____________

f. Which angles are **ACUTE**?________________

g. Which angles are **OBTUSE**?______________

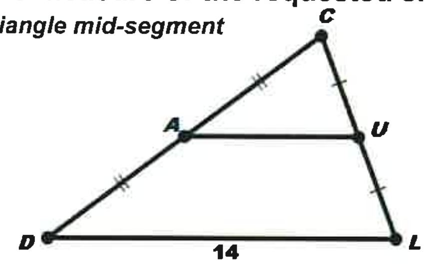
16. Explain why the reason the triangles are similar and find the measure of the requested side.

A)



What is the measure of TU?

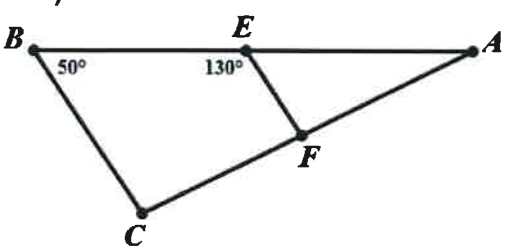
B) ***Triangle mid-segment***



What is the measure of AU?

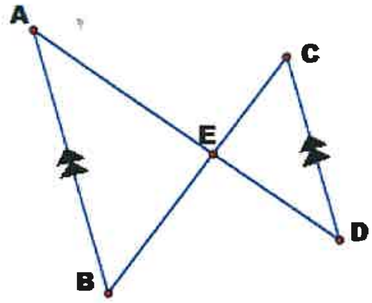
17. Using (SSS, AA, SAS) which triangles can you determine must be similar? (explain why)

A)



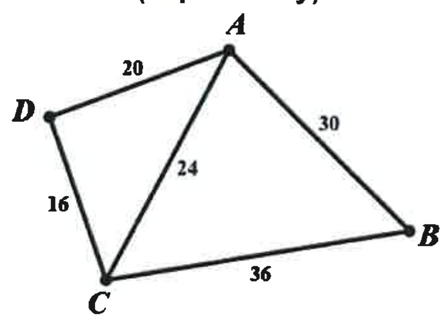
circle one		
Similar?	YES	NO
SSS	AA	SAS

B)



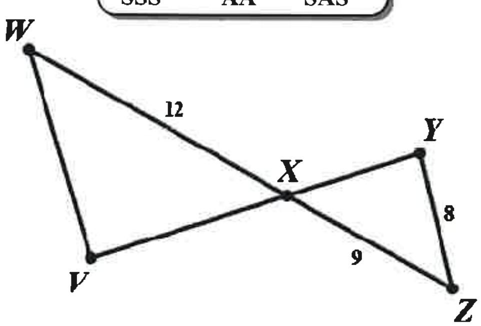
circle one		
Similar?	YES	NO
SSS	AA	SAS

C)



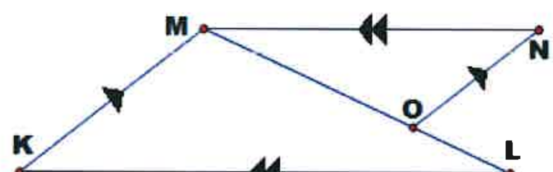
circle one		
Similar?	YES	NO
SSS	AA	SAS

D)



circle one		
Similar?	YES	NO
SSS	AA	SAS

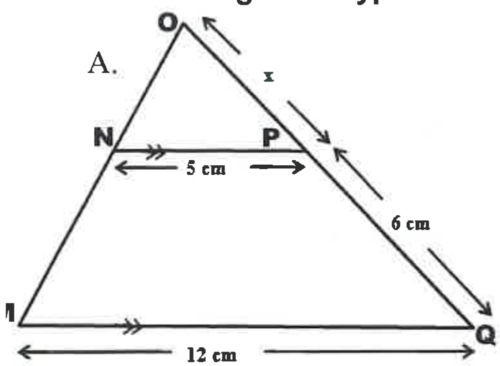
E)



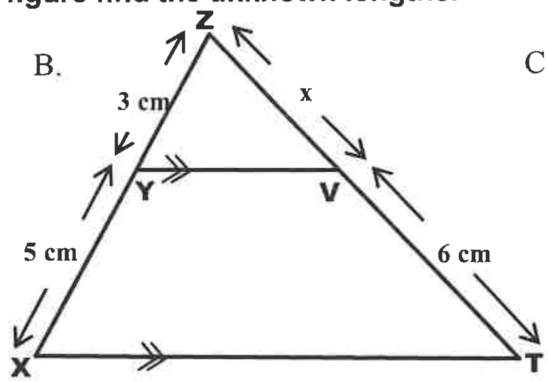
circle one		
Similar?	YES	NO
SSS	AA	SAS

18. Using some type of similar figure find the unknown lengths.

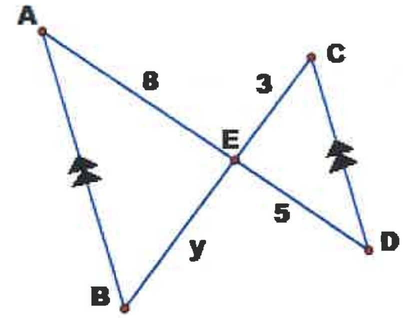
A.



B.

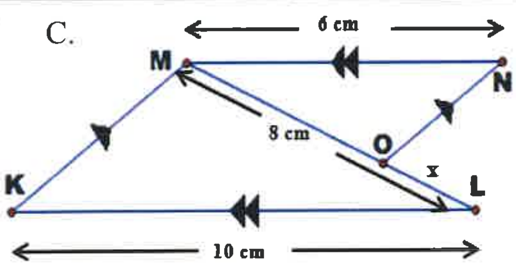


C.

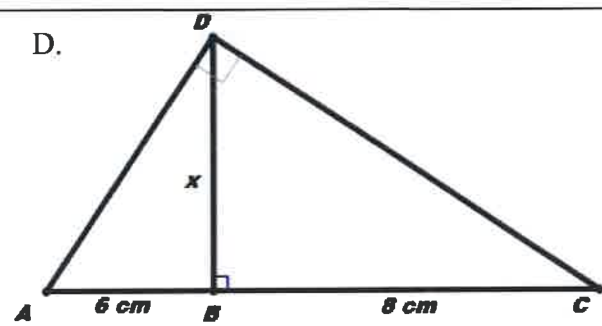


22a. $x =$	22b. $x =$	22c. $y =$

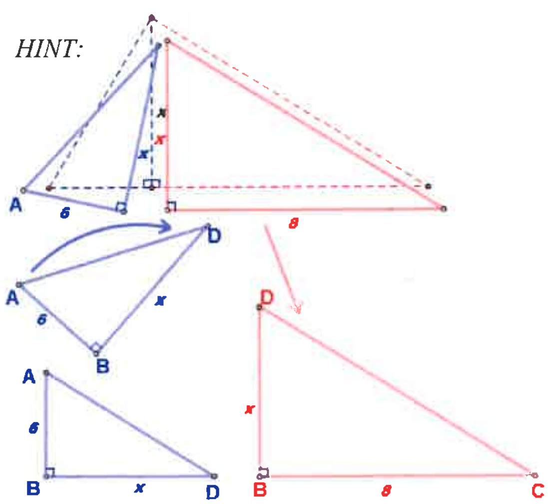
C.



D.



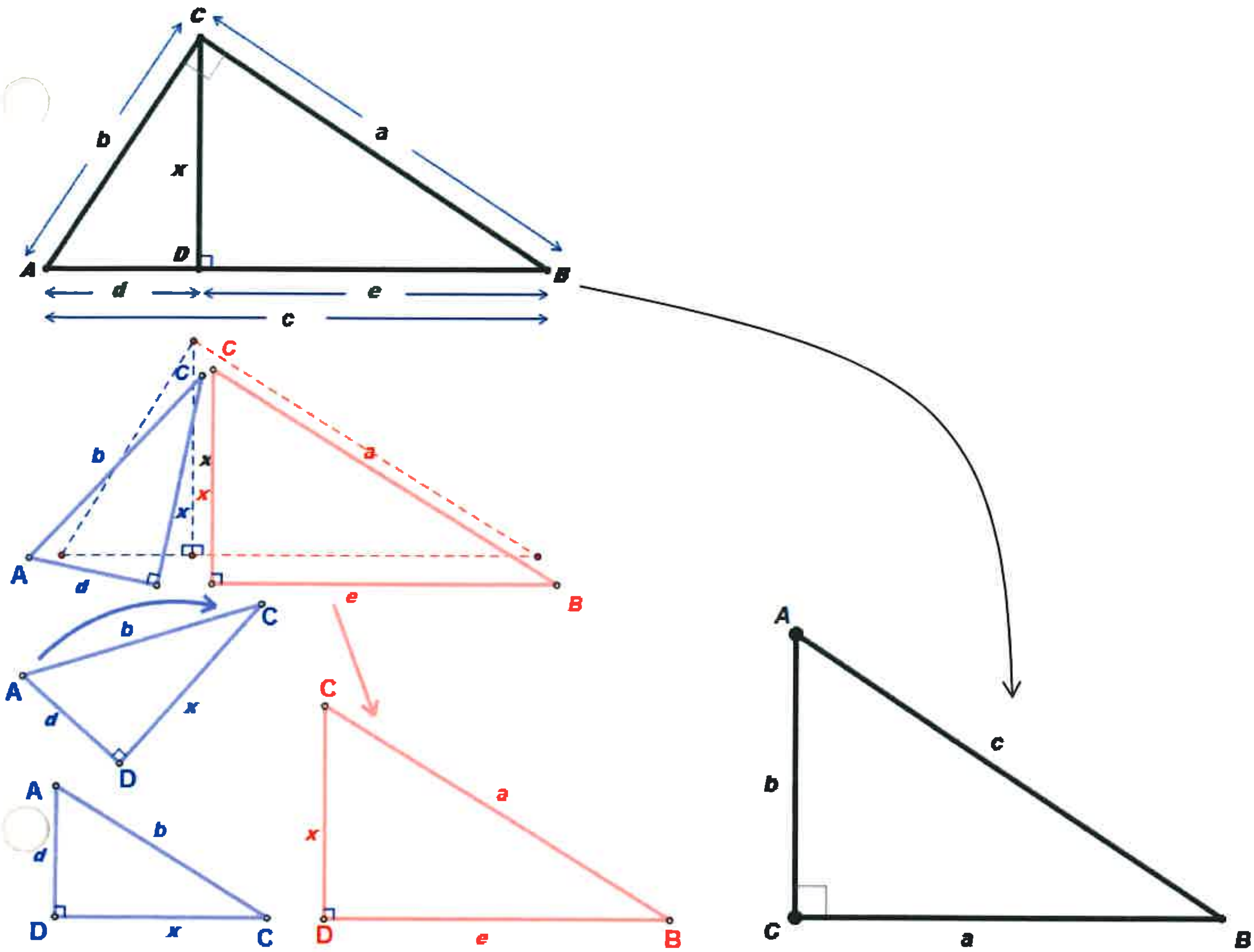
HINT:



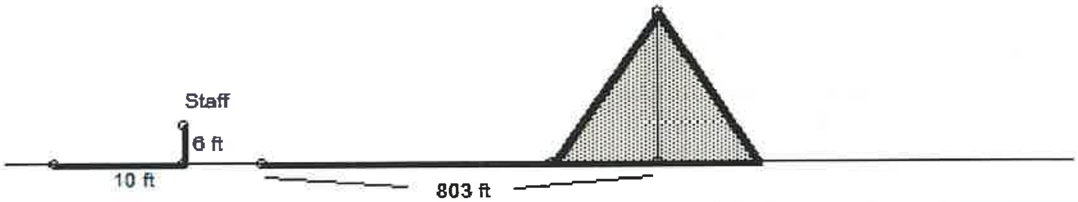
22c. $x =$

22d. $x =$

Prove the Pythagorean Theorem ($a^2 + b^2 = c^2$) using similar right triangles:



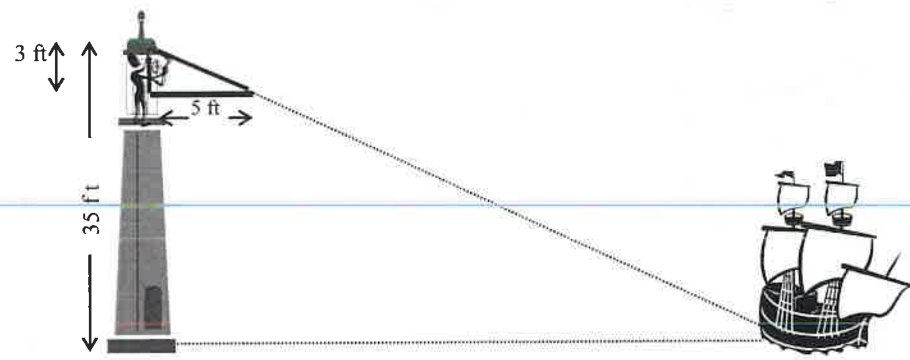
10. Thales was one of the first to see the power of the property of ratios and similar figures. He realized that he could use this property to measure heights and distances over immeasurable surfaces. Once, he was asked by a great Egyptian Pharaoh if he knew of a way to measure the height of the Great Pyramids. He looked at the Sun, the shadow that the pyramid cast, and his 6 foot staff. By the drawing below can you figure out how he found the height of the pyramid?



Height of Pyramid:

---

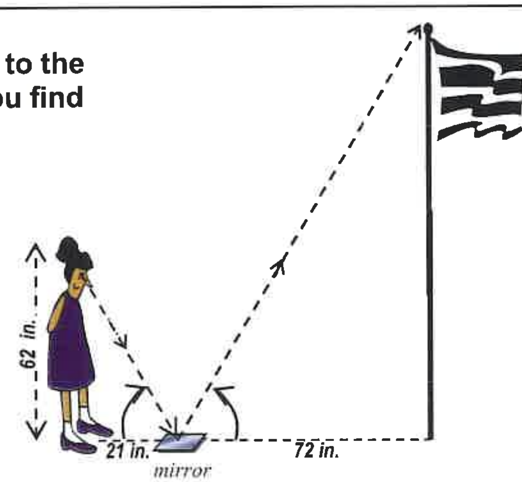
11. Using similar devices he was able to measure ships distances off shore. This proved to be a great advantage in war at the time. How far from the shore is the ship in the diagram?



Distance from Shore:

---

12. Using a mirror you can also create similar triangles (Thanks to the properties of reflection similar triangles are created). Can you find the height of the flag pole?

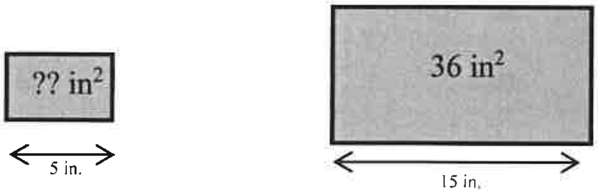


Height of the Flag Pole:

---

13. Use your knowledge of special right triangles to measure something that would otherwise be immeasurable.

14. **Find the unknown area based on the pictures below.**

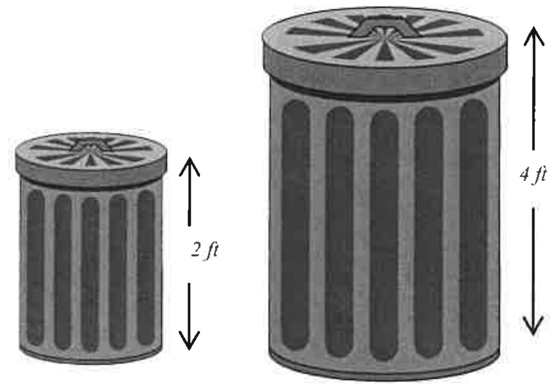


Scale Factors	Length	Area

Area of the small square:

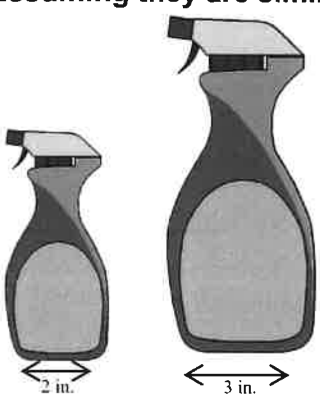
15. **If the small can holds 20 gallons how much will the big trashcan hold (assuming they are similar shapes)**

Scale Factors	Similarity Ratios	Length	Area	Volume



Volume of the Large Trashcan:

16. **If the smaller spray bottle holds 37 fl. oz. , then how much does the larger one hold assuming they are similar shapes?**

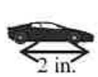


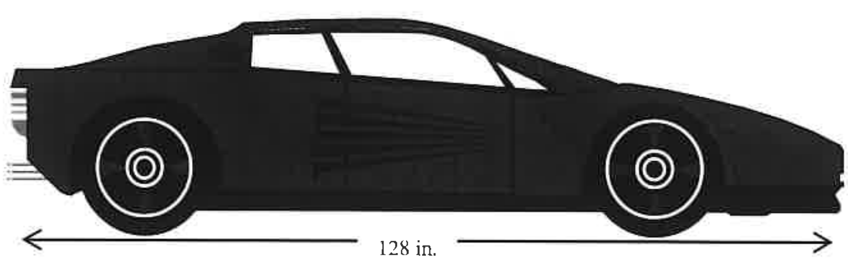
Scale Factors	Similarity Ratios	Length	Area	Volume

Volume of the Large Spray Bottle:

17. **The smaller of the two cars is a Matchbox car set at the usual $\frac{1}{64}$ th scale (the length) and it takes 0.003 fluid ounces to paint the car. If the smaller is a perfect scale of the actual car and the ratios of the paint remains the same then how many gallons of paint will be needed for the real car? (128 fl. oz = 1 gallon)**

Scale Factors	Similarity Ratios	Length	Area	Volume

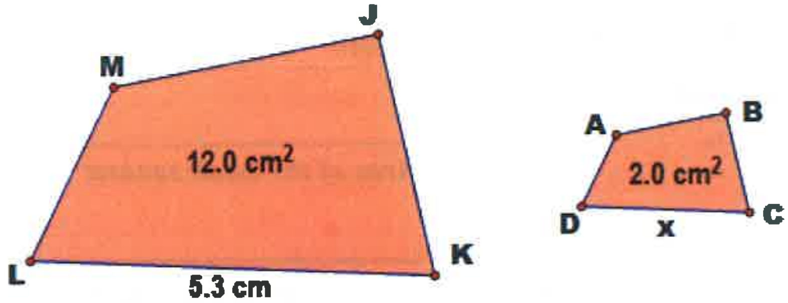




Amount of Paint Needed:

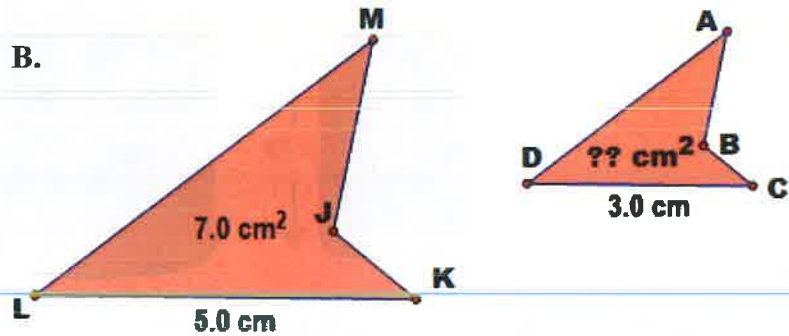
18. If the following are similar determine the length of the unknown sides.

A.



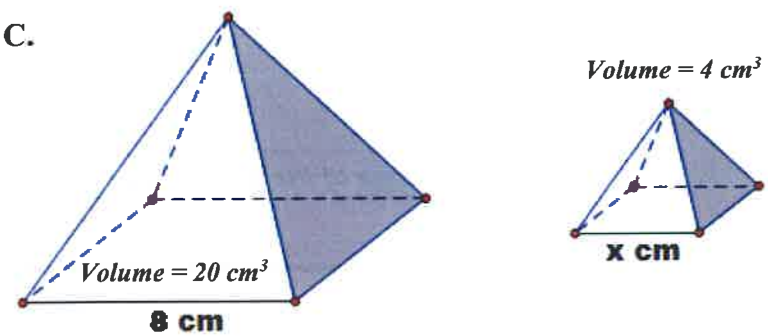
18a.
x =

B.



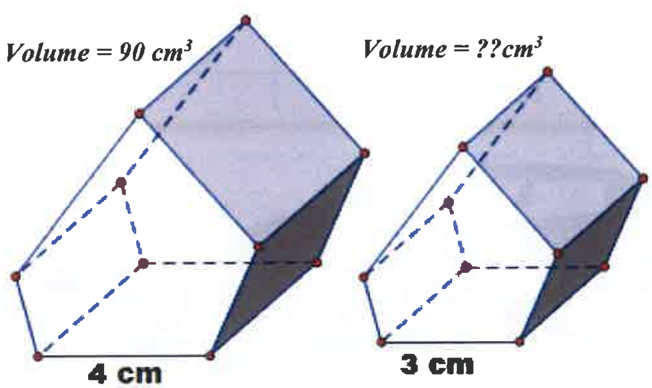
18 b.
V =

C.



18c.
x =

D.



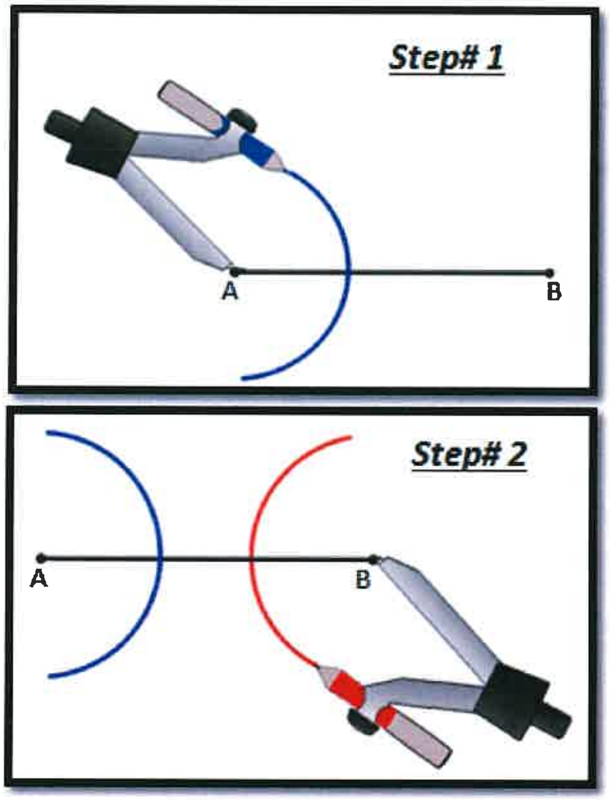
18d.
V =

Name: ________________________ Class: ___________________ Date: __________ ID: A

02-02-Sample Quiz-Basic Constructions

Multiple Choice
Identify the choice that best completes the statement or answers the question.

____ 1. Eric was attempting to construct a perpendicular bisector to the segment $\overline{AB}$ with a compass and straight edge.



Which of the below statements explains what Eric may have done wrong?

a. Eric should have started by putting the compass needle point at the midpoint of the segment $\overline{AB}$.

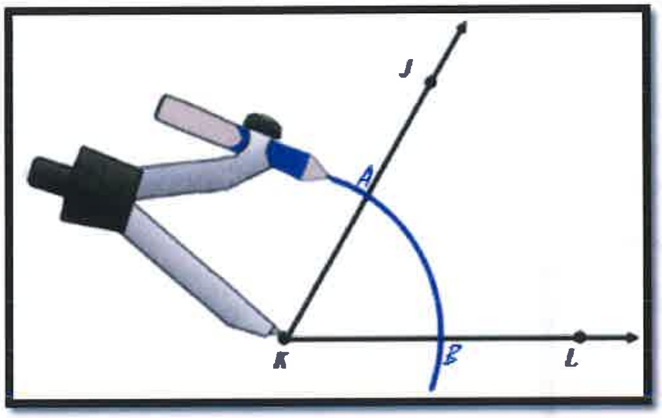
b. On the second step, Eric should have placed the compass needle point where the first arc intersected the segment $\overline{AB}$.

c. Erick just needed to open the compass more to create arcs that have a radius of more than half the length of the segment $\overline{AB}$.

d. Erick didn't do anything wrong he just needs to connect the opposite endpoints of each arc to finish the construction.

Name: ____________________ ID: A

____ 2. Janice is attempting to construct the angle bisector of $\angle JKL$. Her first step is shown below.



What would be the next step?

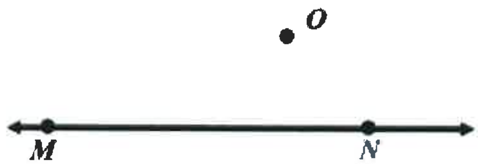
a. Create another arc with the same radius as the first but placing the compass needle point at point L.

b. She would need to first open the compass so that the compass needle point is at the point K and the pencil end of the compass is at intersection point A.

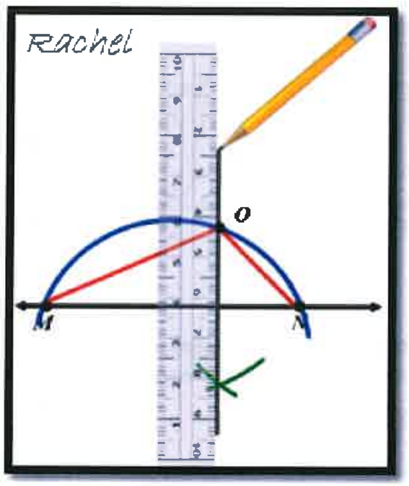
c. She would need to use a straight edge and draw a line through point L and point A. She would also need to draw line through point B and point J.

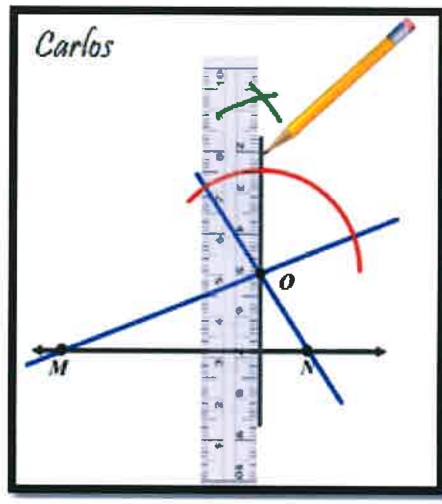
d. She would need to first open the compass so that the compass needle point is at the intersection point A and the pencil end of the compass is at intersection point B.

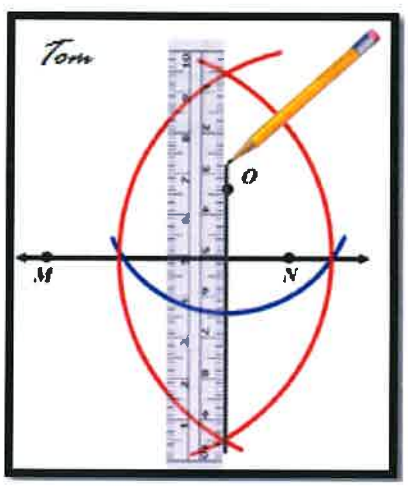
3. The students in Ms. Logan's class were asked to construct a line perpendicular to line $\overleftrightarrow{MN}$ through point O.

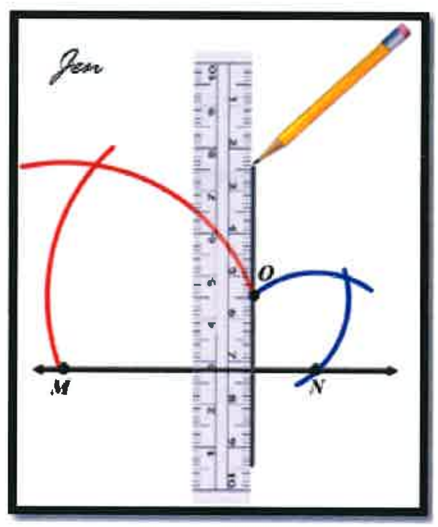


Below are four different student responses. Which student appears to have done the construction correctly?

a. 

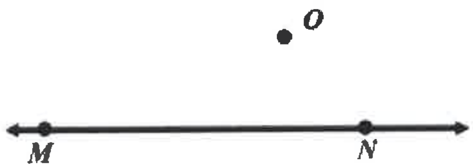
c. 

b. 

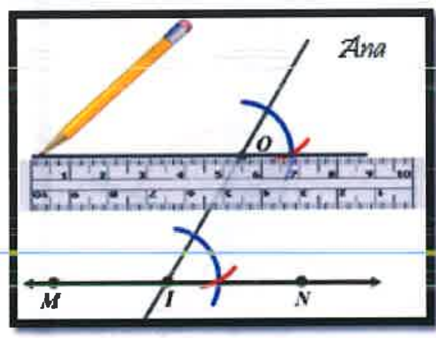
d. 

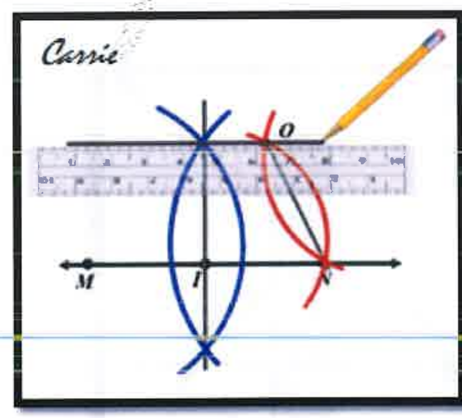
Name: ____________________ ID: A

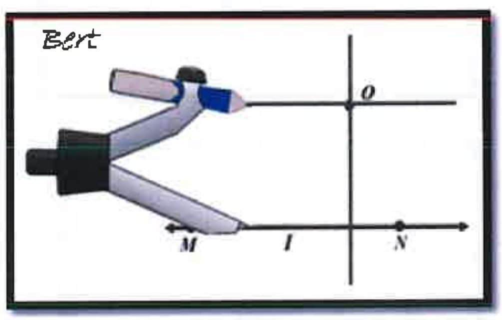
____ 4. The students in Mr. Martin's class were asked to construct a line parallel to line $\overleftrightarrow{MN}$ through point O.

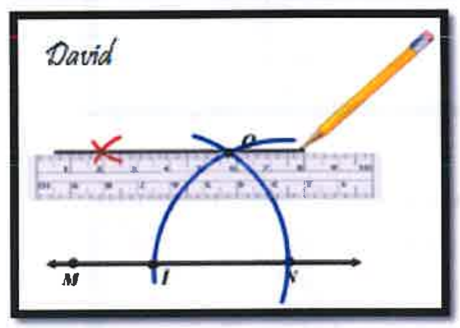


Below are four different student responses. Which student appears to have done the construction correctly?

a.


c.


b.


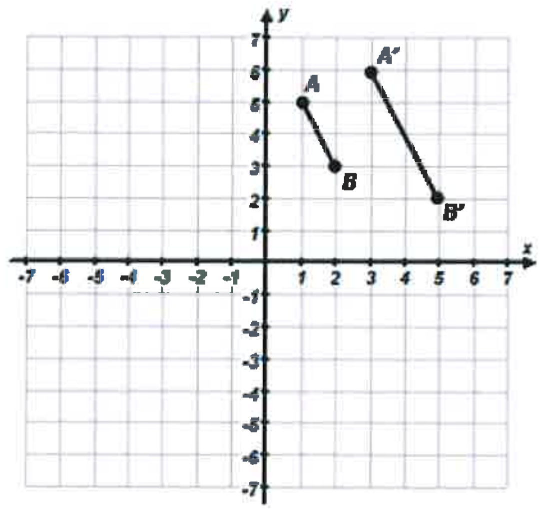
d.


Name: ______________________ Class: __________________ Date: __________ ID: A

02-03-Sample Quiz-Dilations

Multiple Choice
Identify the choice that best completes the statement or answers the question.

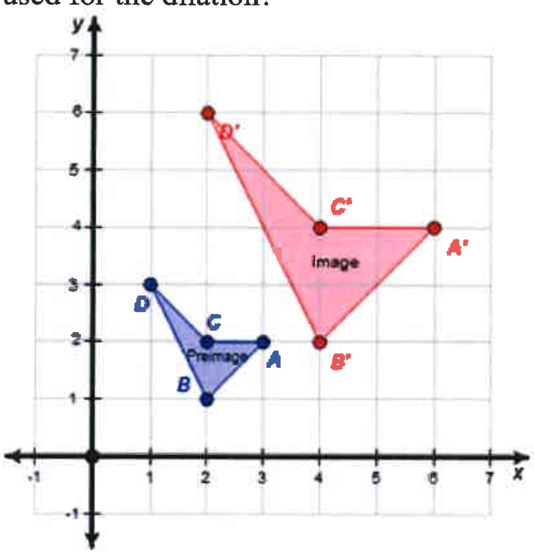
____ 1. The segment $\overline{AB}$ is dilated to create the segment $\overline{A'B'}$.



What are the coordinates of the **center** of the dilation?

a. $(5,7)$
b. $(8,1)$
c. $(0,4)$
d. $(-1,4)$

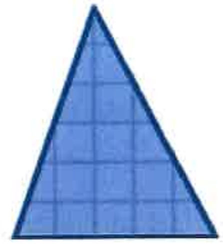
____ 2. In the diagram shown below quadrilateral ABCD is dilated from the origin. What **scale factor** was used for the dilation?



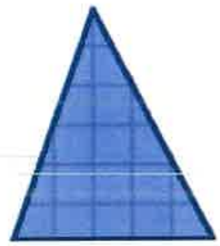
a. Scale Factor = 0.5
b. Scale Factor = 1.5
c. Scale Factor = 2
d. Cannot be determined

Name: ________________________ **ID: A**

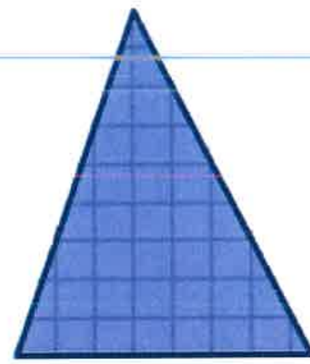
____ 3. If the triangle at the right were to be dilated using a scale factor of $\frac{1}{2}$, which is the only triangle below that could represent the transformed image?



a. 

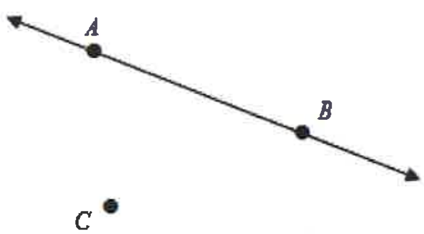
c. 

b. 

d. 

____ 4. If you were to dilate line $\overleftrightarrow{AB}$ using point C as the center of dilation, which of the below would best describe the newly created image?

(You may assume that the scale factor is not 1.)



a. A line segment

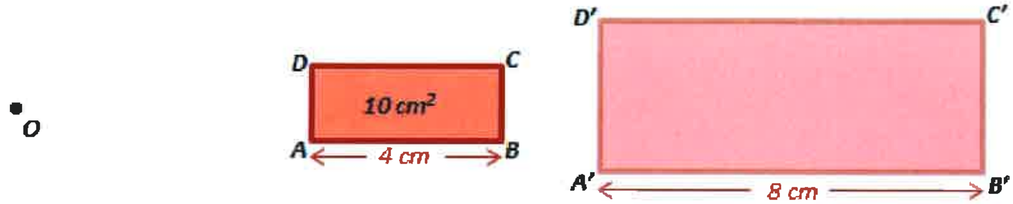
b. A ray

c. A line that is parallel to line $\overleftrightarrow{AB}$

d. A line that is in the same place as line $\overleftrightarrow{AB}$

Name: ________________________ ID: A

____ 5. A rectangle ABCD is dilated from point O to create the rectangle A'B'C'D'.



Given **AB = 4 cm**, **A'B' = 8 cm**, and the **area of ABCD = 10 cm²**, determine the **ratio of the areas** between the two similar shapes.

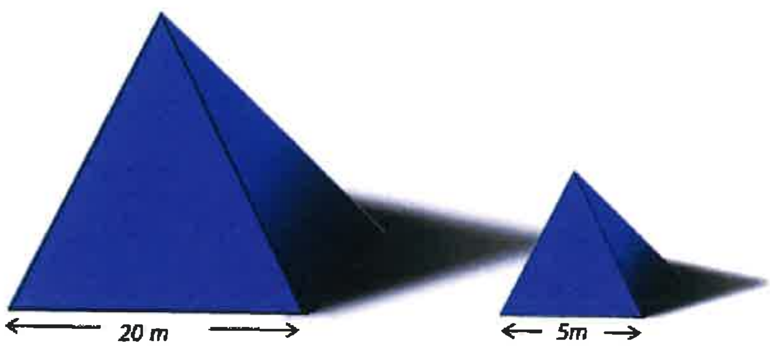
a. $\frac{2}{1}$

b. $\frac{4}{1}$

c. $\frac{8}{1}$

d. $\frac{16}{1}$

____ 6. One pyramid is dilation of the other.



Which would best represent the **ratio of the volumes** between the two similar pyramids?

a. 1:2

b. 1:4

c. 1:16

d. 1:64

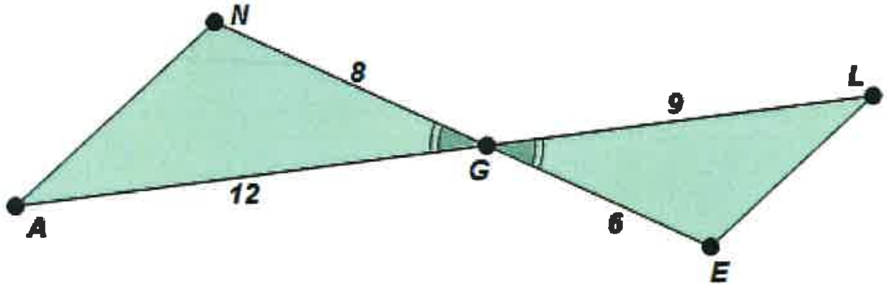
Name: ________________________ **Class:** ___________________ **Date:** __________ **ID: A**

02-04-Sample Quiz-Similar Figures

Multiple Choice

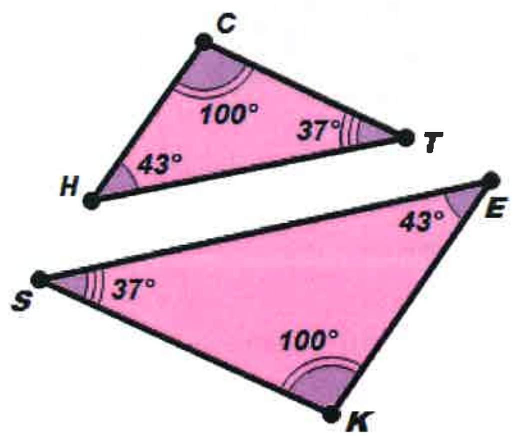
Identify the choice that best completes the statement or answers the question.

____ 1. If the triangles shown in the diagram are similar, which would be a correct similarity statement with an appropriate reason?



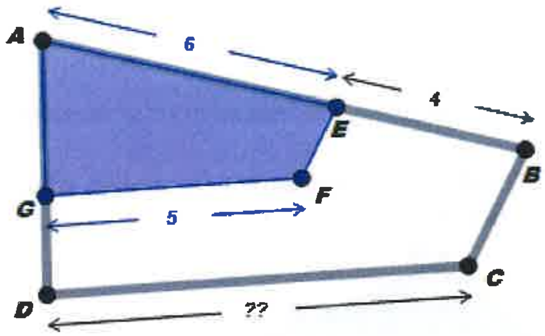
a. $\triangle ANG \sim \triangle LEG$ because **SSS** (proportional sides)
b. $\triangle ANG \sim \triangle LEG$ because **SAS** (proportional sides)
c. $\triangle ANG \sim \triangle GLE$ because **SSS** (proportional sides)
d. $\triangle ANG \sim \triangle GLE$ because **SAS** (proportional sides)
e. The triangles are **NOT** similar.

____ 2. Which of the below is the only correct similarity statement for the two triangles?



a. $\triangle SKE \sim \triangle THC$
b. $\triangle SKE \sim \triangle TCH$
c. $\triangle SKE \sim \triangle CHT$
d. $\triangle SKE \sim \triangle HCT$

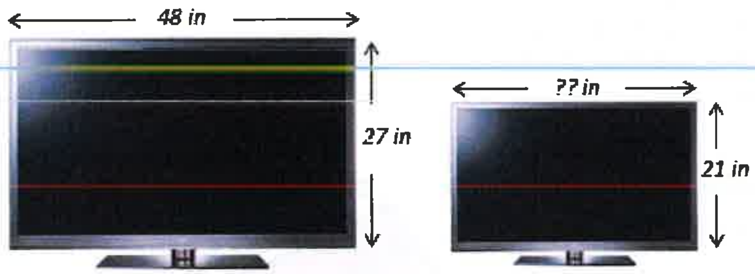
____ 3. The quadrilateral $\square ABCD$ is dilated using the center point A to create quadrilateral $\square AEFG$



Determine the length of $\overline{DC}$ based on the measures provided in the diagram.

a. 5.0 *units*
b. 7.5 *units*
c. $7.\overline{6}$ *units*
d. $8.\overline{3}$ *units*

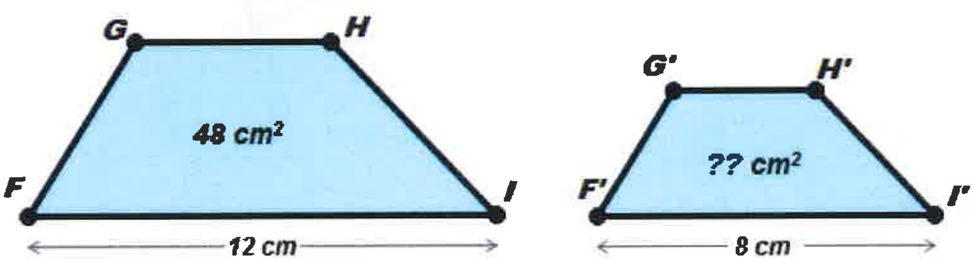
____ 4. A TV manufacturer makes two models of a particular TV that are nearly identical except one is a dilation of the other.



If it is a proper dilation what should the approximate length of the smaller TV be?

a. 35.2 *inches*
b. 36.5 *inches*
c. $37.\overline{3}$ *inches*
d. $39.\overline{6}$ *inches*

____ 5. A quadrilateral FGHI is similar to quadrilateral F'G'H'I'.

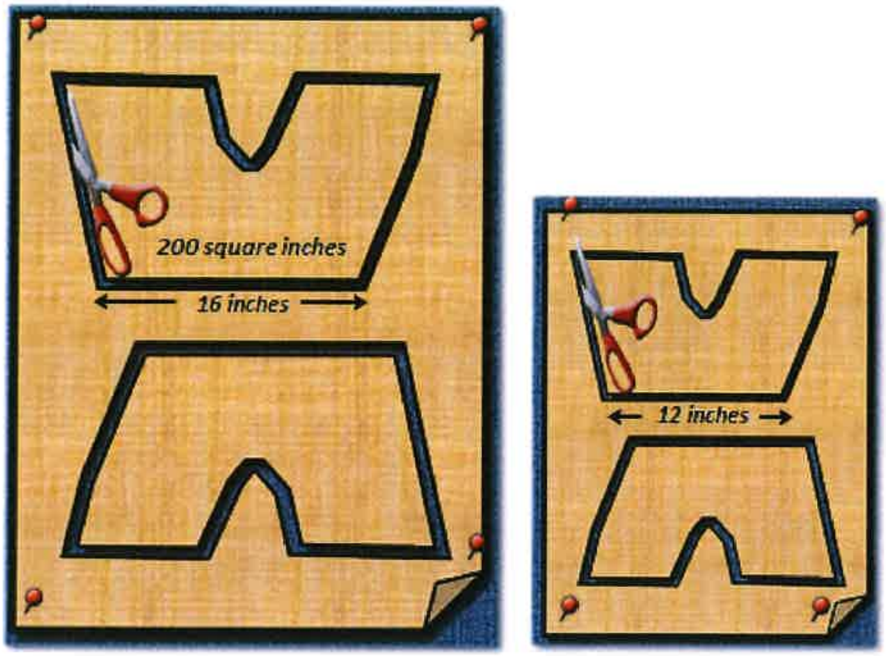


Given **FI = 12 cm**, **F'I' = 8 cm**, and the **area of FGHI = 48 cm²**, determine the **area of F'G'H'I'**.

a. 12.5 cm^2
b. 24 cm^2
c. $21.\overline{3}$ cm^2
d. 32 cm^2

Name: ________________________ **ID: A**

____ 6. Two panels of denim are to be cut out to create a pair of shorts. One set of shorts are being created for an adult and one set is for a child.

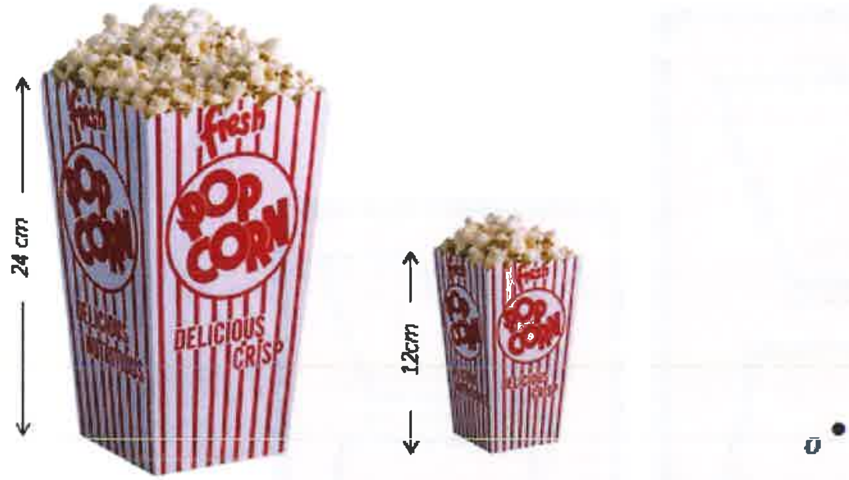


The child size is simply a dilation of the adult size pattern. The first adult panel measures 16 inches across and uses an **area** of 200 square inches of denim fabric. The first child size panel measures 12 inches across. How much denim fabric will be required for the first child size panel?

a. 80 square inches
b. 112.5 square inches
c. 120.5 square inches
d. 150 square inches

Name: ________________________ **ID: A**

____ 7. A regular popcorn at a movie theater is 24 cm tall and holds approximately 600 popped popcorn kernels.



If a child's size is simply a dilated version of the regular container as shown above, how many popped popcorn kernels should the child's size hold? *(Hint: use the scale factor for* ***volume****)*

a. 50 popped popcorn kernels
b. 75 popped popcorn kernels
c. 150 popped popcorn kernels
d. 300 popped popcorn kernels

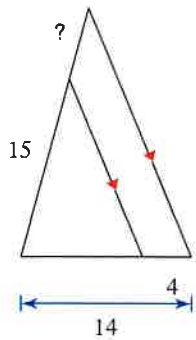
Kuta Software - Infinite Geometry Name________________________

Proportional Parts in Triangles and Parallel Lines

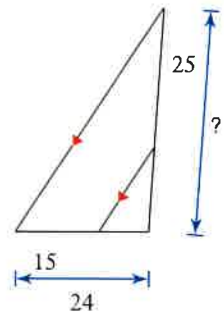
Date________________ Period____

Find the missing length indicated.

1)



2)



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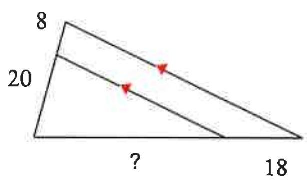
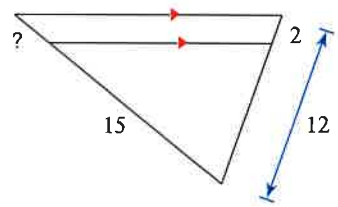
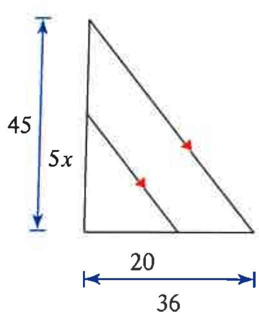


4)

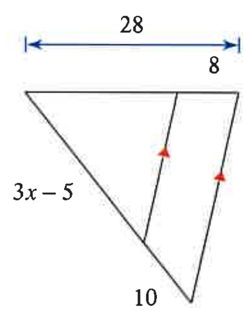


Solve for *x*.

5)

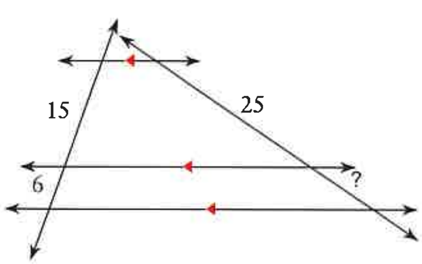


6)



Find the missing length indicated.

7)



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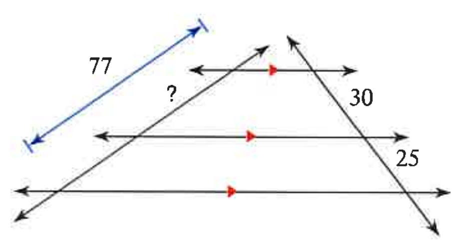

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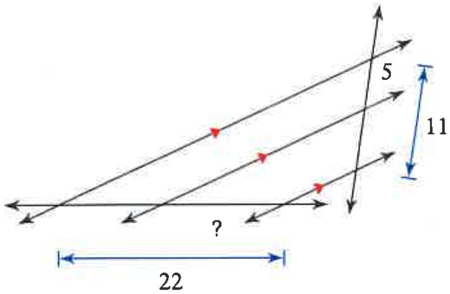

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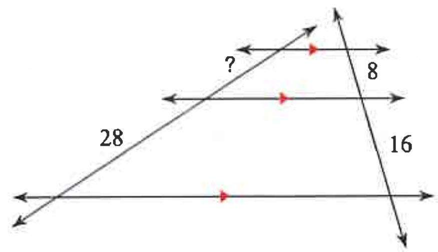
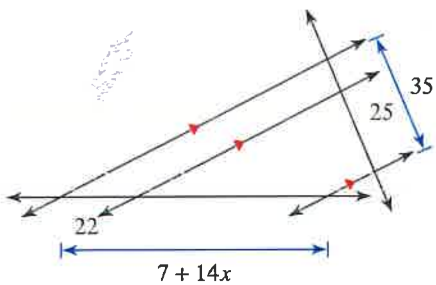
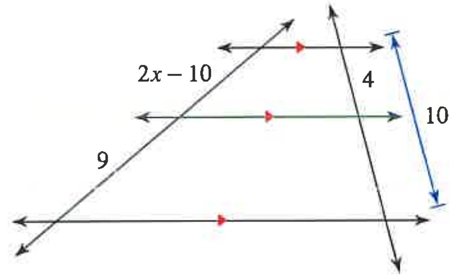


Solve for x.

11)

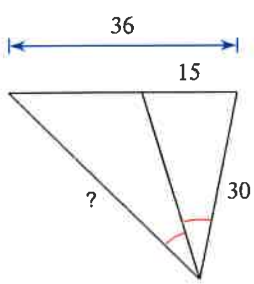


12)

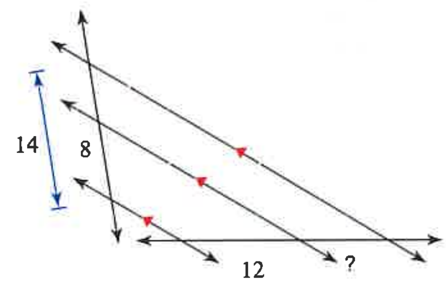


Find the missing length indicated.

13)



14)



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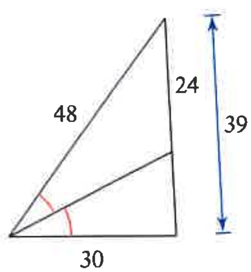
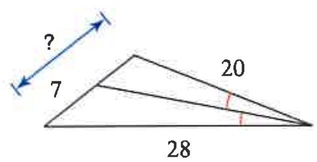
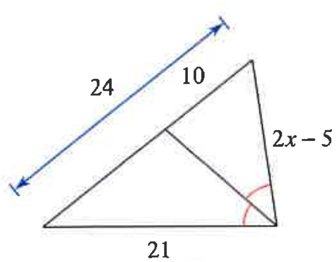


16)

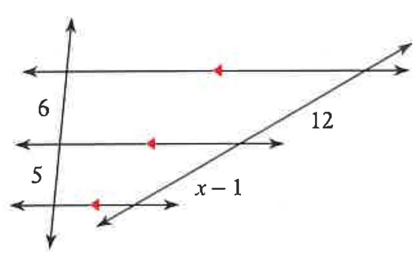


Solve for x.

17)



18)



Kuta Software - Infinite Geometry

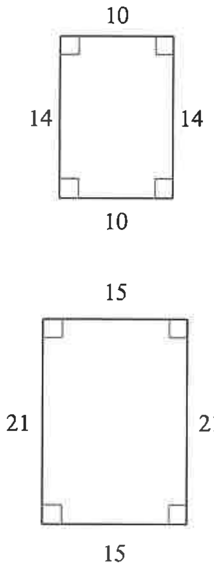
Name________________________

Similar Polygons

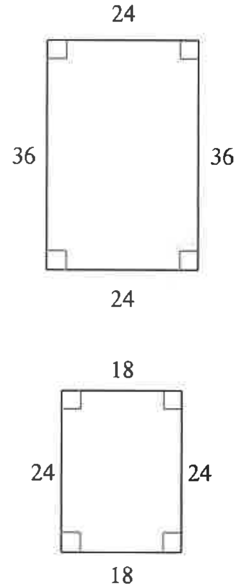
Date________________ Period____

State if the polygons are similar.

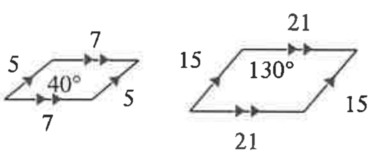
1)



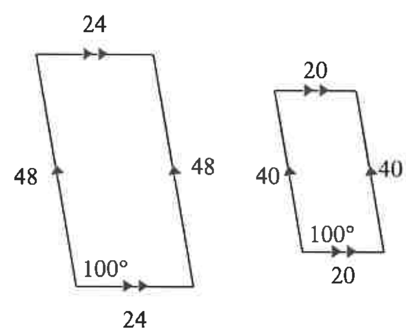
2)



3)



4)



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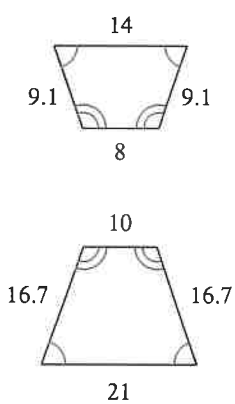
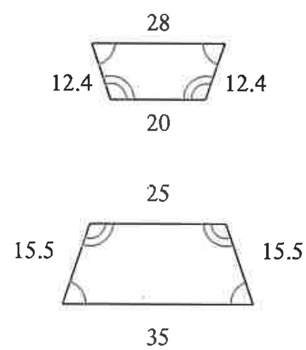
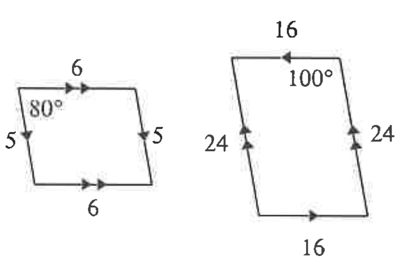


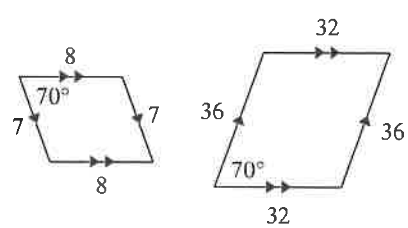
6)



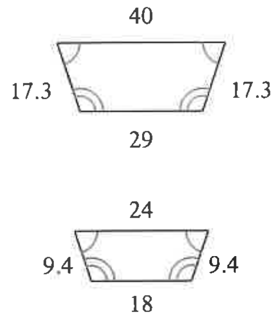
7)



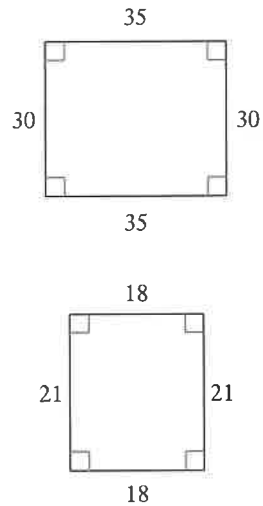
8)



9)

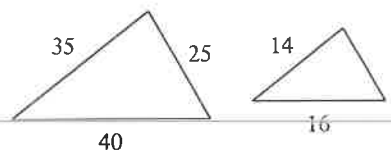


10)

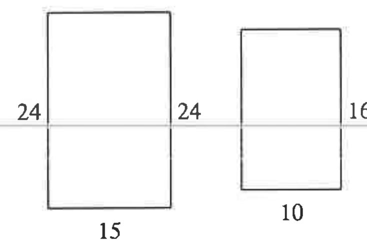


The polygons in each pair are similar. Find the scale factor of the smaller figure to the larger figure.

11)



12)



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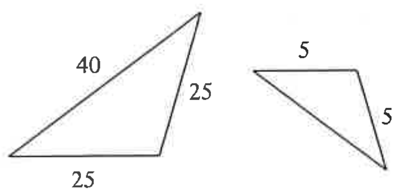

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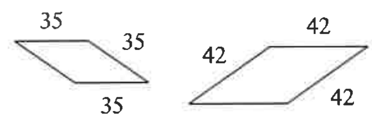

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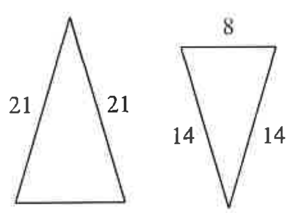

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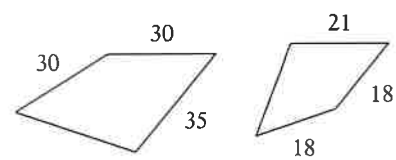
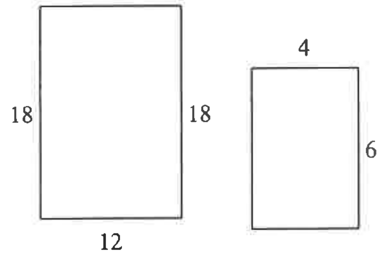
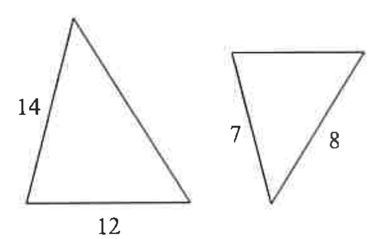


17)



18)



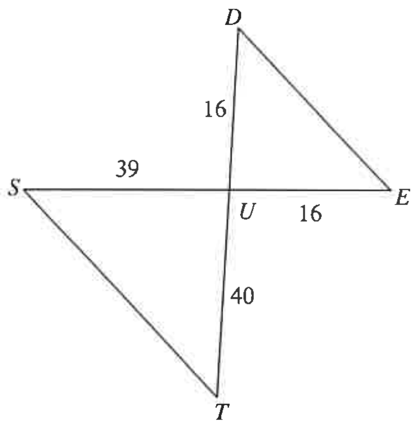
Kuta Software - Infinite Geometry   Name________________________

Similar Triangles

Date________________ Period____

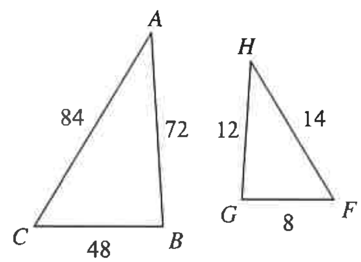
State if the triangles in each pair are similar. If so, state how you know they are similar and complete the similarity statement.

1)



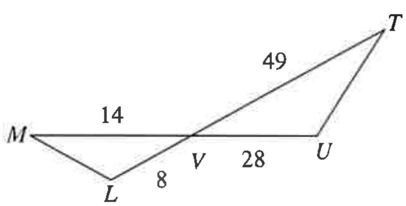
$\Delta UTS \sim$ ______

2)



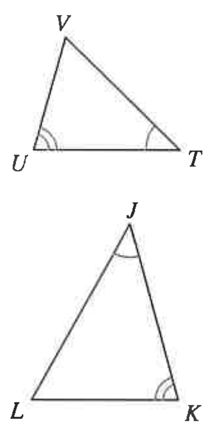
$\Delta CBA \sim$ ______

3)



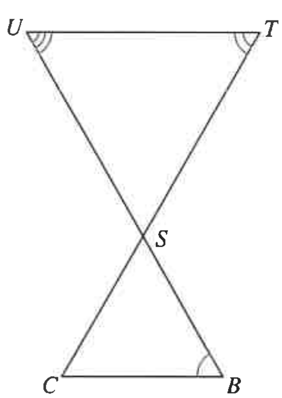
$\Delta VUT \sim$ ______

4)



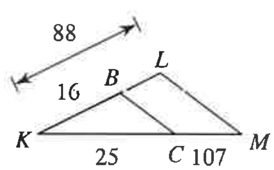
$\Delta JKL \sim$ ______

5)



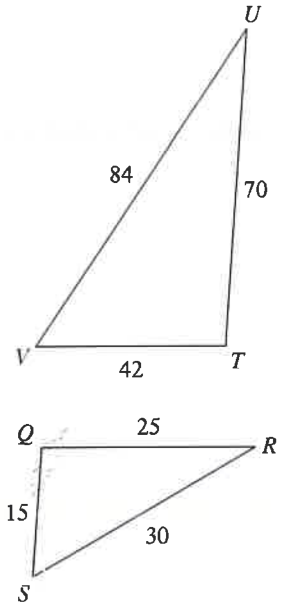
$\Delta STU \sim$ ______

6)



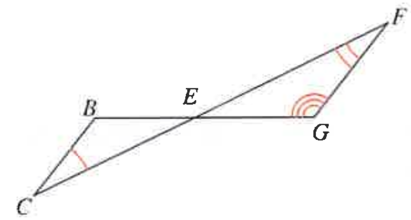
$\Delta KLM \sim$ ______

7)



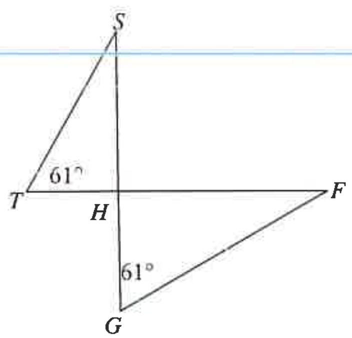
$\Delta TUV \sim$ ______

8)



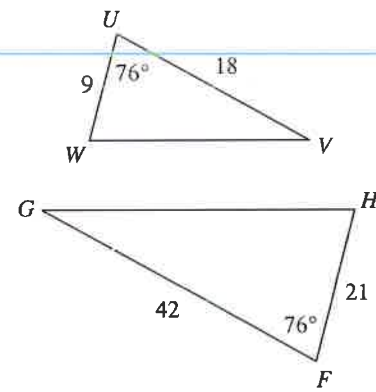
$\Delta EFG \sim$ ______

9)



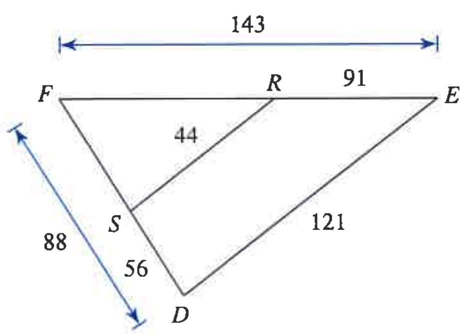
$\Delta HGF \sim$ ______

10)



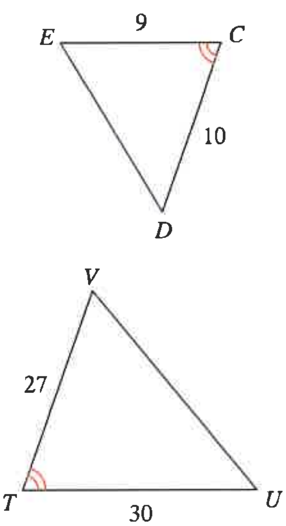
$\Delta FGH \sim$ ______

11)



$\Delta FED \sim$ ______

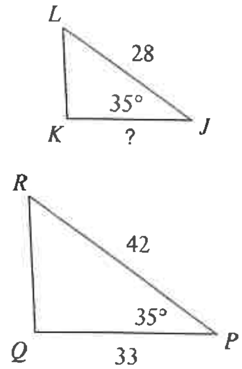
12)



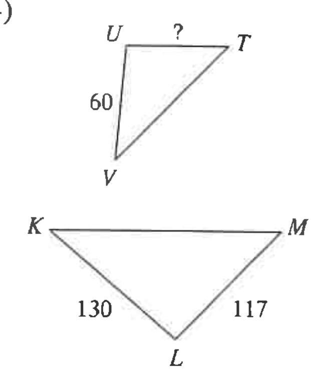
$\Delta TUV \sim$ ______

Find the missing length. The triangles in each pair are similar.

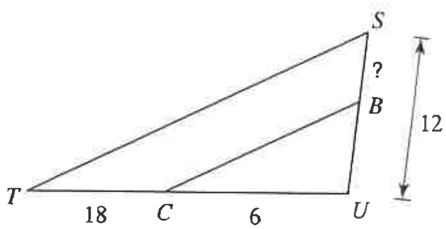
13)



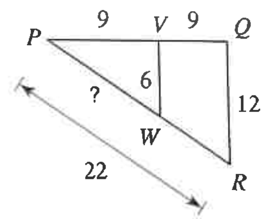
14)



15)

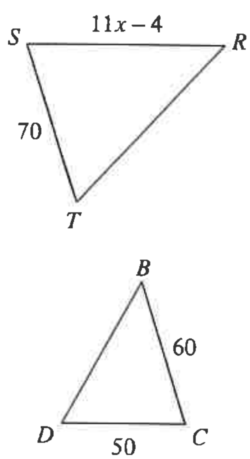


16)

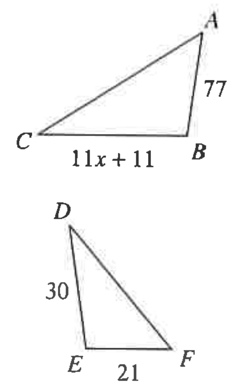


Solve for x. The triangles in each pair are similar.

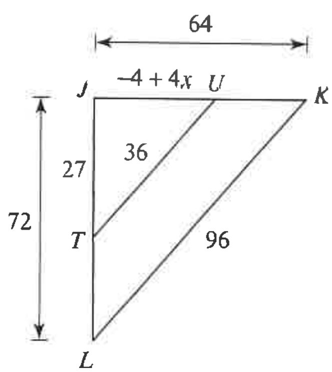
17)



18)



19)



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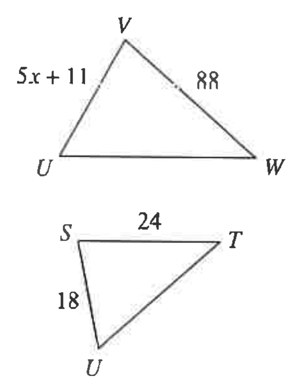


Kuta Software - Infinite Geometry

Name________________________________

Solving Proportions

Date________________ Period____

Solve each proportion. Leave your answer as a fraction in simplest form.

1) $\frac{6}{2} = \frac{4}{p}$

2) $\frac{4}{k} = \frac{8}{2}$

3) $\frac{n}{4} = \frac{8}{7}$

4) $\frac{5}{3} = \frac{x}{4}$

5) $\frac{m}{5} = \frac{7}{2}$

6) $\frac{7}{4} = \frac{r}{5}$

7) $\frac{7}{6} = \frac{5}{x}$

8) $\frac{6}{5} = \frac{2}{5n}$

Solve each proportion. Round your answers to the nearest hundredth.

9) $\frac{7.7}{3.6} = \frac{2.3}{b}$

10) $\frac{v}{4.9} = \frac{5.4}{6.1}$

11) $\frac{6.3}{x} = \frac{2.56}{9.3}$

12) $\frac{3.4}{x} = \frac{2.17}{7.7}$

Solve each proportion. Leave your answer as a fraction in simplest form.

13) $\frac{9}{8} = \frac{k+6}{6}$

14) $\frac{2}{10} = \frac{4}{a-3}$

15) $\frac{10}{p+2} = \frac{4}{3}$

16) $\frac{4}{6} = \frac{8}{x-1}$

17) $\frac{m}{8} = \frac{m+7}{9}$

18) $\frac{n}{n+1} = \frac{3}{5}$

19) $\frac{9}{4} = \frac{r-10}{r}$

20) $\frac{x+6}{x} = \frac{10}{7}$

21) $\frac{n-9}{n+5} = \frac{7}{4}$

22) $\frac{6}{b+9} = \frac{4}{b+5}$

23) $\frac{8}{3} = \frac{v-9}{7v+4}$

24) $\frac{8}{5x-4} = \frac{6}{x+5}$

Critical thinking questions:

25) Do you think that a person's age and the amount they eat each day are basically in proportion?

Kuta Software - Infinite Geometry

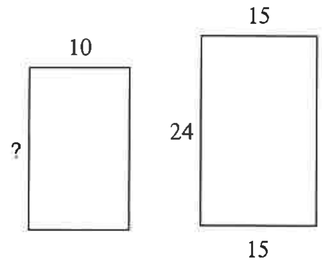
Name________________________

Using Similar Polygons

Date________________ Period____

The polygons in each pair are similar. Find the missing side length.

1)



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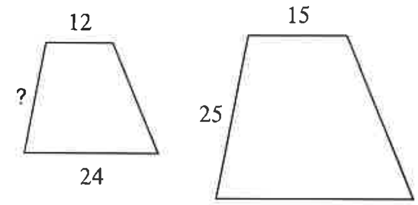

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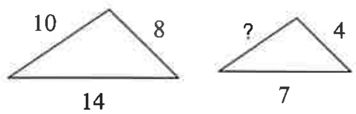

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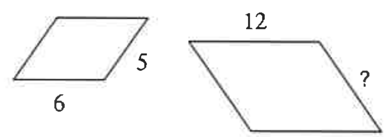

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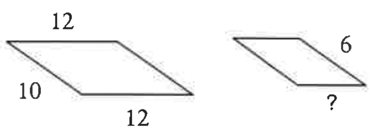

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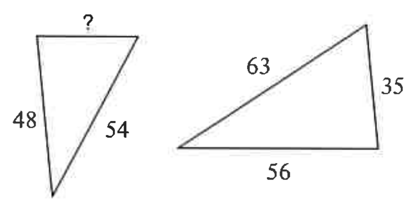

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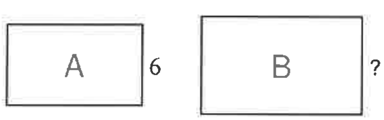
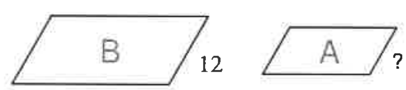


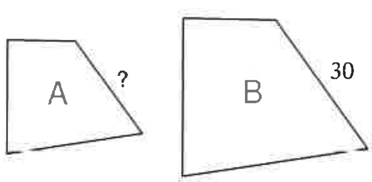
scale factor from A to B = 2 : 7

8)



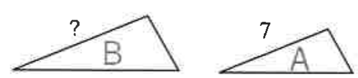
scale factor from A to B = 2 : 3

9)



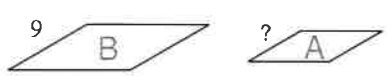
scale factor from A to B = 5 : 6

10)



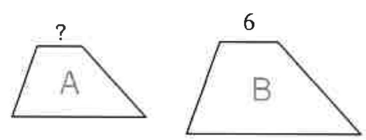
scale factor from A to B = 1 : 7

11)



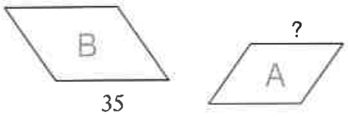
scale factor from A to B = 2 : 3

12)



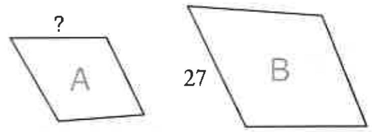
scale factor from A to B = 1 : 2

13)



scale factor from A to B = 6 : 7

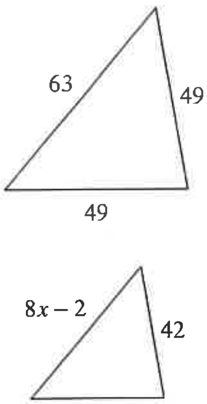
14)



scale factor from A to B = 1 : 3

Solve for *x*. The polygons in each pair are similar.

15)



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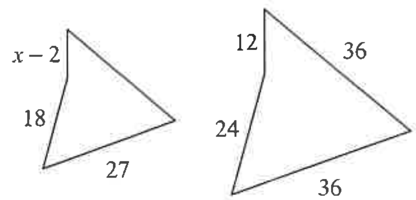

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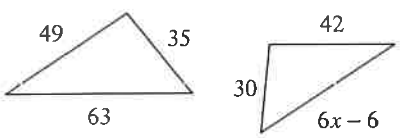

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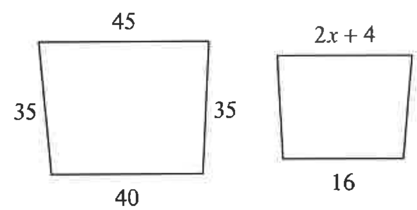
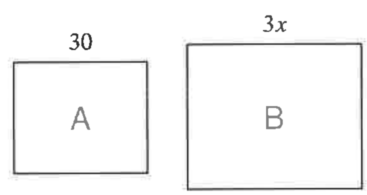


19)

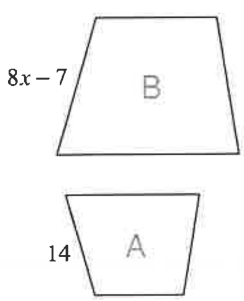
scale factor from A to B = 5 : 6

20)



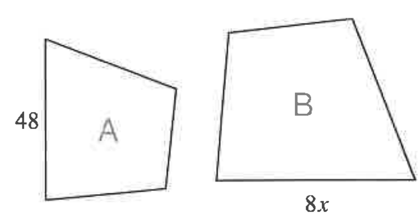
scale factor from A to B = 5 : 6

21)



scale factor from A to B = 2 : 7

22)



scale factor from A to B = 6 : 7

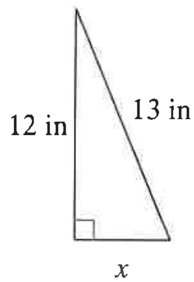
Kuta Software - Infinite Geometry

Name________________________________

The Pythagorean Theorem and Its Converse

Date________________ Period____

Find the missing side of each triangle. Round your answers to the nearest tenth if necessary.

1)


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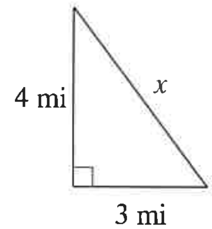

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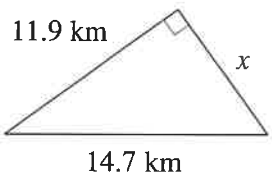

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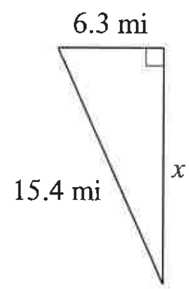
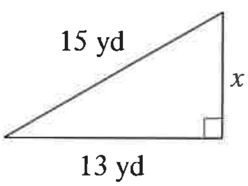
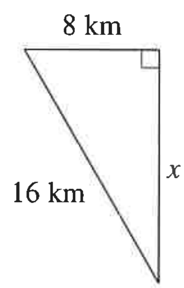


Find the missing side of each triangle. Leave your answers in simplest radical form.

5)


6)


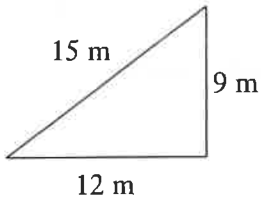
Find the missing side of each right triangle. Side c is the hypotenuse. Sides a and b are the legs. Leave your answers in simplest radical form.

7) $a = 11$ m, $c = 15$ m

8) $b = \sqrt{6}$ yd, $c = 4$ yd

State if each triangle is a right triangle.

9)



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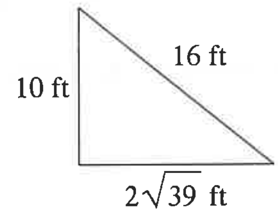

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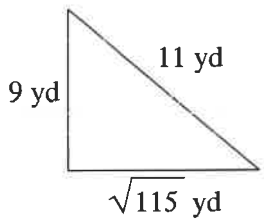
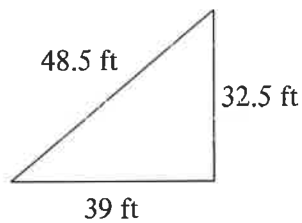


12)



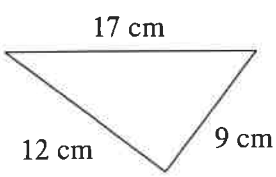
State if the three sides lengths form a right triangle.

13) 10 cm, 49.5 cm, 50.5 cm

14) 9 in, 12 in, 15 in

State if each triangle is acute, obtuse, or right.

15)



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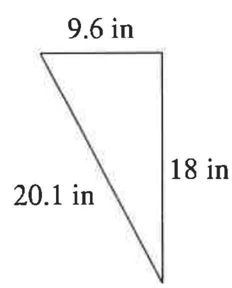

State if the three side lengths form an acute, obtuse, or right triangle.

17) 6 mi, $2\sqrt{55}$ mi, 17 mi

18) 4.8 km, 28.6 km, 29 km